Hanwha Investment&Securities

INVESTMENT STRATEGY TEAM | OCTOBER 19, 2026

Silver Lining

2027 Annual Outlook

Economy · Rates · Credit · Equities · Quant

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Summary

Economy
Rising Hurdles

- For now, this is a phase in which growth (manufacturing rebound, government-led investment, AI demand) can withstand the burdens (interest rates and prices)
- Korea and the US hold a relative advantage, while China and the Eurozone have limited recovery capacity. Attention is needed on a cyclical rebound alongside semiconductors
- The KRW/USD exchange rate is forecast to fall to the 1,200 level in the second half, reflecting the favorable supply-demand environment

Rates
Sluggish Box-Range

- Korea 3Y KTB: 3.60-4.10% / Korea 10Y KTB: 4.00-4.50%, US 2Y: 4.20-5.00% / US 10Y: 4.80-5.60%
- As we judge that clear upside and downside pressures will be absent, we expect both the Korean and US bond markets to move in a sluggish box range
- We forecast solid growth and a slow disinflation trend. Accordingly, monetary policy will remain at the terminal rate for a considerable period

Credit
Tight & Steady

- 2027 credit issuance is KRW 551tn, a decrease from 2026 on reduced bank bond issuance. For public agency bonds, LH net issuance, and for specialized credit finance bonds, the largest-ever refinancing, are the supply variables
- Considering the KTB level, credit spreads are at a low level. After the end of the hiking cycle, we expect them to confirm a peak in 1H and narrow only slightly through year-end
- We prefer bank bonds, whose issuance incentive has weakened. The lower the rating, the larger the credit spread tightening, but the concentration of corporate bond issuance early in the year and non-prime investor sentiment need to be monitored

Summary (continued)

Equities

A KOSPI Re-rating Formula Easier Than You Think

- We judge that political and geopolitical uncertainty and macro gravity will peak out in early next year. Given the characteristics of this cycle, the "bad is good" market will be extended
- Confidence in sustained growth like Taiwan's will be difficult to form. However, the elevated ROE level needs to be reflected in valuation, and the formula for doing so is not difficult
- In 2027, the KOSPI is expected to be low in the first half and high in the second half, while the 12MF P/B is expected to be high in the first half and low in the second half. We present a 2027 KOSPI band of 6,000-9,300pt and a target of 8,500pt

Quant

A Picture Where Bio Becomes the Story

- Considering the US IT sector weight, the slope of the semiconductor index and earnings momentum, it will be difficult for semiconductors to show the same upward slope as before
- We believe the true beneficiary of AI development is bio: technology toward building brains, building robots, and people who do not fall ill. We judge it could become a new engine
- Attention is needed on the secondary battery and bio sectors, which alternate between second and third place behind No. 1 semiconductors

I

Economy

Rising Hurdles

Hye-yoon Lim · hylim@hanwha.com · 3772-7728
Gyu-ho Choi · choi.gh@hanwha.com · 3772-7720

[Overview] Korea, US > Japan > China > Eurozone

- Relative advantage for Korea and the US; weaker preference for China and the Eurozone
- Korea has the advantage in terms of the semiconductor cycle and fiscal conditions. The US is attractive as purchasing power is maintained and it is at the center of the AI investment cycle
- China faces a difficult domestic demand recovery, and the Eurozone sees a limited impact from the manufacturing rebound
- By industry, keep in mind a semiconductor peak-out after 2Q. From the perspective of passing the trough, cyclical appeal may come to the fore

[Table 1] Summary of Economic Assessments for Major Countries

Country	Economic Phase	Growth Outlook	Prices	Funding Cost	Rate Burden Relative to Growth	Remarks
Korea	↗	↗↗	↘	↗	↗	Advantage in terms of the semiconductor cycle and fiscal capacity; 2027 export growth of around 20%
US	↗	↗	↘	↗↗	↗	Still preferred as it leads the AI investment cycle; sharp rise in funding costs is a burden
China	→	↘	↗	↗	↗	Competitiveness in advanced manufacturing confirmed; domestic demand recovery is not easy; considering the debt burden, capacity for economic stimulus is limited
Eurozone	↗	↘	↘	↗	↗↗	Easing inflation is favorable, but the absence of growth drivers limits the upside; France's fiscal conditions have deteriorated relative to other advanced countries
Japan	↗	→	↗	↗	↗↗	Expected to benefit from the manufacturing and investment recovery cycle; narrowing growth-interest rate gap is a burden

Note: Economic phase is judged based on the S&P Global PMI; growth outlook on changes in economic growth forecasts; funding cost on interest expense and maturities; rate burden relative to growth on the gap between growth and interest rates, etc.
Source: Hanwha Research Center

[Overview] Growth Still Has the Upper Hand

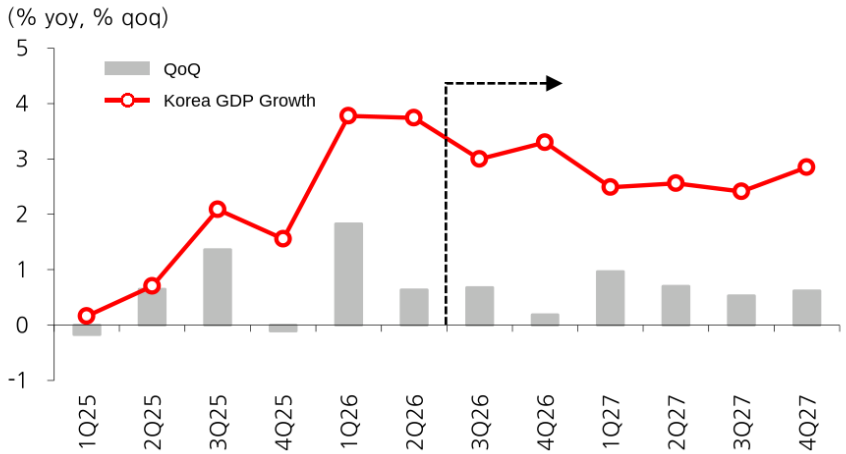
- In 2027, the global economy is expected to be led by a manufacturing rebound, government-led investment and AI demand
- Prices and interest rates are peaking out. Although the burdens of rising prices and high interest rates are accumulating, this is still a phase in which growth exceeds the burden
- Korea: solid exports and growing investment / US: still in the 2% range. Consumption slows, investment is solid
- China: a resilient domestic demand recovery is not easy / Eurozone: the price burden eases, but the economic rebound is slow

[Table 2] Economic Growth and Exchange Rate Forecasts for Major Countries

Category	Country	1Q27	2Q27	3Q27	4Q27	2026	2027
Economic Growth (% yoy)	Korea	2.5	2.6	2.4	2.9	3.5	2.6
	US	2.2	2.2	2.1	2.1	2.3	2.1
	China	4.3	4.2	4.0	4.0	4.5	4.1
	Eurozone	1.4	1.1	1.0	1.0	1.0	1.1
Exchange Rates (Period Average)	KRW/USD	1,350	1,300	1,280	1,270	1,430	1,300
	Dollar Index	102	100	99	98	100	100
	USD/EUR	1.11	1.11	1.12	1.13	1.15	1.12
	CNY/USD	6.70	6.68	6.66	6.65	6.79	6.67
	JPY/USD	158	160	158	156	158	158

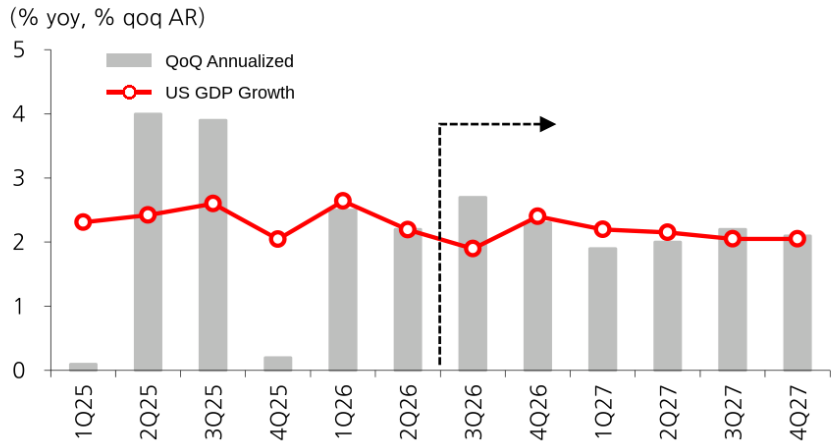
Source: Hanwha Research Center

Korea: Growth Led by Exports and Investment



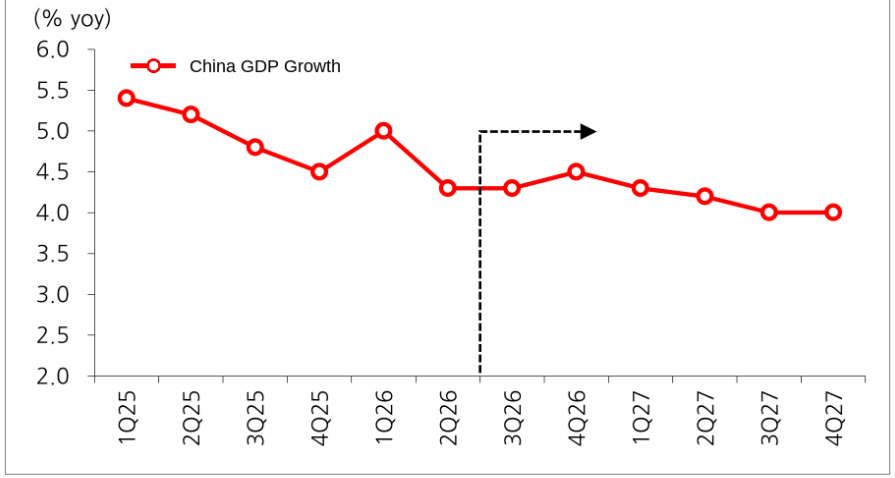
Source: Bank of Korea, Hanwha Research Center

US: 2% Growth Even as It Slows



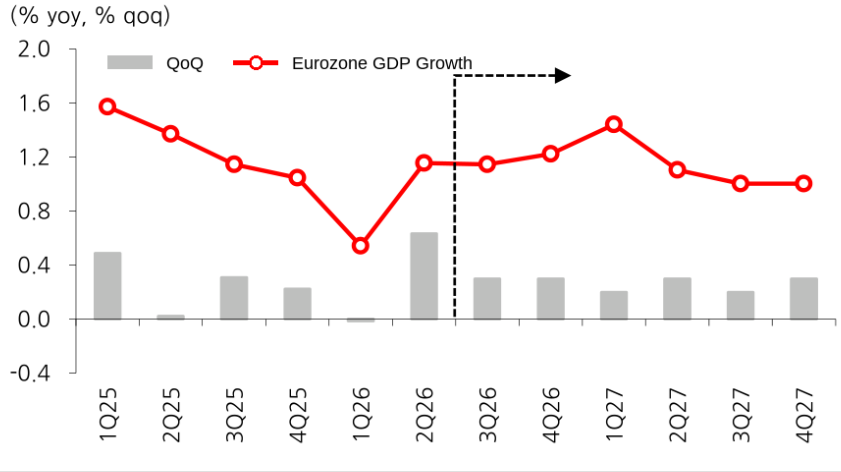
Source: Bureau of Economic Analysis, Hanwha Research Center

China: Domestic Demand Recovery Is Not Easy



Source: National Bureau of Statistics, Hanwha Research Center

Eurozone: Easing War Burden vs. Sluggish Recovery

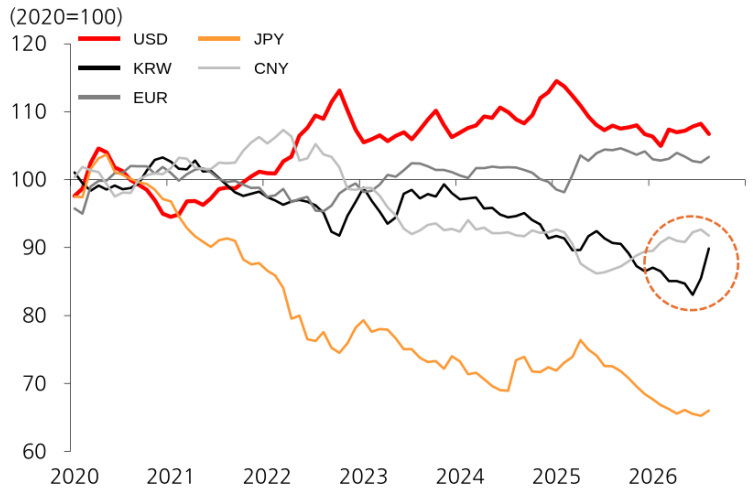


Source: Eurostat, Hanwha Research Center

[Overview] KRW > USD > JPY

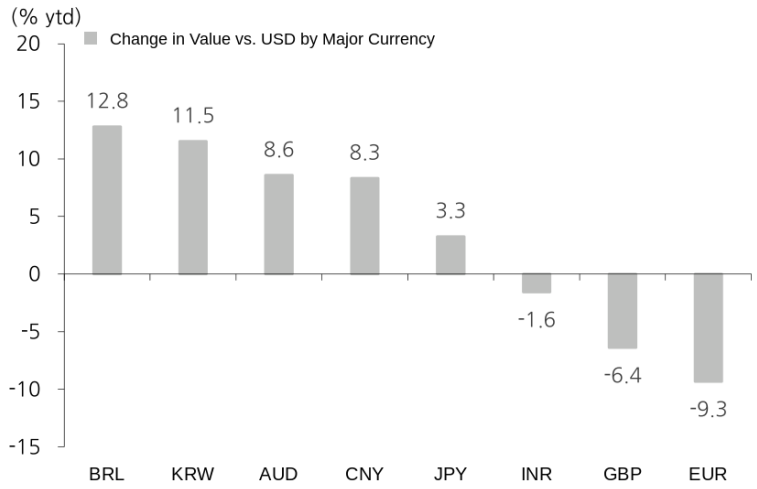
- Even if global inflation slows and the Fed stops raising rates, demand for the dollar will continue, so downward pressure on the dollar is expected to be limited
- The won is expected to see continued upward pressure due to supply-demand effects from strong semiconductor earnings
- By contrast, East Asian currencies other than the won, and the euro, may be relatively weak owing to economic and fiscal concerns and monetary policy uncertainty
- The annual currency value trend is forecast as KRW > USD > JPY

2026: The Won Stands Out Among Major Currencies



Note: Real effective exchange rate basis
Source: Bank for International Settlements, Hanwha Research Center

YTD: Strength in 3 East Asian Currencies vs. Weak Euro

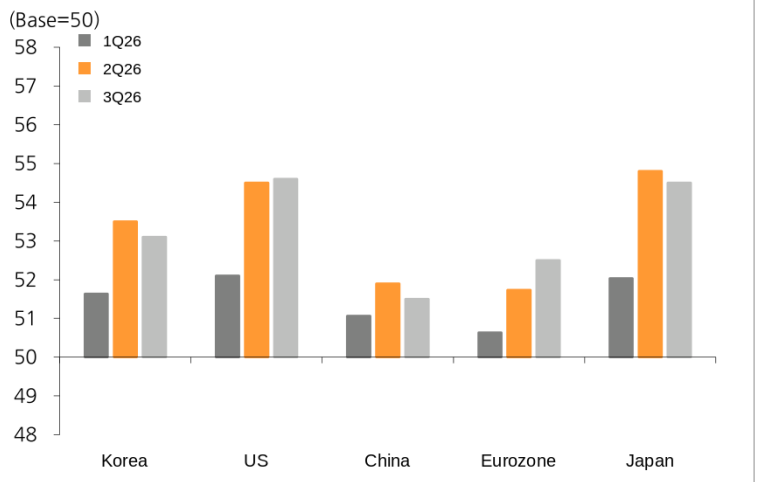


Note: As of Oct 14
Source: Bloomberg, Hanwha Research Center

Manufacturing Again Next Year

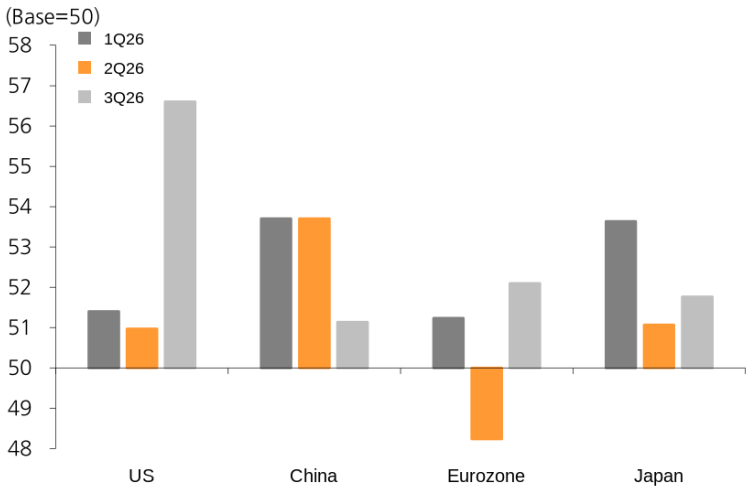
- The global manufacturing recovery is strengthening, while services are mostly slowing. The trend of manufacturing leading growth is expected to continue
- The September manufacturing PMIs of Korea, the US, China and the Eurozone rose from the previous three months, suggesting the manufacturing rebound continues
- Excluding some European countries and Brazil, the 3Q manufacturing PMI of most countries is above 50
- By contrast, the September services PMIs of China, the Eurozone and Japan fell from the previous three months

Manufacturing PMI: Improvement in Major Economies' Manufacturing Continues



Note: S&P Global Manufacturing PMI
Source: S&P Global, Hanwha Research Center

Services PMI: Mostly Slowing

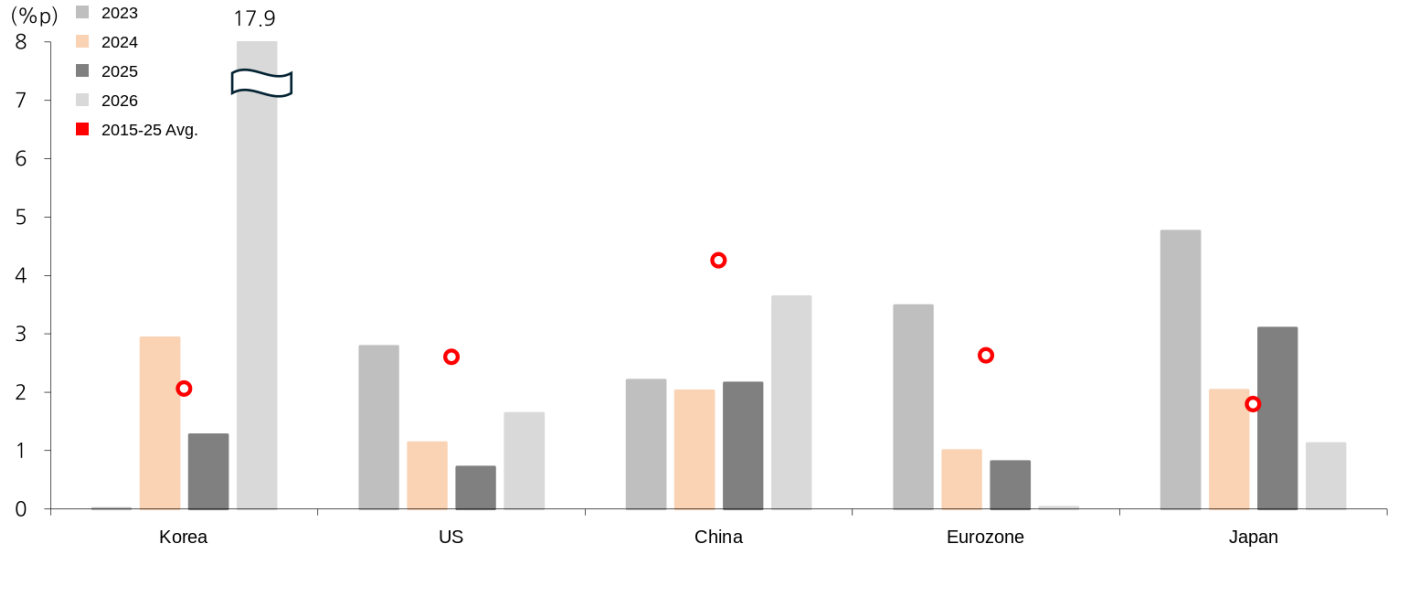


Note: S&P Global Services PMI. Korea not available
Source: S&P Global, Hanwha Research Center

g (Growth) > r (Interest Rate)

- Although high interest rates are cited as one of the biggest risks to the world economy, it is hard to say the interest rate level is excessive relative to growth in major countries
- This year, the gap between nominal growth (g) and the 10-year yield (r) widened in Korea, the US and China. The difference from the historical average is also not large
- The growth-interest rate gap narrowed in the Eurozone and Japan. The Eurozone was affected by slowing growth, and Japan by a steep rise in interest rates
- The interest rate burden relative to growth is small in Korea, the US and China

g (Growth) - r (Interest Rate): Rate Burden Relative to Growth Is Not Excessive

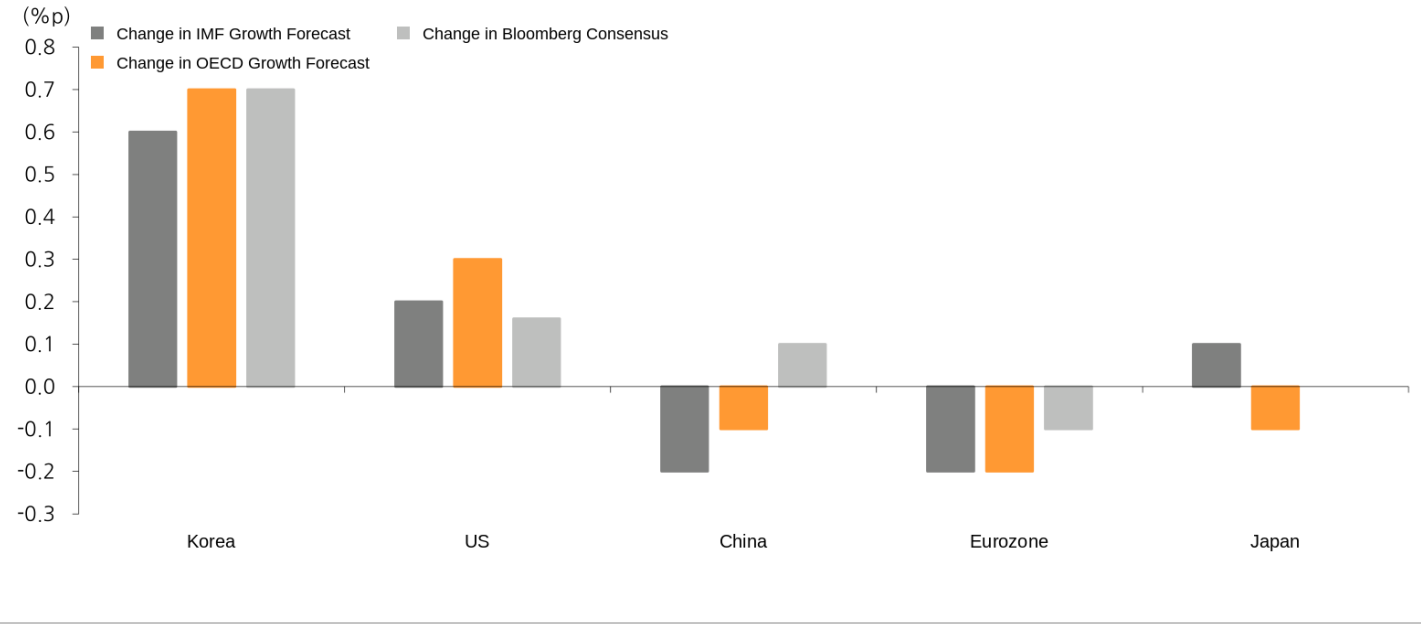


Note 1: Gap between nominal growth and 10-year yield
Note 2: 2026 nominal growth is 1H; 10-year yield is the Jan-Sep average
Source: Bank of Korea, Bureau of Economic Analysis, National Bureau of Statistics, Eurostat, Economic and Social Research Institute, Bloomberg, Hanwha Research Center

Stronger Expectations for Korea and the US

- 2027 economic growth forecasts for Korea and the US were revised up, reflecting expectations of improved growth
- In the second half, the IMF, OECD and Bloomberg forecasts for Korea and the US were all revised up
- By contrast, growth expectations for China, the Eurozone and Japan weakened
- Considering the trend in major economic indicators such as the PMI and the interest rate level relative to growth, the relative advantage of the Korean and US economies may continue

Growth Forecasts Revised Up for Korea and US, Down for Eurozone

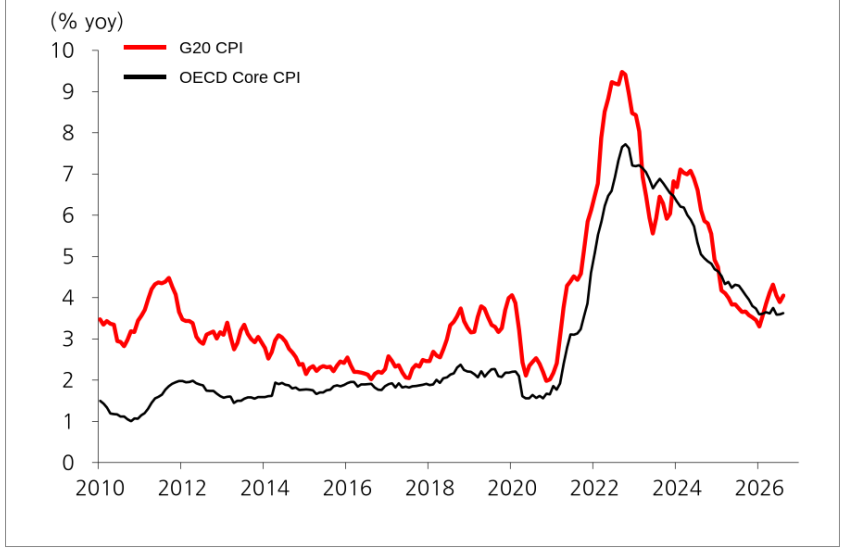


Note 1: Change in 2027 GDP growth forecast
Note 2: IMF is Oct 2026 vs. Oct 2025 forecast; OECD is Sep 2026 vs. Dec 2025 forecast; Bloomberg is Sep consensus vs. end-2025
Source: IMF, OECD, Bloomberg, Hanwha Research Center

Inflation Peak-Out (1): Annualized Rates of Increase Decline

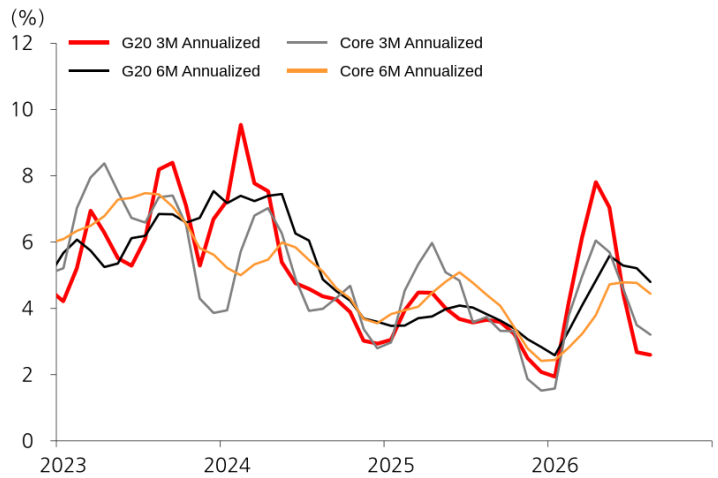
- The decline in the rates of increase of G20 headline CPI and OECD core CPI reflects easing upward price pressure
- The 3-month and 6-month annualized rates of increase of headline and core CPI have been falling since May or June
- The simultaneous decline in the 3-month and 6-month annualized rates supports the outlook for slowing inflation
- Although prices in major countries are still above central bank targets, the slowdown in the rate of increase is gradually becoming clearer
- However, the relatively slow core inflation slowdown suggests that the downside may be limited

Global Inflation Momentum Slows



Source: OECD, Hanwha Research Center

Recent Trend Supports Outlook of Inflation Peak Having Passed

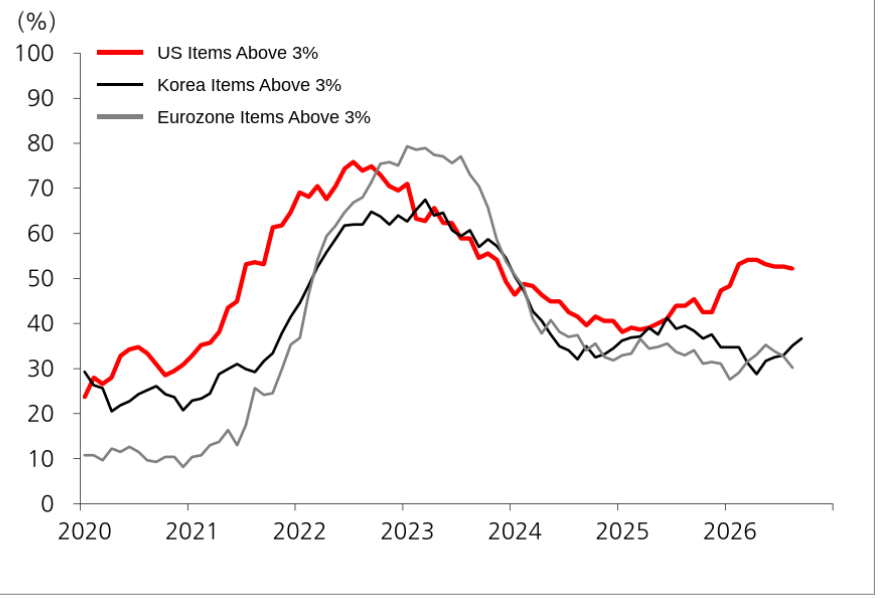


Source: OECD, Hanwha Research Center

Inflation Peak-Out (2): Fewer Items Above 3%

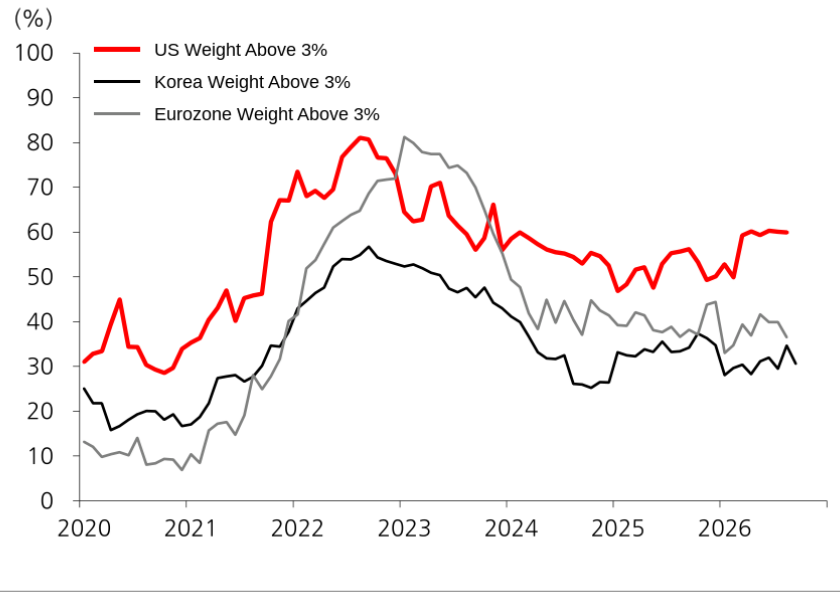
- A decrease in the number of items with an inflation rate above 3% is also a basis for the outlook of slowing inflation
- The number of US items above 3% decreased, suggesting that prices for most items are stabilizing
- The number of Korean items above 3% increased, but fell on a weighted basis. Prices of heavily weighted items such as telecommunication costs slowed
- Both the share and the weight of Eurozone items above 3% peaked out, which means the inflation slowdown is spreading

Share of Items with Inflation Above 3% Shrinks



Source: Bureau of Economic Analysis, Bank of Korea, Eurostat, Hanwha Research Center

Weighted Share Above 3% Also Shrinks

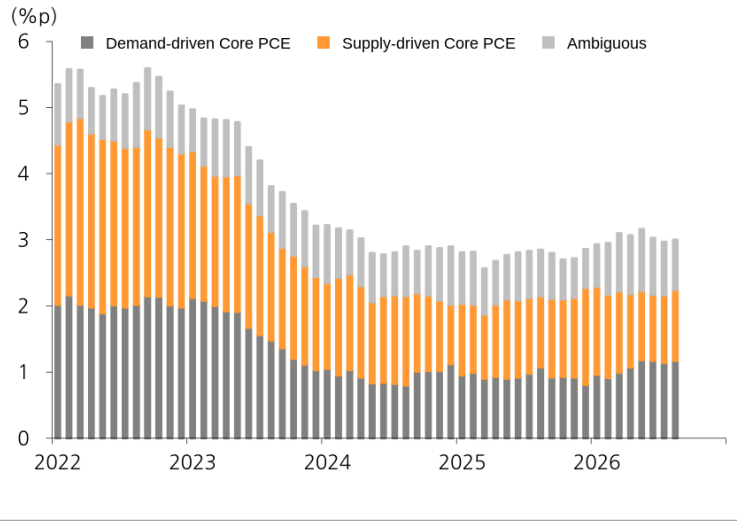


Source: Bureau of Economic Analysis, Bank of Korea, Eurostat, Hanwha Research Center

US: Blocked Upside and Downside

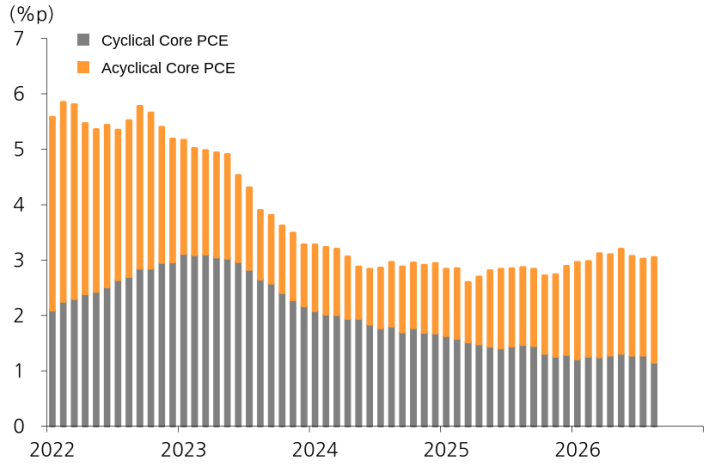
- US disinflation continues, but the pace is expected to be gradual
- This year, the contribution of the supply side to prices has declined steadily. The contribution of the demand side stagnated after rising in 1H
- Cyclical inflation, which tends to strengthen when the economy recovers, is slowing. By contrast, the contribution of acyclical inflation has rebounded
- The upward trend in prices in 2H is generally slower than in 1H. However, the trend in the demand and acyclical sectors reflects the possibility that the downside for prices is limited

Supply-Side Contribution to Inflation Falls, Demand-Side Contribution Stalls



Source: FRB of San Francisco, Hanwha Research Center

Decline in Cyclical Inflation Contribution Reflects Moderate Economic Slowdown

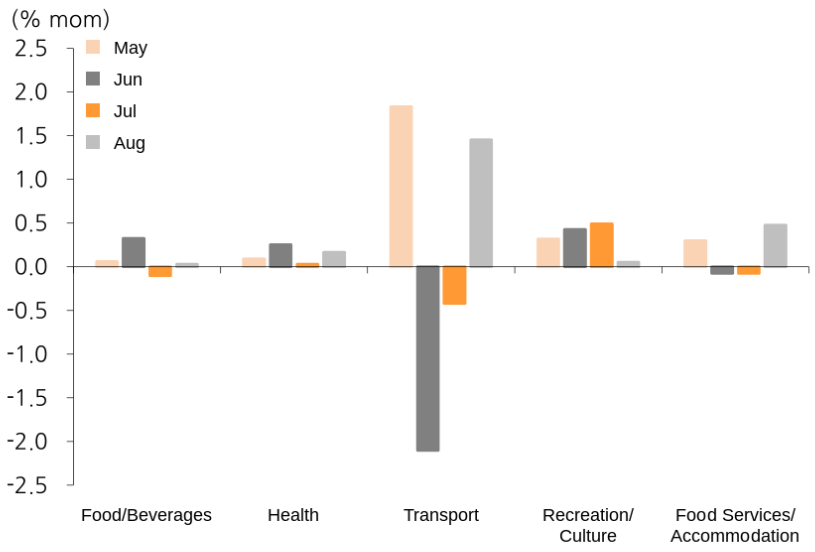


Note: Contribution to YoY change in Core PCE prices
Source: FRB of San Francisco, Hanwha Research Center

Eurozone: Limited Spread of Rising Inflation

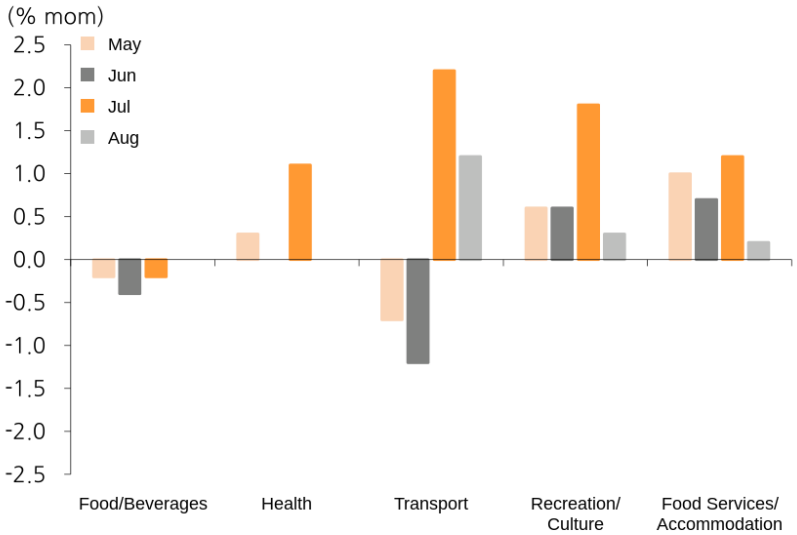
- Recent Eurozone inflation reflects rising fuel prices and the burden of services inflation, but it is also confirmed that the spread of the rise is passing its peak
- In August, month-on-month increases in transport, recreation/culture and food/accommodation prices fell. The September HICP flash estimate showed a stronger rise
- Given the slow demand recovery and the decline in the 6-month annualized rate of increase, the trend is highly likely to be a slowdown

US: Non-Energy Inflation Burden Eases



Note: PCE price basis
Source: Bureau of Economic Analysis, Hanwha Research Center

Eurozone: Services Inflation Strengthens, Including Recreation/Culture and Dining Out, Not Just Transport

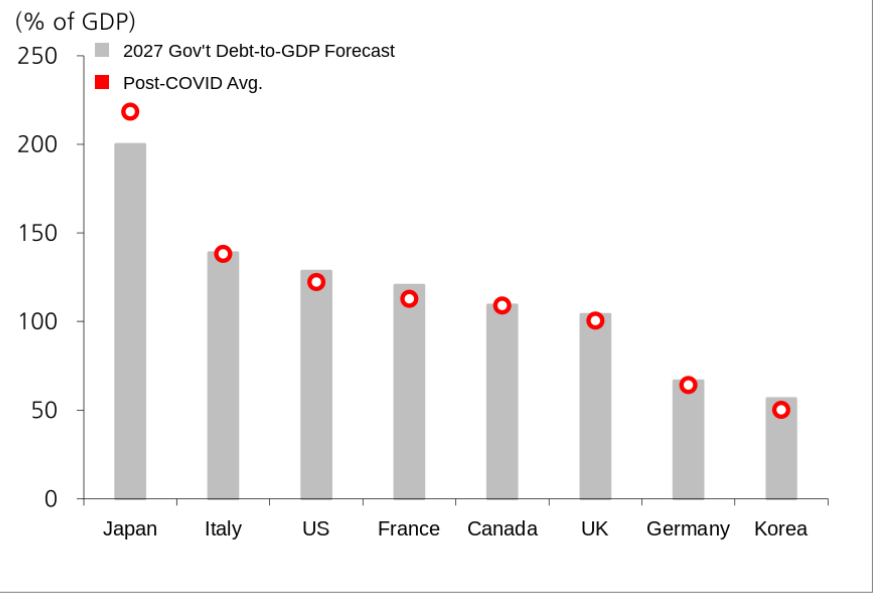


Source: Eurostat, Hanwha Research Center

Growing Debt Burden

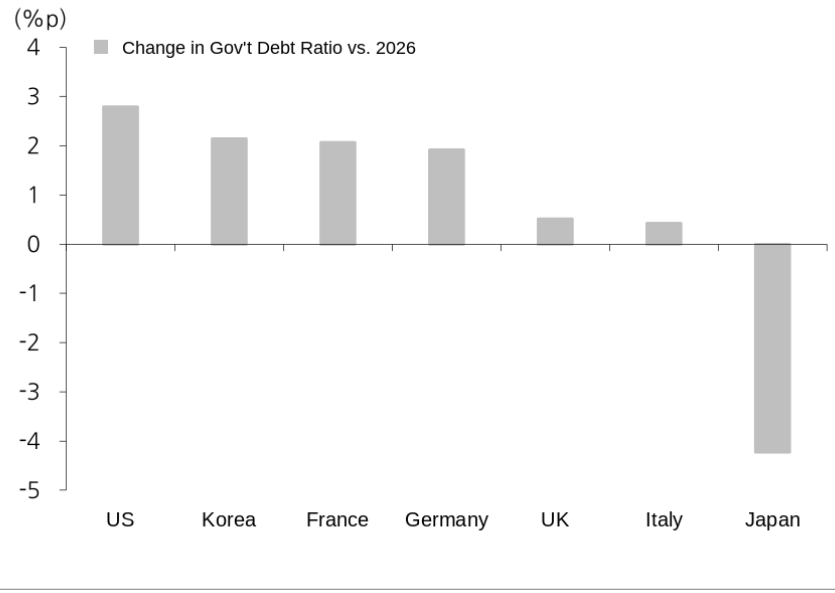
- The trend of expanding government debt burdens continues. In most countries, the 2027 government debt-to-GDP ratio is likely to rise compared with this year
- Excluding Japan, major countries' government debt ratios exceed the 2021-25 average
- France and Korea have higher government debt ratios than during COVID
- Although a tipping point cannot be judged from the absolute ratio, the debt burden continues to grow

Government Debt-to-GDP Ratio Higher Than Average



Source: IMF, Hanwha Research Center

Government Debt Ratios Keep Rising

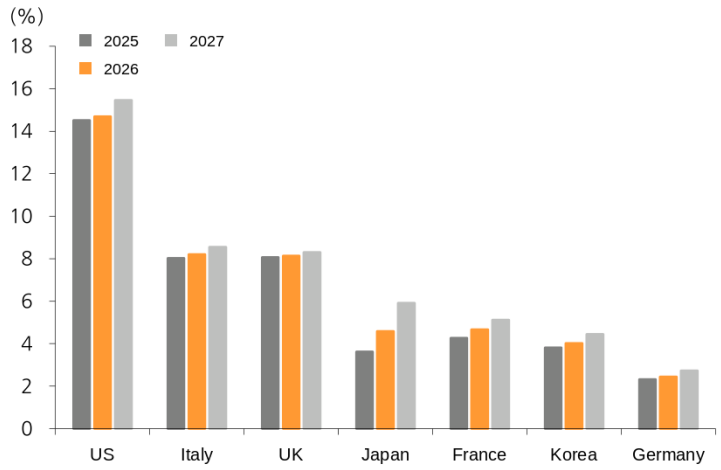


Source: IMF, Hanwha Research Center

Rising Interest Expenditure

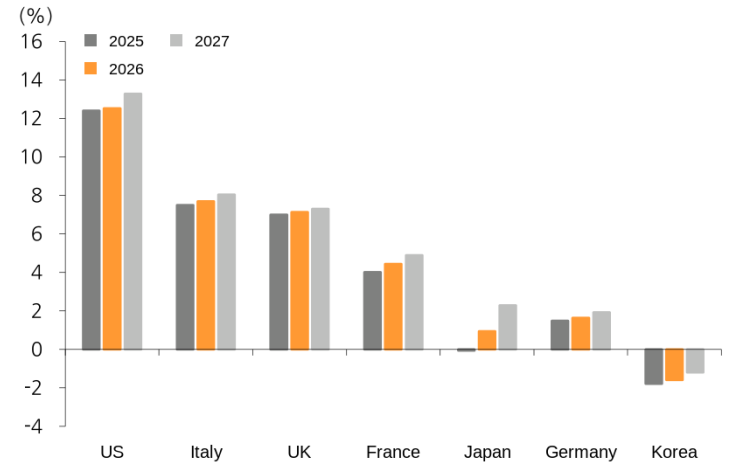
- As the size of government debt grows, the interest burden expands as well
- The gross interest/gross revenue ratio is rising sharply in the US and Japan, meaning the burden of debt payments is increasing
- The gap between net and gross interest is small in the US, Italy and France, meaning the effect of interest income offsetting expenditure is relatively small
- However, fiscal capacity cannot be judged from interest indicators alone. The size of maturities, refinancing rates and net borrowing must also be checked

Gross Interest/Gross Revenue: Interest Burden on Debt Rising Relatively Fast in US and Japan



Note: Gross interest expense / gross revenue. Interest payment burden on debt
Source: OECD, Hanwha Research Center

Net Interest/Gross Revenue: Offsetting Effect of Interest Income Is Relatively Small in US, Italy and France

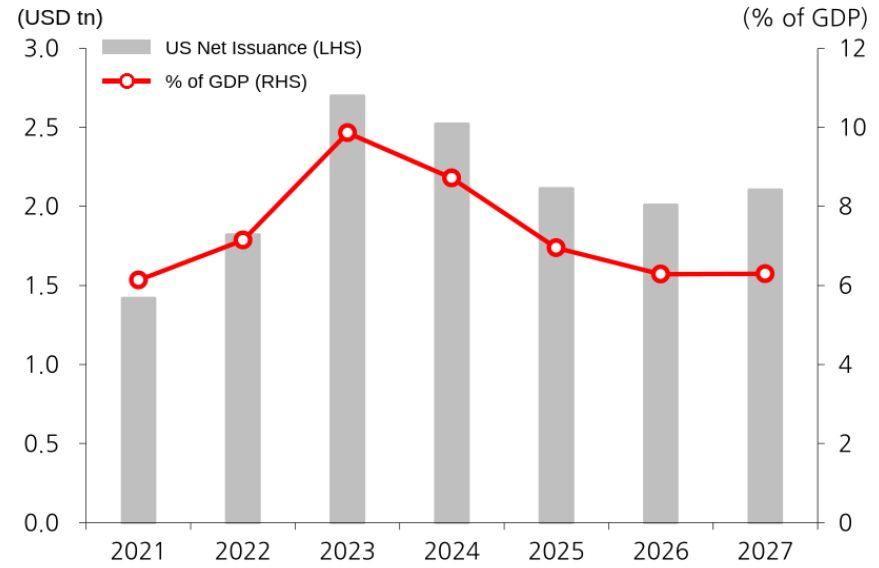


Note: Net interest expense / gross revenue. Fiscal burden remaining after offsetting with interest income
Source: OECD, Hanwha Research Center

Above-Average Net Issuance

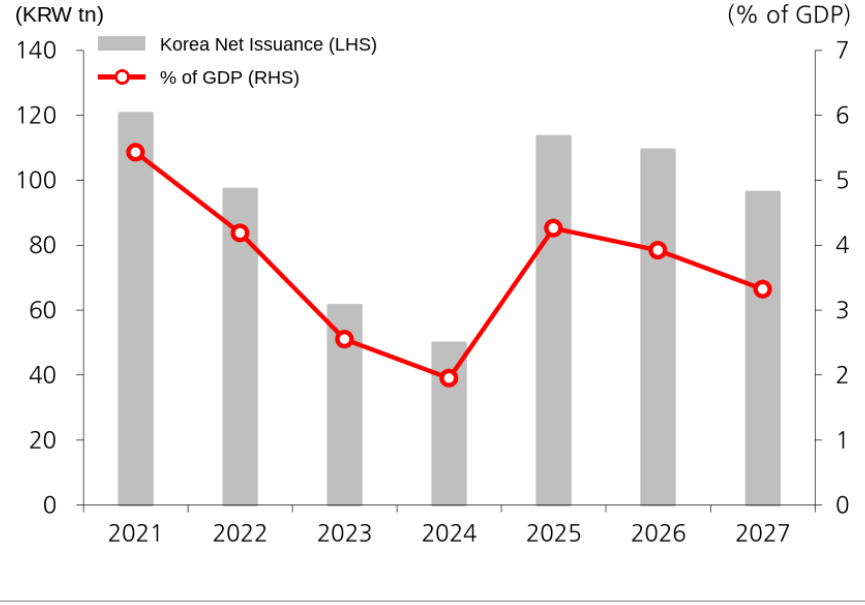
- As the expansionary fiscal stance continues, net government bond issuance also exceeds the post-COVID average
- Net government bond issuance in the US and the UK in 2027 is expected to exceed this year's level
- Next year's net increase in Korean Treasury bonds is expected to fall below KRW 100tn (KRW 96.2tn), but is still higher than the 2020s average

US Net Treasury Issuance Expected to Increase Slightly



Note: 2926 and 2027 are Primary Dealer forecasts
Source: US Department of the Treasury, Hanwha Research Center

Korea Net Treasury Bond Issuance to Decline Somewhat but Stays Above Post-COVID Average

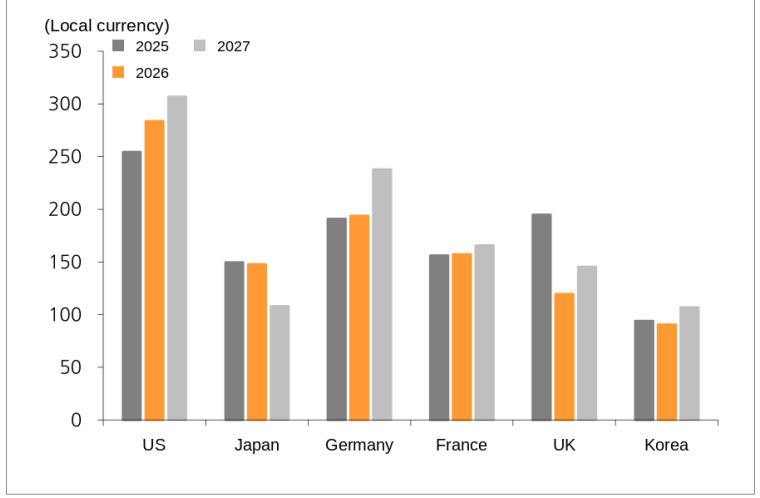


Source: Ministry of Finance and Economy, Bank of Korea, Hanwha Research Center

Refinancing Burden Eases Somewhat

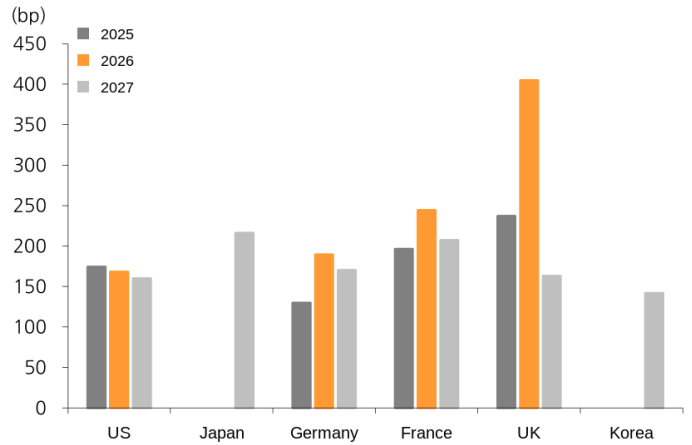
- It is difficult to judge the burden from the size of government bond maturities alone, because the refinancing burden varies with the gap between new funding rates and existing rates
- In 2027, major countries' government bond maturities are generally expected to exceed this year's level. Because interest rates have risen, the interest burden at refinancing expands
- However, the gap between new funding rates and the coupon rates of maturing bonds is generally expected to narrow next year
- Considering maturities and interest rate differences, the additional interest burden of France and the UK actually shrinks. The increase in interest in the US and Germany is also not large

Maturing Government Bonds Rise in Major Economies



Note 1: Size of maturing government bonds in major economies
Note 2: US in USD 10bn, Japan in JPY tn, Germany/France in EUR bn, UK in GBP bn, Korea in KRW tn
Source: Statistics by country, Hanwha Research Center

Burden of New Funding Costs Eases

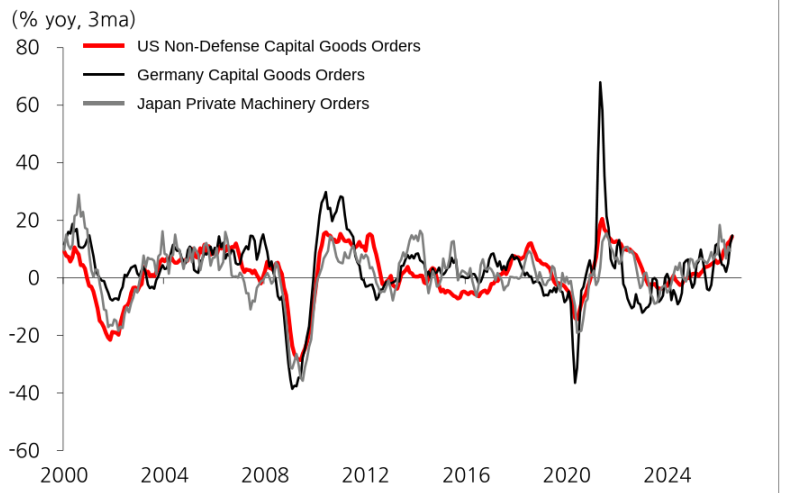


Note 1: New funding yield minus coupon of maturing bonds
Note 2: 2026 funding yield assumed at annual average 10-year yield; 2027 at average 10-year yield of the latest six months
Source: Statistics by country, Hanwha Research Center

Capex Cycle

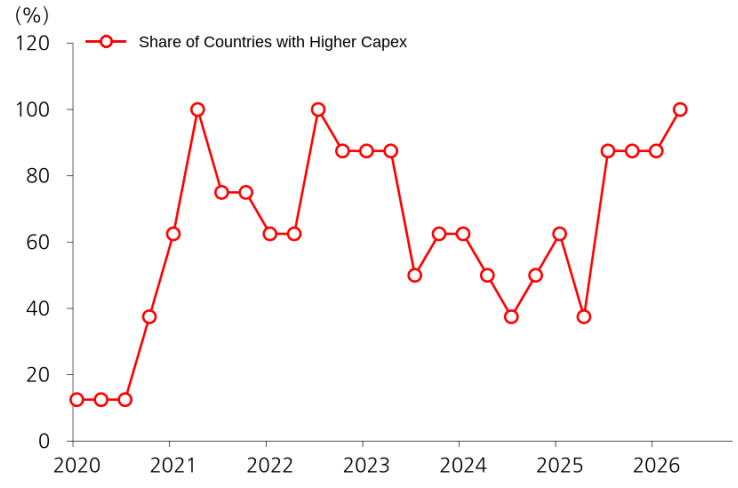
- The rise in investment growth rates in major countries suggests that capex demand is recovering
- US and German capital goods orders and Japanese private machinery orders, which are classified as capex leading indicators, show a strengthening uptrend
- Not only the US, Germany and Japan but also other major countries such as Korea and the UK saw capex increase in 1H
- Considering the expansionary fiscal stance and the trend toward expanded government-led investment, the capex expansion trend is expected to continue

Rise in Capital Goods Orders Reflects Improving Investment Demand



Source: U.S. Census Bureau, Statistisches Bundesamt, Economic and Social Research Institute, Hanwha Research Center

Capex Growth in Major Economies Continues in 1H

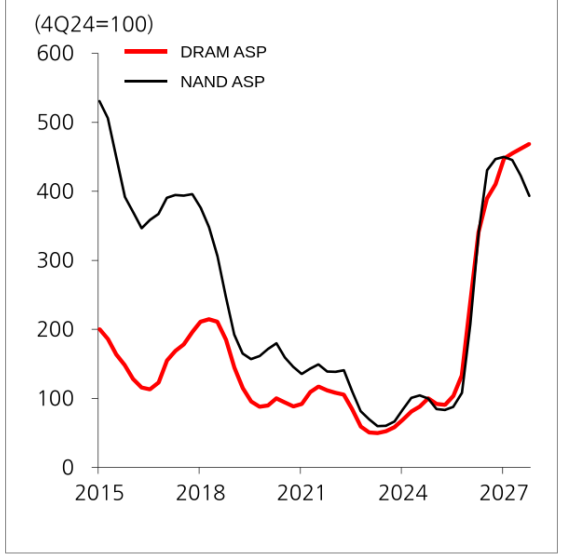


Note 1: YoY growth of real investment in machinery and equipment
Note 2: Based on Korea, US, Germany, France, Italy, UK, Canada and Australia, which provide time series on a consistent basis
Source: OECD, Hanwha Research Center

Semiconductors: Growth Rate Peaks Out

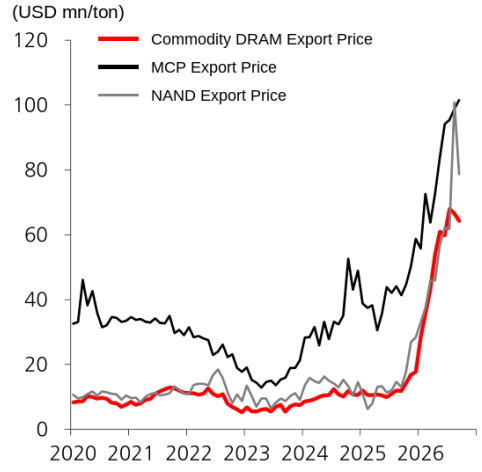
- Excess demand for memory semiconductors is expected to be maintained in 2027 as well. However, the slowing price increase and the narrowing gap between production and demand must be kept in mind
- According to OMDIA, the DRAM ASP (Average Selling Price) growth rate has fallen to around 1.5% QoQ since 2Q
- The NAND ASP is expected to decline gradually from 1Q next year (OMDIA)
- Korea's semiconductor export growth rate is also expected to decline, peaking in 4Q this year

2Q27: Memory Price Increases May Slow or Decline



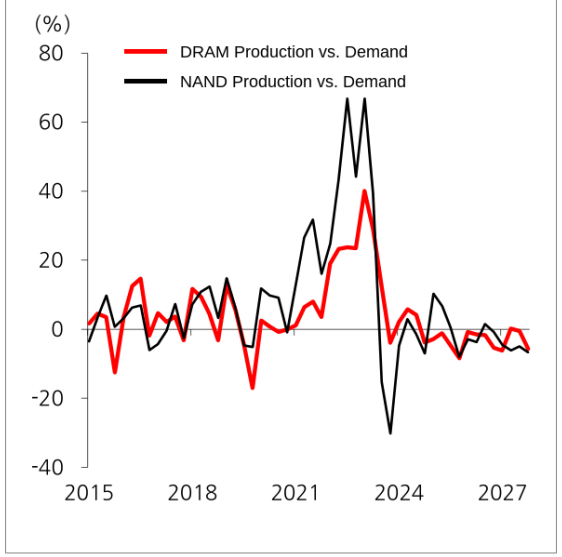
Note: 3Q26 onward are forecasts
Source: OMDIA, Hanwha Research Center

Korea Semiconductor Export Prices: Rate of Increase Gradually Slows



Note 1: MCP (Multi Chip Package) is high-capacity/high-value-added memory such as HBM and DDR5
Note 2: Korea memory semiconductor export unit price by weight
Source: Korea Customs Service, Hanwha Research Center

Excess Demand Expected to Persist Next Year

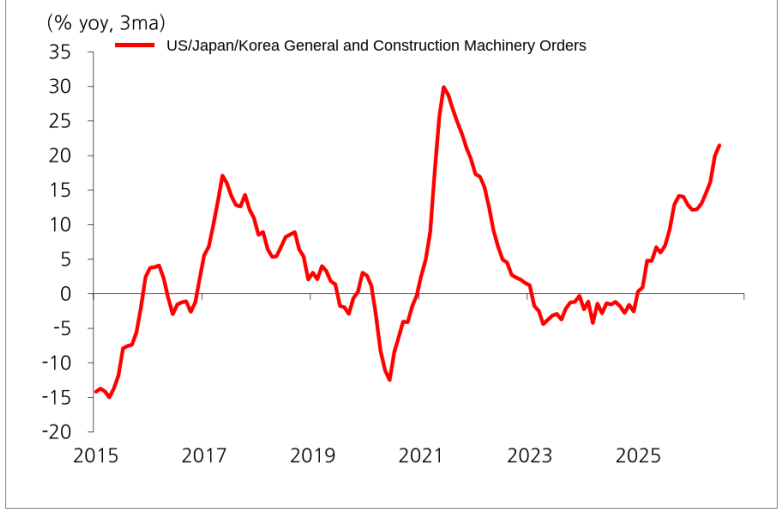


Note: 3Q26 onward are forecasts
Source: OMDIA, Hanwha Research Center

Industrials: Growing Orders

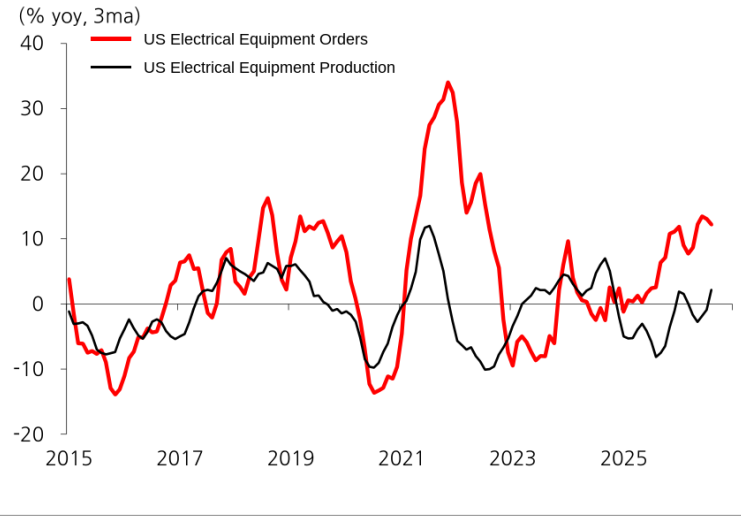
- Reflecting the investment expansion stance, the demand recovery for industrials such as machinery and equipment is also expected to continue
- Orders are strengthening not only for semiconductor equipment and power equipment, which are closely tied to AI demand, but also for general and construction machinery
- Korea's recent export performance of general machinery and construction machinery has also passed its trough and is increasing gradually
- This suggests that industrial demand is improving overall

Machinery Orders Growth Rises in Major Economies



Source: U.S. Census Bureau, Ministry of Data and Statistics, Economic and Social Research Institute, Hanwha Research Center

AI-Related Power Equipment Demand Is Solid

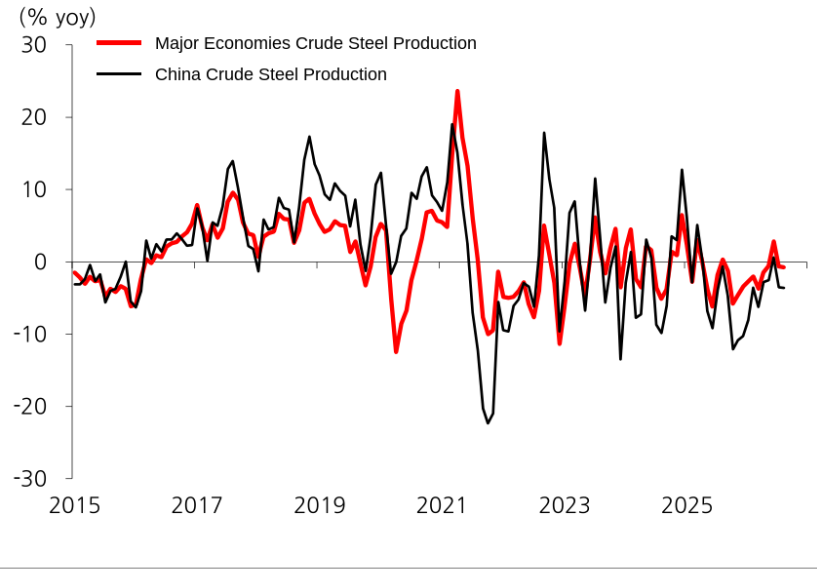


Source: U.S. Census Bureau, Federal Reserve Board, Hanwha Research Center

Materials: The Trough Has Passed

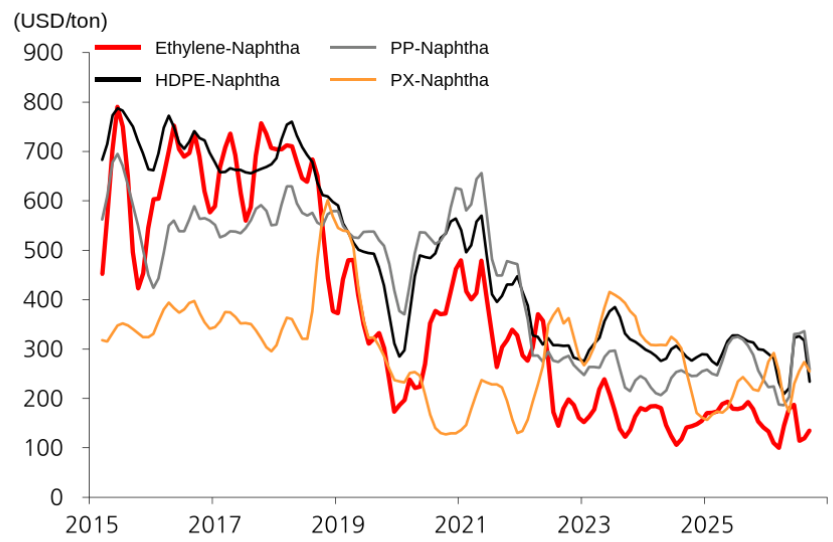
- Materials such as steel, refining and chemicals are weaker than industrials, but demand is expected to increase reflecting the manufacturing recovery
- For steel, an increase in East Asian exports, a rebound in crude steel production and easing of the inventory burden are confirmed
- For chemicals, the narrowing of spread contraction and easing of the inventory burden are positive changes
- If this is a cycle in which global manufacturing and investment recover, demand for materials is also expected to rebound gradually

Crude Steel Production Expected to Rise Moderately



Note: Covers about 70 countries for which World Steel provides production data
Source: World Steel Association, Hanwha Research Center

Key Chemical Product Spreads: Contraction Eases

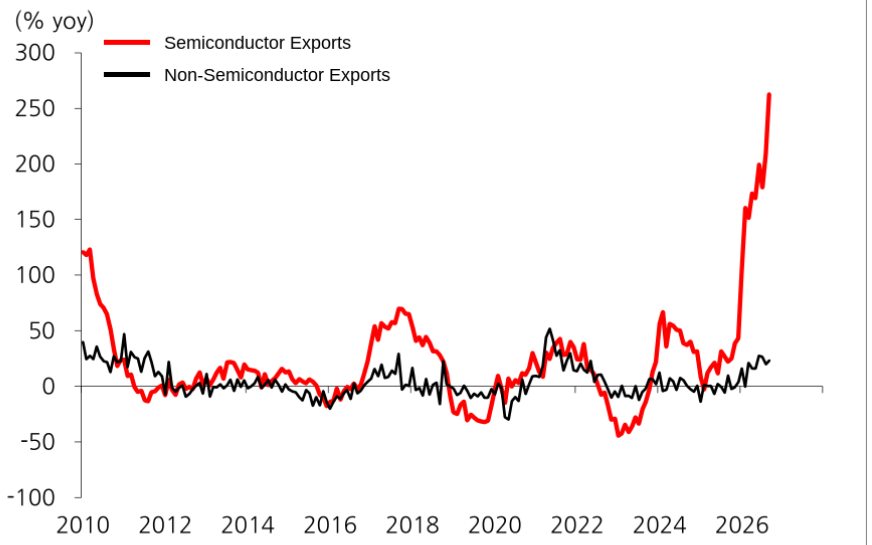


Source: Cischem, Hanwha Research Center

KRW/USD Exchange Rate: KRW 1,250-1,400

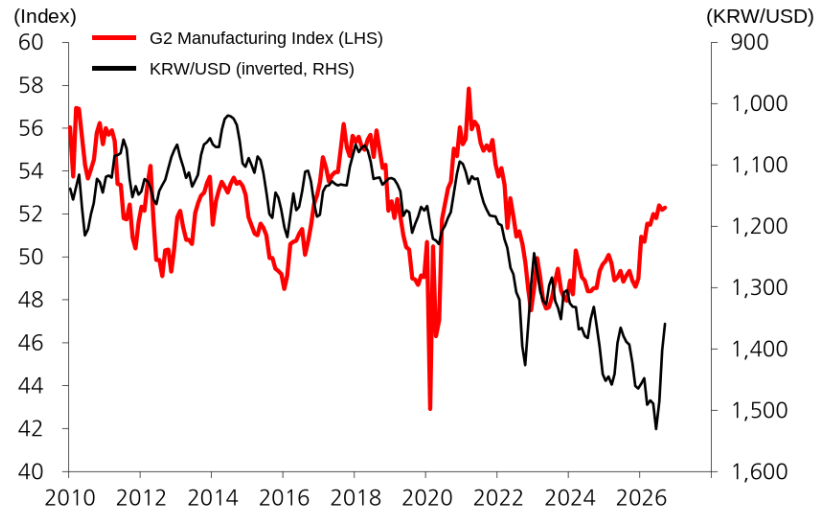
- The KRW/USD exchange rate is expected to decline toward year-end within a band of KRW 1,250-1,400
- We judge that semiconductor exports will be solid next year as well and that the global manufacturing cycle will also be fine
- Accordingly, dollar supply in the domestic FX market is likely to keep expanding, centered on semiconductor companies
- The size of seasonal dollar selling (quarter-ends, just before holidays, etc.), which did not significantly affect the market in the past, is also growing and will steadily weigh on the exchange rate

While Semiconductor Export Performance Keeps Improving



Source: Korea Customs Service, Ministry of Trade, Industry and Resources, Hanwha Research Center

Improving Global Manufacturing Also Acts as Pressure for Won Strength

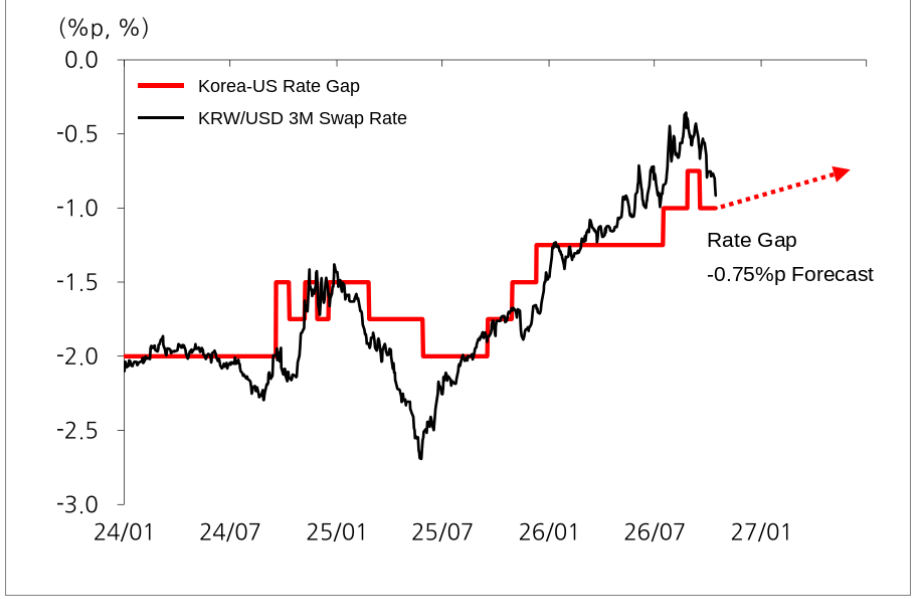


Source: Institute of Supply Management, NBS, Bloomberg, Hanwha Research Center

Heed Domestic and External Risks

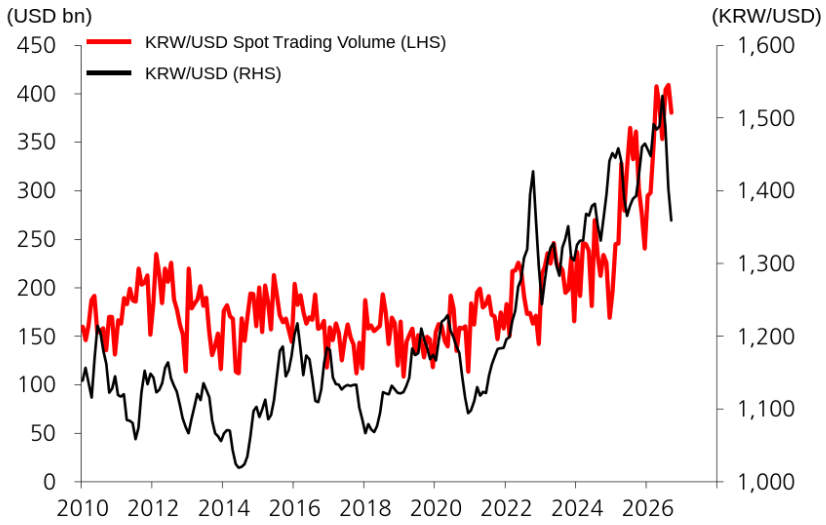
- Attention is needed to an external environment that will not improve much and to rising volatility in the domestic FX market
- Next year the Fed is expected to hold the policy rate, and the Korea-US rate gap is expected to remain around -0.75%p
- The yen is also highly likely to continue weakening, so it is hard to say the external environment improves greatly for the won
- Therefore, if the domestic semiconductor cycle weakens contrary to expectations, the floor of the KRW/USD exchange rate may rise and volatility may strengthen, which must be kept in mind

Korea-US Rate Gap Expected to Hold at -0.75%p in 2027



Source: Bloomberg, Yonhap Infomax, Hanwha Research Center

Spot FX Trading Volume Rising, Market Volatility to Persist

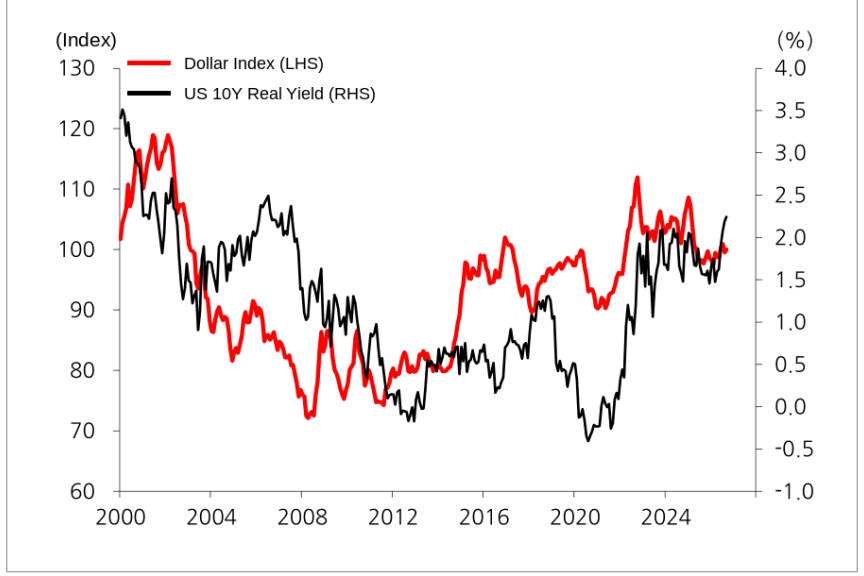


Note: Volume is monthly total; exchange rate is monthly average
Source: Bloomberg, Yonhap Infomax, Hanwha Research Center

Dollar: Gradual Weakness

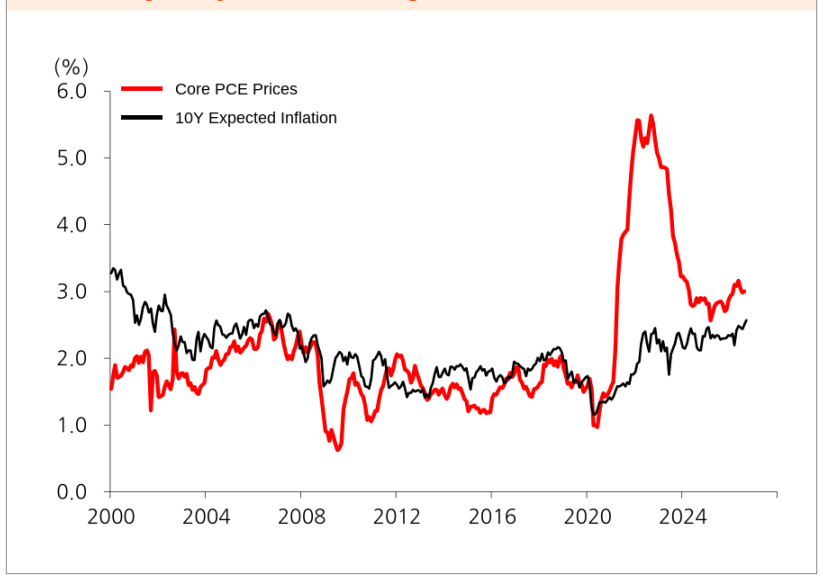
- The dollar index is expected to fluctuate between 97 and 103pt
- The possibility that the Fed holds its policy rate within the year is a source of downward pressure on the dollar
- However, it is not easy for US inflation to stabilize quickly, and global geopolitical risks also appear difficult to resolve completely
- Because upward pressure on US real interest rates also continues, the dollar's value is highly likely to be maintained above a certain level

Rising US Real Rates to Limit Dollar Weakness



Source: Bloomberg, Federal Reserve Board, Hanwha Research Center

US Inflation Also Unlikely to Stabilize Quickly at the 2% Target

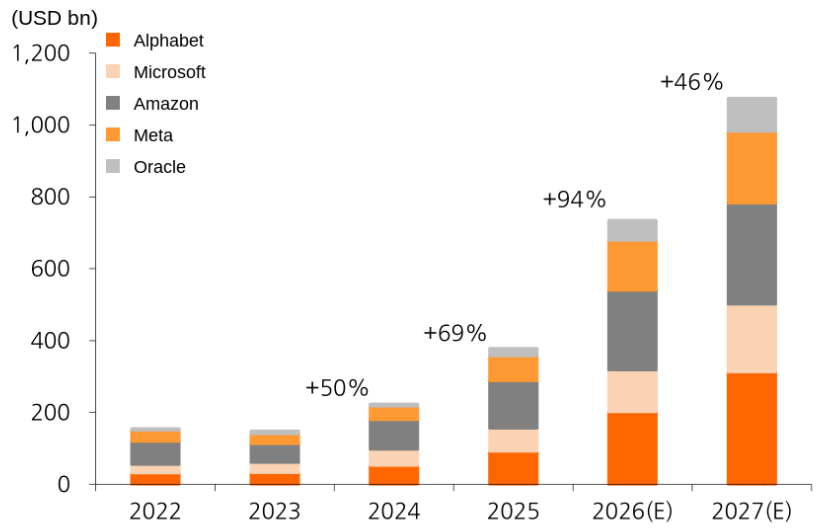


Source: Bureau of Economic Analysis, Federal Reserve Board, Hanwha Research Center

Companies Need Dollars

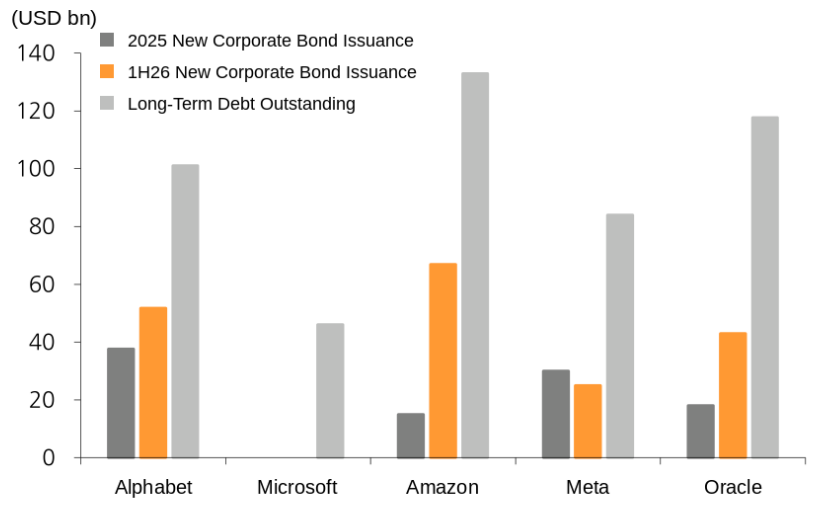
- Demand for the dollar is expected to continue, centered on hyperscalers
- As major hyperscalers' capital expenditure (CapEx) expands next year as well, corporate bond issuance is highly likely to continue
- The resulting dollar demand is expected to support US real interest rates and the dollar's value at the same time
- Downward pressure on the dollar will be limited

Hyperscaler Capex Expected to Expand About 46% Next Year as Well



Note: Non-GAAP basis; 2026-27 are Bloomberg forecasts
Source: Bloomberg, Hanwha Research Center

Corporate Bond Issuance by These Companies Is Also Rising

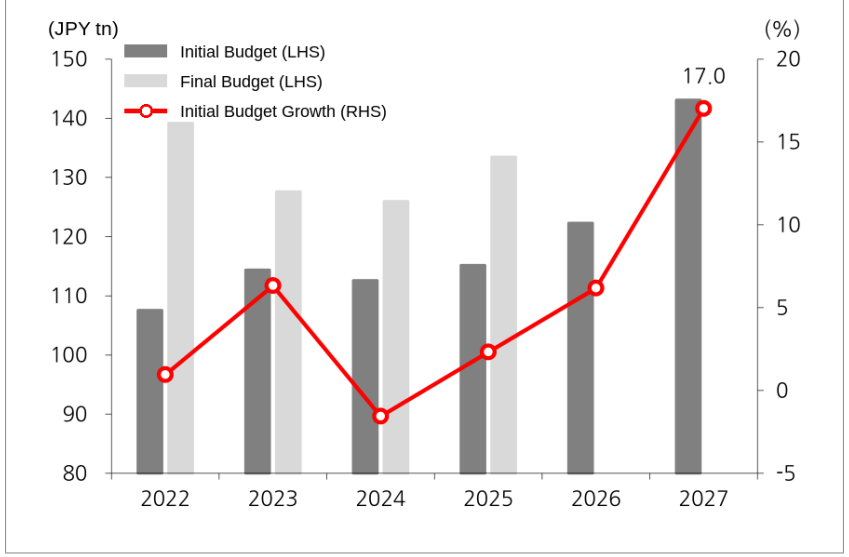


Note: Includes USD and non-USD bonds. Oracle as of end-Aug 2026; others as of end-June
Source: U.S. Securities and Exchange Commission, Hanwha Research Center

Lack of Yen-Strengthening Catalysts

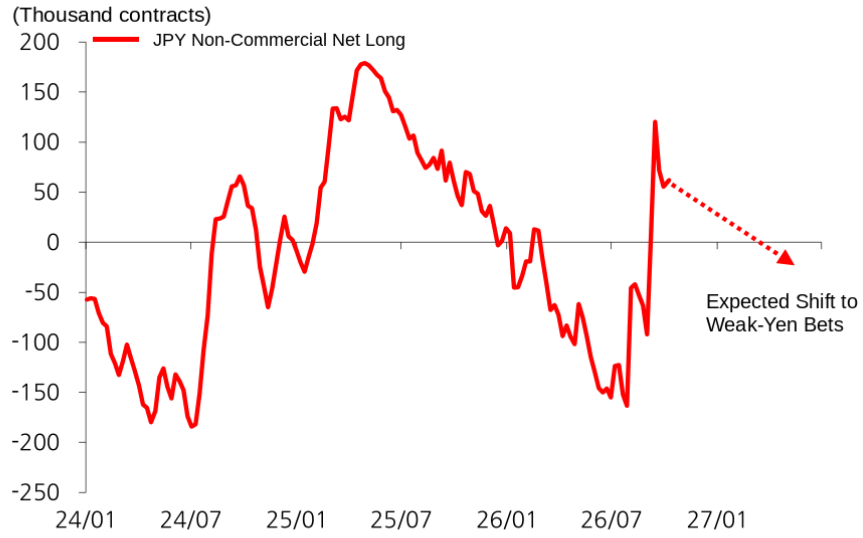
- The JPY/USD exchange rate is expected to weaken in the range of JPY 155-162
- The Japanese government submitted its FY2027 budget request at JPY 143.1tn, up 17.0% from last year's initial budget
- The government explained that the increase is to avoid compiling a supplementary budget, but fiscal concerns are likely to continue next year
- Japan's pace of monetary policy normalization is also highly uncertain, so we judge that market bets on the yen may gradually weaken

Japan Government's FY2027 Budget Request Is 17% Above Last Year's Initial Budget



Note: Fiscal-year basis; FY2027 is the budget request (preliminary estimate)
Source: Ministry of Finance, Hanwha Research Center

Market's Yen Bets May Revert from Strength to Weakness



Source: Bloomberg, Hanwha Research Center

II

Rates

Sluggish Box-Range

Sung-soo Kim · sungsoo.kim@hanwha.com · 3772-7616

[Bond Market Outlook] Sluggish Box-Range

- 2027 Korea Treasury and US Treasury yield ranges. The supply-demand environment will structurally keep the floor for long-term yields firm
  - Korea 3Y KTB: 3.60-4.10% / Korea 10Y KTB: 4.00-4.50%, US 2Y: 4.20-5.00% / US 10Y: 4.80-5.60%
- Considering a relatively short policy rate hiking cycle, we expect a bull steepening of the curve. We judge there is relatively more room for short-term yields to fall
- As we judge that clear upside and downside pressures will be absent, we expect both the Korean and US bond markets to move in a sluggish box range
  - We forecast solid growth and a slow disinflation trend. Accordingly, monetary policy will remain at the terminal rate for a considerable period
  - The bond supply environment will continue to act as a burden, but dip-buying and carry demand exist

[Table 3] Key Interest Rate and Spread Forecasts

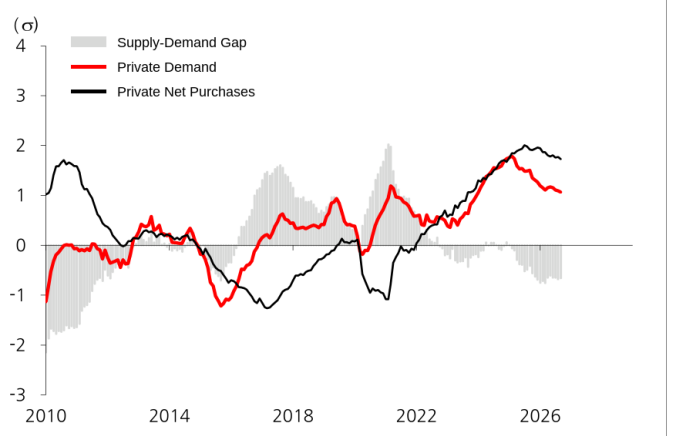
Item (%, bp)	2026 Full Year	1Q27	2Q27	3Q27	4Q27	2027 Full Year
Korea Policy Rate	3.25	3.50	3.50	3.50	3.50	3.50
Korea 3Y KTB	3.59	4.00	3.90	3.75	3.63	3.83
Korea 10Y KTB	4.00	4.35	4.30	4.18	4.10	4.23
10Y-3Y Spread	41.6	35.0	40.0	43.0	47.0	41.0
US Policy Rate	4.25	4.25	4.25	4.25	4.25	4.25
US 2Y	4.02	4.85	4.90	4.65	4.30	4.68
US 10Y	4.50	5.30	5.25	5.13	5.00	5.18
10Y-2Y Spread	47.8	45.0	35.0	48.0	70.0	50.0

Note 1: Policy rates are period-end; spreads and government bond yields are period averages
Note 2: 2026 yields are based on the 10/15 close (KST)
Source: Hanwha Research Center

[The Problem Is Supply-Demand] Demand Has Increased, Supply Has Increased More

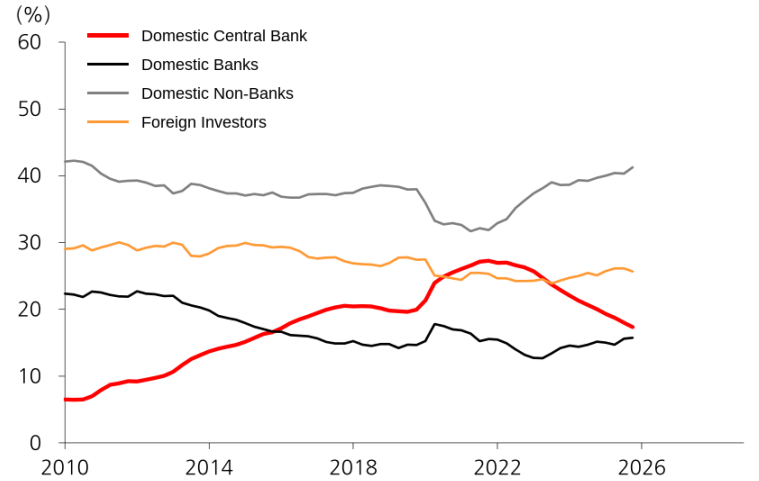
- Over the last 12 months, private demand for government bonds in major advanced countries (G7) has clearly strengthened compared with the 2010s
- However, the volume that the private sector must absorb (the portion of newly issued government bonds that the market must take because central banks do not) has increased even more
  - The supply-demand gap peaked at 2.0σ in February 2021, turned negative in 2H 2022, and widened to -0.7σ in 2025
- The demand-dominant phase created by central bank purchases has ended

G7 Treasury Supply-Demand Gap: Private Demand - Private Net Purchases Trend



Note 1: Demand index reflects auction bid-to-cover and rate concession ahead of auctions
Note 2: Supply index is coupon issuance converted to 10-year equivalent (gross supply) minus central bank net purchases, i.e. private net purchases
Note 3: Units are standard deviations (σ), with each country's 2010-2019 average set to 0
Note 4: A negative gap (demand minus private net purchases) means supply dominance
Source: Ministries of Finance and central banks, Yonhap Infomax, IMF, Hanwha Research Center

G7 Treasury Holdings by Holder

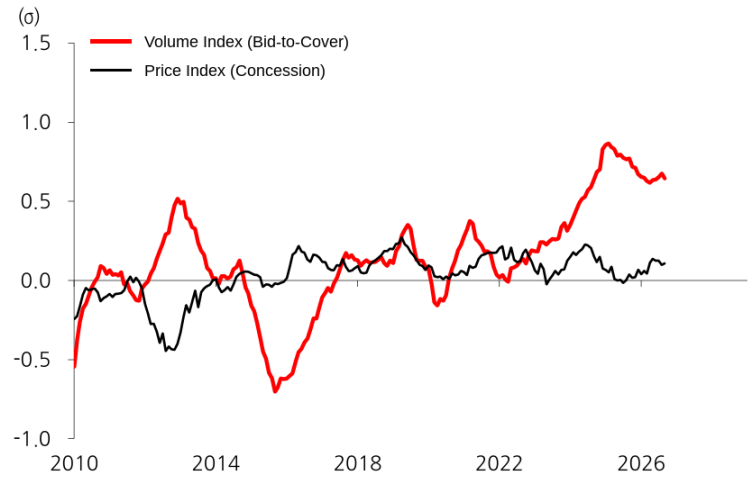


Note: General government debt basis; weighted average by country Treasury outstanding
Source: IMF, Hanwha Research Center

[The Problem Is Supply-Demand] Unable to Fill the Void Left by Central Banks

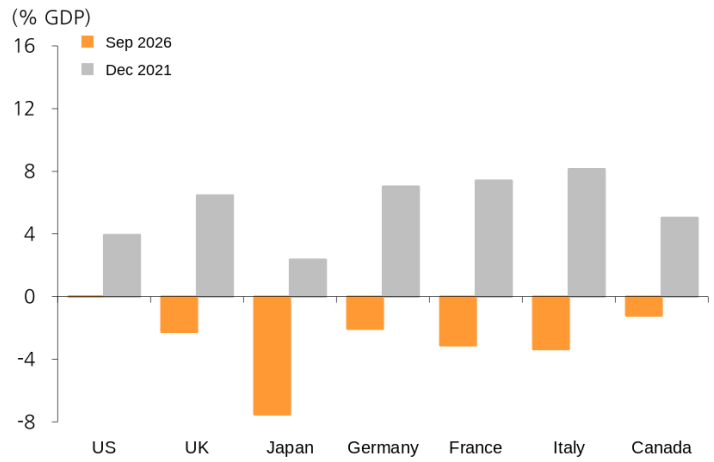
- The supply-demand burden stems more from the halt of central bank purchases than from increased government bond issuance
  - Over the last 12 months, the G7 total government bond supply index is 0.6σ, while the private net purchase index excluding central bank purchases is 1.7σ
- The central bank share of the government bond market fell from 27.3% in 2021 to 17.4% in 2025. Over the same period, the domestic non-bank share increased (31.8% → 41.3%)
  - Of the increase in government bond issuance in 2025 (6.8% of GDP), 4.4%p was absorbed by domestic non-bank investors (pension funds, insurers, funds, households)
- Demand also needs to be examined in detail. The volume index reflecting bid-to-cover ratios is 0.6σ on the G7 average. The price index reflecting auction concessions is 0.1σ, a normal level
  - Investors crowd into auctions thanks to the elevated yield level, not because demand is strong enough to buy while receiving less interest

G7 Treasury Auction Demand: Volume and Price



Note: 12-month average of 7-country average, excluding central bank purchases
Source: Ministries of Finance, Yonhap Infomax, Hanwha Research Center

Central Bank Net Treasury Purchases (Trailing 12-Month Cumulative)

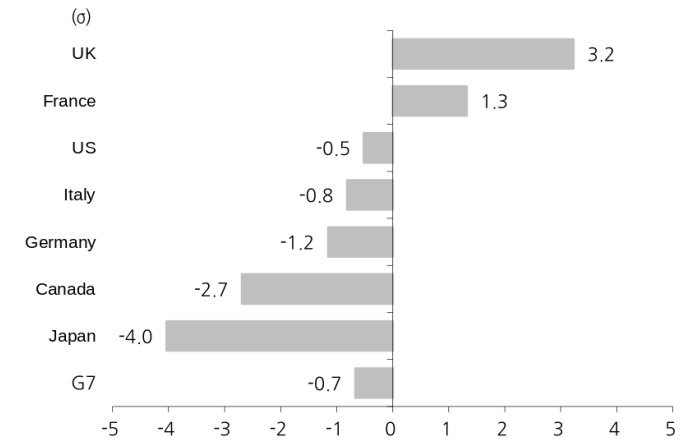


Note: Eurozone is country-level net purchases under PSPP and PEPP; September partly estimated
Source: Central banks, IMF, FRED, Hanwha Research Center

[The Problem Is Supply-Demand] Either Way, It Is Hard for Yields to Come Down

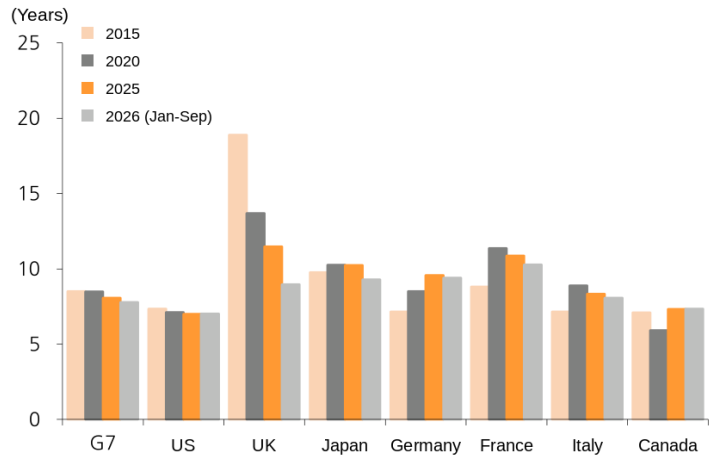
- Governments have responded to the supply-demand burden by shortening maturities. This has a greater effect of pulling forward the refinancing burden than of stabilizing long-term yields
  - The weighted average maturity of G7 coupon bond issuance shortened from 8.9 years at end-2021 to 7.7 years, and in the UK it nearly halved from 16.4 years to 8.6 years
- Unless central banks return, the supply-dominant supply-demand environment will not change easily
- Efforts to control long-term yields are not working, and central banks are unanimously saying there will be no more asset purchases

Treasury Supply-Demand Gap by Country (Trailing 12-Month Average)



Note: Average of Oct 2025 to Sep 2026; September private net purchases estimated
Source: Ministries of Finance and central banks, Yonhap Infomax, IMF, Hanwha Research Center

Weighted Average Maturity of Coupon Bond Issuance

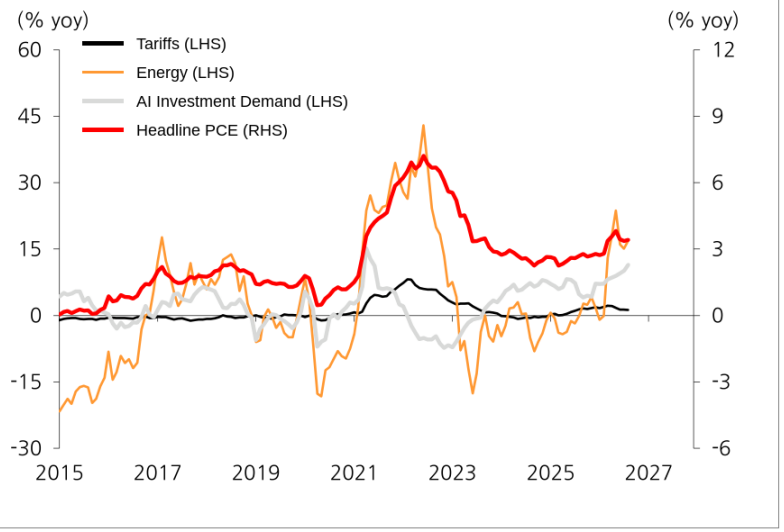


Note: Issuance-weighted; remaining maturity as of issue date
Source: Ministries of Finance and Debt Management Offices, Bank of Canada, Hanwha Research Center

[US] Fundamentals: Inflation, Services Remain the Problem Even as Upside Catalysts Weaken

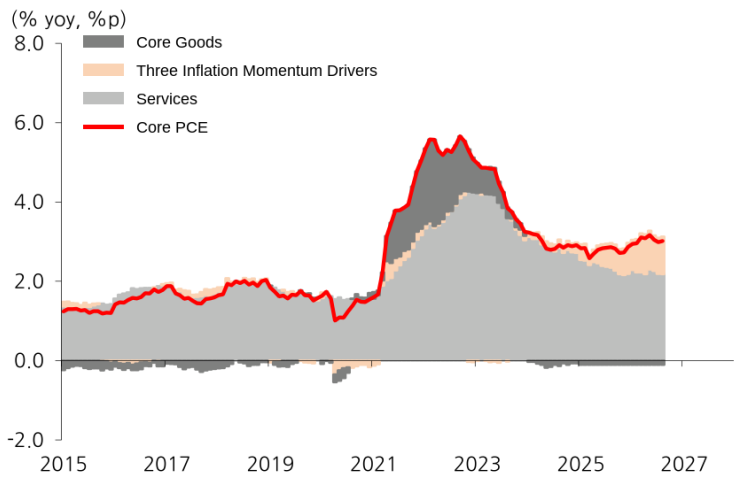
- All three factors driving inflation identified by the Fed are expected to see weaker upward pressure in 2027. However, the speed and degree of weakening differ by factor
  - Tariffs: may disappear around 1Q 2027. Following the Supreme Court ruling, the effective tariff rate fell from the 12% range to the 7% range
  - Energy prices: on a futures price basis, act as disinflationary pressure from 2Q 2027. They are an inflationary factor through 1Q only because of the base effect
  - AI investment demand: base effects from share prices and memory prices (disinflation) vs. growing demand pressure spreading to power, construction and technical labor (inflationary)
- Given solid growth and consumption, services inflation excluding portfolio management and airfares will remain around 3.0% throughout 2027
- The three inflation momentum factors are weakening at different speeds. Services inflation appears to be entrenched above the target. All of these imply slow disinflation

Three Inflation Momentum Drivers and Headline PCE Inflation



Note: Tariffs are core goods (excl. software); energy is PCE energy; AI is portfolio management and software prices
Source: Bureau of Economic Analysis, FRED, Yonhap Infomax, Hanwha Research Center

Services Inflation Pressure Expected to Remain in Effect in 2027

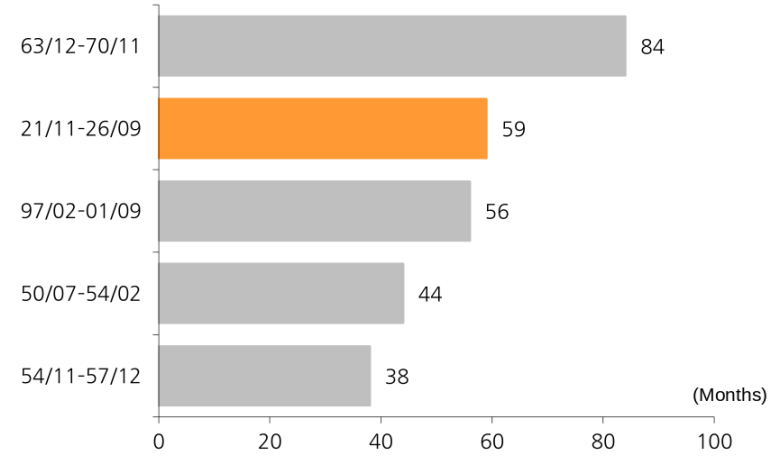


Source: Bureau of Economic Analysis, Hanwha Research Center

[US] Fundamentals: Low Unemployment Rate, a Changed Yardstick for Employment

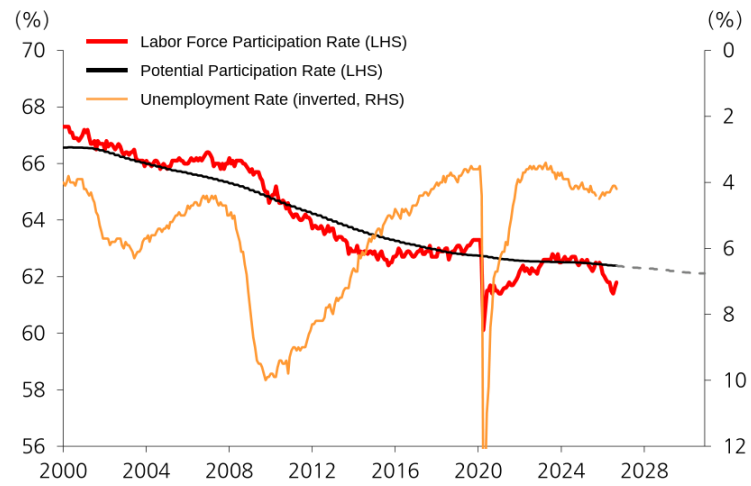
- The unemployment rate is the core employment indicator officially recognized by the Fed. In terms of the unemployment rate, the full-employment period is the second longest on record
  - Chair Warsh, Jackson Hole: "The labor market is consistent with full employment" / September FOMC: "The unemployment rate is consistent with full employment"
- Low employment growth is attributable to a supply decline, not evidence of an economic slowdown. "Low hiring + low firing" is a new equilibrium in which demand and supply have fallen together
  - 2025: employment growth (10,000 per month) fell far short of potential labor force growth (140,000), and the unemployment rate rose from 4.0% (January) to 4.5% (November)
  - 2026: the two numbers converged to 68,000 and 80,000, and the unemployment rate fell to 4.2% in September. Even if employment growth is only half of 2024's, the unemployment rate does not rise. The monthly job growth break-even (BEP) that maintains the current unemployment rate is only 50,000-60,000 in 2027

Second-Longest Full-Employment Period Since 1950



Note: Duration of periods when the unemployment rate was below the CBO natural rate of unemployment since 1950
Source: U.S. Bureau of Labor Statistics, CBO, FRED, Hanwha Research Center

Slowing Labor Demand Offset by Slowing Labor Supply

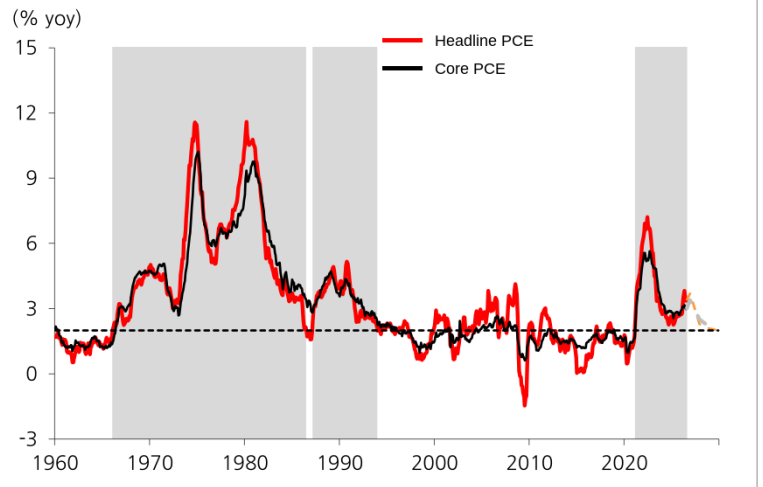


Note: Potential participation rate is the trend excluding cyclical factors
Source: U.S. Bureau of Labor Statistics, CBO, FRED, Hanwha Research Center

[US] Monetary Policy: A Credibility Problem If Inflation Is Not Contained This Time

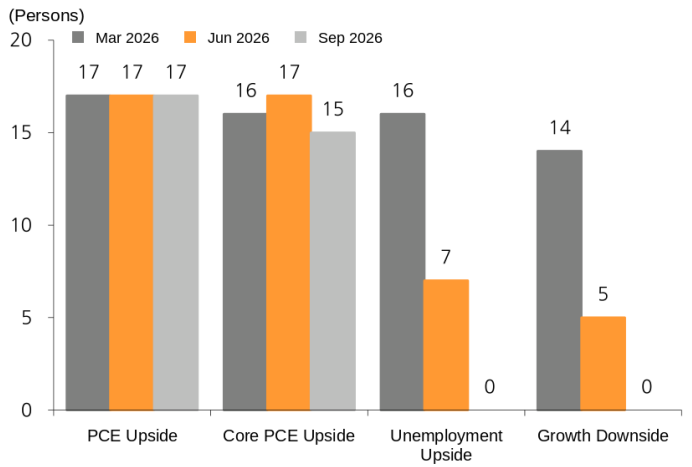
- The Fed's terminal policy rate is forecast at 4.25%, with an additional hike in December 2026 followed by a hold through year-end
- Since 2000, the policy rate was cut despite growth above potential and inflation above target only in 2003 and 2024
  - 2003: deflation concerns + growth slowdown (economic rebound in 2H) / 2024: inflation progress + employment slowdown concerns
  - On a September SEP basis, inflation is above target, growth is above potential, and employment remains effectively at full employment. The long-run policy rate is also on an uptrend
- The Fed's perception of the degree of monetary tightening and the credibility issue has also changed
  - Monetary policy is at an appropriate level → not very restrictive / The inflation trend needs to be watched → failing to contain it for over five years is a matter of credibility

Third-Longest Failure to Reach the Inflation Target on Record



Note: Shaded areas are periods when headline PCE exceeded 2% for 66 or more consecutive months (1966.2~1986.6, 1987.3~1993.12, 2021.3~)
Source: Bureau of Economic Analysis, the Fed, FRED, Hanwha Research Center

Inflation Is Serious, Growth and Employment Are Not a Concern

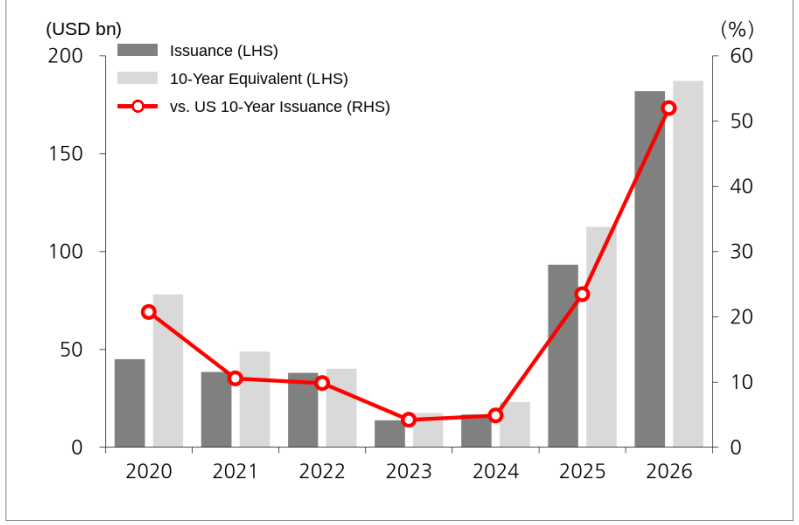


Note: Number of participants who see risks tilted in that direction. Unemployment upside means risk of higher unemployment. Participants: 19 in March, 18 in June and September
Source: the Fed, Hanwha Research Center

[US] Supply-Demand: Fiscal Plus AI

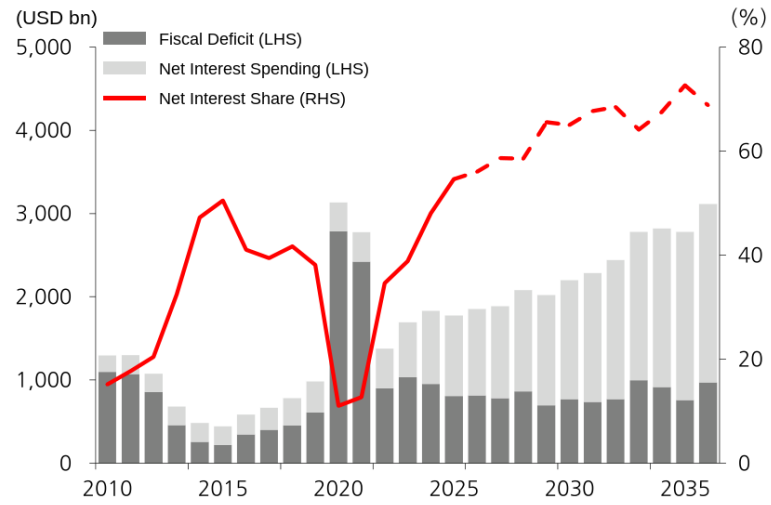
- The supply-demand environment is a factor that cements the era of 5% US Treasury yields
  - Supply from AI and fiscal policy, and demand from the changed stance of foreign investors, are acting as factors strengthening the downward rigidity of yields
- The amount of AI company corporate bonds issued in 2026, converted into US 10-year Treasury issuance, is USD 187.1bn. This is about 52% of US 10-year issuance over the same period
- The fiscal deficit from 2025 to 2036 is expected to increase by USD 1.34tn. Of this, USD 1.17tn (87.3%) is the increase in net interest expenditure
  - A vicious cycle has formed in which the higher the interest rate, the larger the interest expenditure and the resulting deficit, and the more government bond supply grows accordingly

AI Company Corporate Bond Issuance



Note 1: Dallas Fed 10-year equivalent method applied
Note 2: Issuance × modified duration ÷ 8 (assumed modified duration of 10-year Treasury), as of issue date
Source: U.S. Department of the Treasury, SEC, Dallas Fed, Hanwha Research Center

US Fiscal Deficit and Net Interest Spending

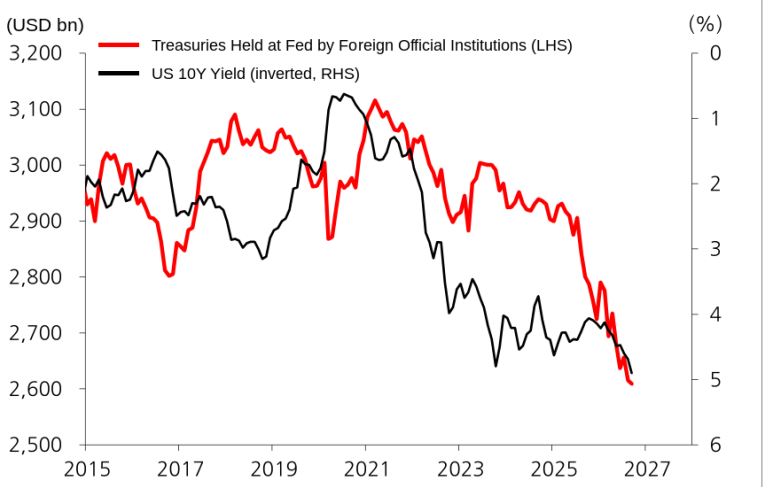


Source: CBO, Hanwha Research Center

[US] Supply-Demand: Shaky Status, Insufficient New Buyers

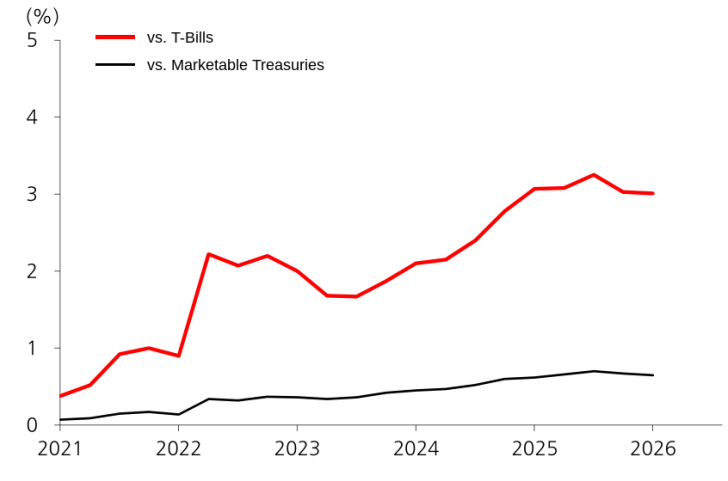
- US Treasuries have weakened investment appeal relative to government bonds that hold other safe-asset status. Their status is also not what it used to be
  - After FX hedging, US Treasury yields have become lower than or similar to the domestic government bond yields of reserve currency countries. The price and investment merit of US Treasuries have declined
  - Foreign official sector Treasuries held at the New York Fed fell by USD 115.8bn from the beginning of the year. On August 19 they recorded the lowest level since March 2011
- Stablecoin issuers are emerging as a new source of demand (the increase in their Treasury holdings is USD 185.4bn, 43% of China's decrease), but
  - The Treasuries they purchase are limited to Treasury bills, and even if the current trend of rising holdings is maintained, absolute holdings are far too small
  - The estimated 2030 holdings (USD 398.3bn) are only 1.3% of total marketable Treasuries outstanding and 6.0% of Treasury bills outstanding

US Treasuries Held by Foreign Official Institutions at the Fed and Yields



Source: the Fed, FRED, Hanwha Research Center

Stablecoin Holdings of US T-Bills as a Share of Total Marketable Treasuries

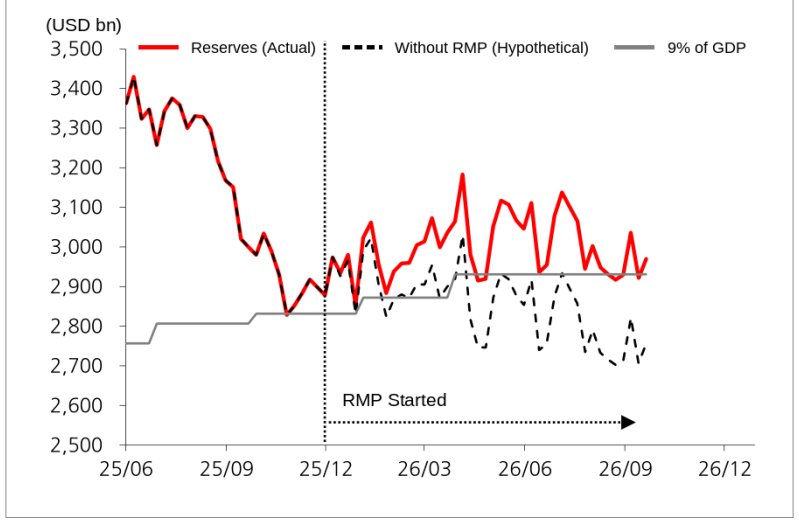


Note: Tether·USDC issuers' holdings of T-Bills and RPs ÷ quarter-end balance
Source: U.S. Department of the Treasury, Tether, Circle, Hanwha Research Center

[Liquidity] A Peaceful Quarter-End Created by RMPs

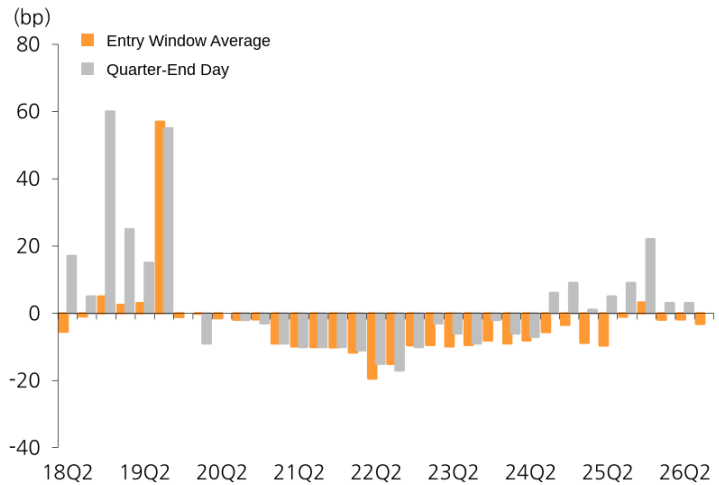
- There are four main reasons the normal short-term funding market environment is maintained even at quarter-end
  - 1) maintaining an adequate level of reserves through RMPs, 2) a stable TGA size, 3) the Fed's absorption of Treasury bills, 4) abolition of the SRF limit, etc.
- RMPs have kept reserves near 9% of GDP. After the end of QT, the reserve range is 8.95-9.78% of GDP
  - If the cumulative RMP limit of USD 215bn is deducted, it would be 8.46%. It can be estimated that this would have been lower than in October 2025 (about 9.07%), when there was a shortage phase
  - This arrangement prevented a further decline in reserves while absorbing Treasury bills, improving funding conditions in the repo market
- The SRF limit was abolished from December 2025. It plays a role in preventing interest rate spikes at quarter-ends and tax payment periods

Reserves Assuming No RMP



Note: Hypothetical reserves = actual reserves − RMP cumulative limit (other conditions assumed equal). 9% level = nominal GDP × 9%
Source: the Fed, New York Fed, FRED, Hanwha Research Center

Quarter-End SOFR-IORB Spread

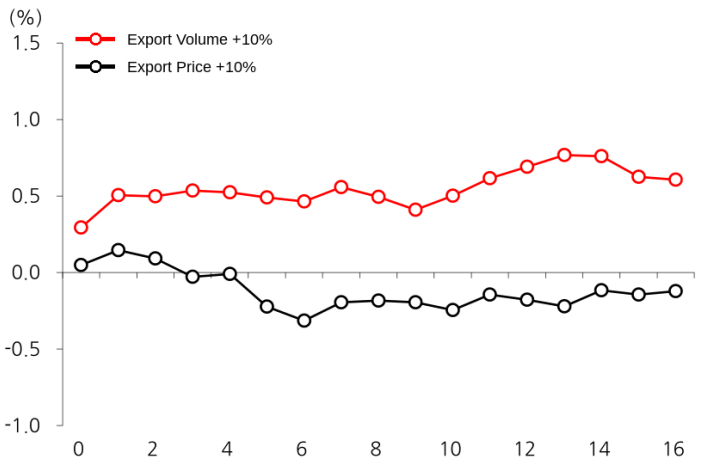


Note: Entry window is 10 to 3 trading days before quarter-end. 3Q26 quarter-end day not yet published as of 9/29
Source: New York Fed, FRED, Hanwha Research Center

[Korea] Fundamentals: Will Semiconductors Lift Potential Growth?

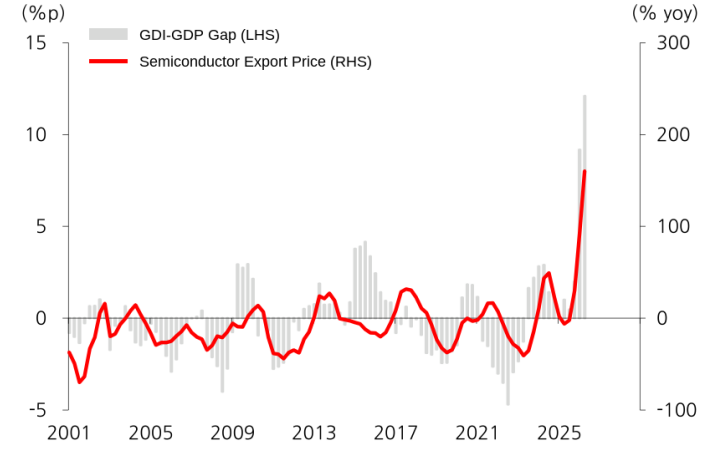
- The semiconductor cycle greatly affects actual growth but is expected to have only a temporary and limited effect on potential growth through the capital channel
  - Potential growth is a matter of production capacity. Labor, capital and productivity determine it
- The rise in estimates during a boom may be an illusion: the experience is that trend growth estimated at the end of an expansion was on average 0.26%p higher than later estimates
- Based on the two major semiconductor companies' capital expenditure plans, the contribution to potential growth is 0.18-0.23%p in 2026 and 0.25-0.31%p in 2027
  - In past CAPEX expansion periods (2017-2018, 2021-2022), the IT contribution was 0.4-0.5%p. Trend growth did not rise at that time
- An increase in GDI can support domestic demand by widening household purchasing power and corporate investment capacity, but in past cases (2009, 2015) the gap closed quickly

Non-IT GDP Change During Semiconductor Export Volume- and Price-Led Cycles



Note: X-axis is quarters elapsed since the semiconductor cycle began
Source: Bank of Korea, FRED, Hanwha Research Center

Real GDI-GDP Growth Gap and Semiconductor Export Price Growth

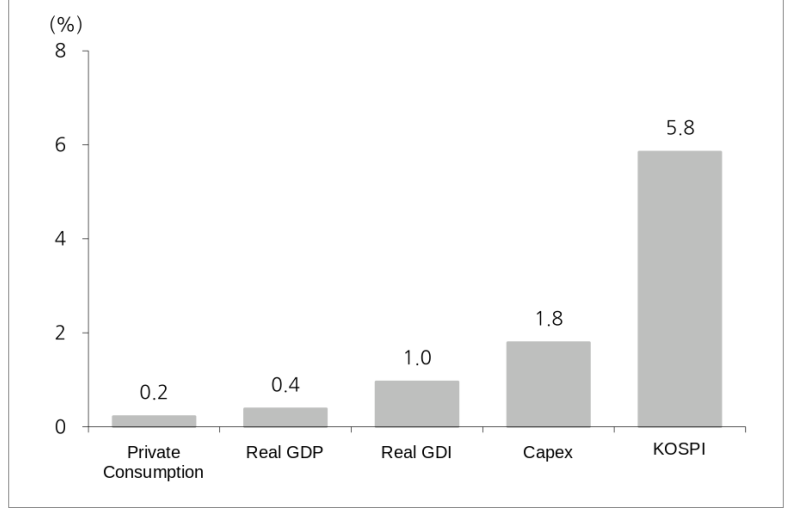


Note: Semiconductor export price is the quarterly average of the export value index ÷ export volume index (seasonally adjusted, calculated)
Source: Bank of Korea, Hanwha Research Center

[Korea] Fundamentals: Will Semiconductors Lift Potential Growth?

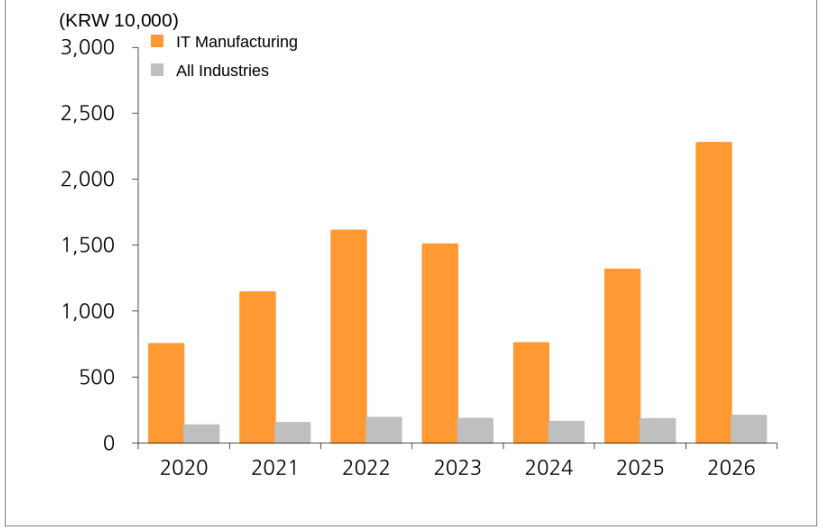
- The effect of the semiconductor industry's strength on consumption is even more limited
  - Since 2000, when semiconductor export value rises 10% in a quarter, the increase in private consumption one quarter later is only 0.2%
  - Semiconductors lift consumption in volume-led expansions. When export volume rises 10%, private consumption increases 0.45% one quarter later, whereas when price rises 10%, it increases 0.19% and then turns to a decline. This is because utilization and employment move together only when volume increases
- Bonuses, the most direct channel, also have a small effect. After-tax bonuses at Samsung Electronics and SK Hynix are estimated at KRW 4.4tn in 2026 and KRW 21.9tn in 2027
  - The resulting increase in consumption is KRW 0.9tn this year and KRW 3.1-4.6tn next year, or about 0.1% and 0.2-0.4% of private consumption

Change in Key Indicators When Semiconductor Exports Rise 10%



Note: Change in level after 1 quarter; analysis period is 2Q00-2Q26
Source: Bank of Korea, Ministry of Data and Statistics, FRED, Hanwha Research Center

Bonus Trends: IT Manufacturing vs. All Industries

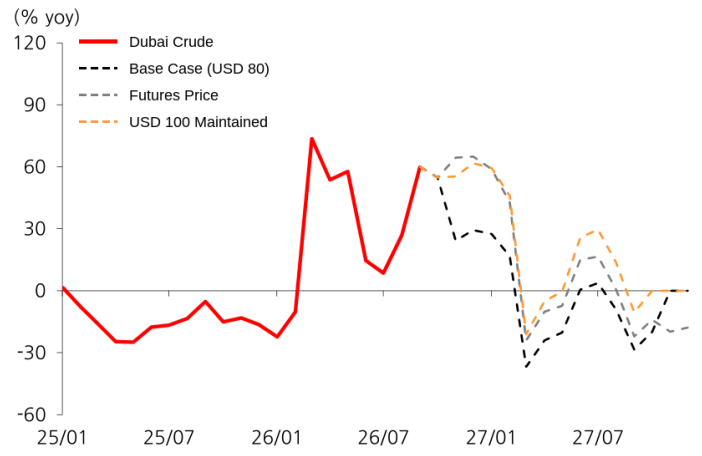


Note: Total special pay in Jan-Feb per regular employee
Source: Ministry of Employment and Labor, Hanwha Research Center

[Korea] Fundamentals: No Incentive to Slow Even as Oil Price Pressure Disappears

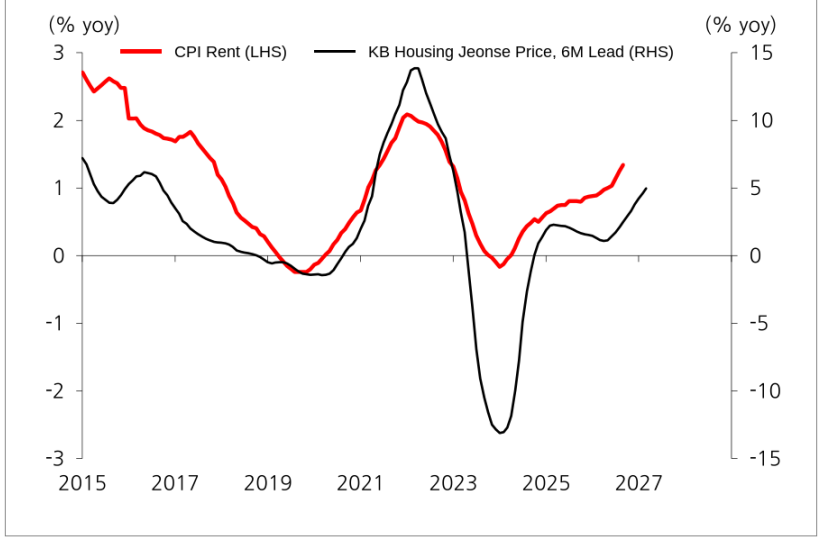
- Prices are expected to see oil prices act as upward pressure through 1Q and then contribute to a slowdown through the base effect. However, there is no clear incentive for a slowdown on the demand side
- Even assuming USD 80 per barrel from November through 1H 2027, the YoY rate is expected to remain positive at 17-29% from November to February 2027
  - The contribution of petroleum products to consumer prices falls from 1.0%p in October to 0.2%p in December, then rises again to 0.2-0.3%p in January-February 2027
- Although we expect semiconductors will not lift consumption meaningfully, there is also no catalyst to curb demand pressure
- What blocks prices from settling at 2% is services and public utility charges
  - Personal services prices and jeonse prices continue to rise. The need to reflect the cost burden of gas and electricity rates frozen for over two years (LNG import prices +26.5%) is increasing, among other factors

Even If Oil Falls to USD 80 Now, It Remains a Positive Factor for Prices Through February



Note: USD basis. Oct-Feb a year ago was USD 61.8-68.5; the March 2026 spike becomes the base month, causing a sharp drop in March 2027
Source: Bank of Korea, Yonhap Infomax, Hanwha Research Center

Jeonse Price Gains Push Up Rent

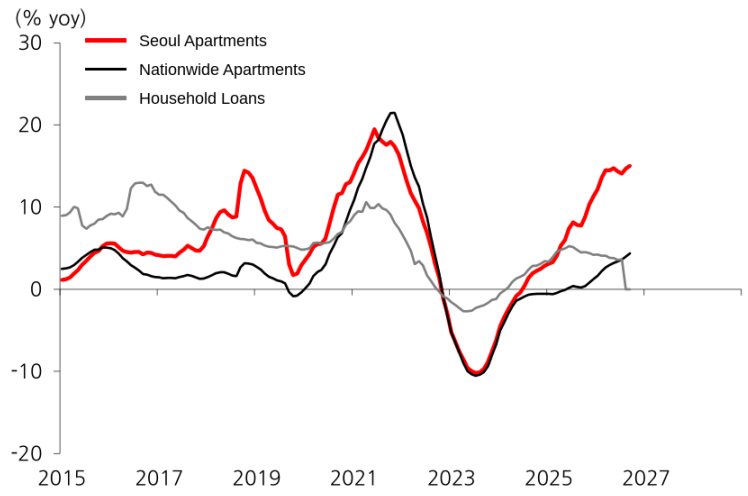


Note: YoY. KB Jeonse price shifted forward six months (through Mar 2027). Correlation since 2005 is 0.77
Source: Ministry of Data and Statistics, KB Kookmin Bank, Bank of Korea, Hanwha Research Center

[Korea] Monetary Policy: No Reason to Cut

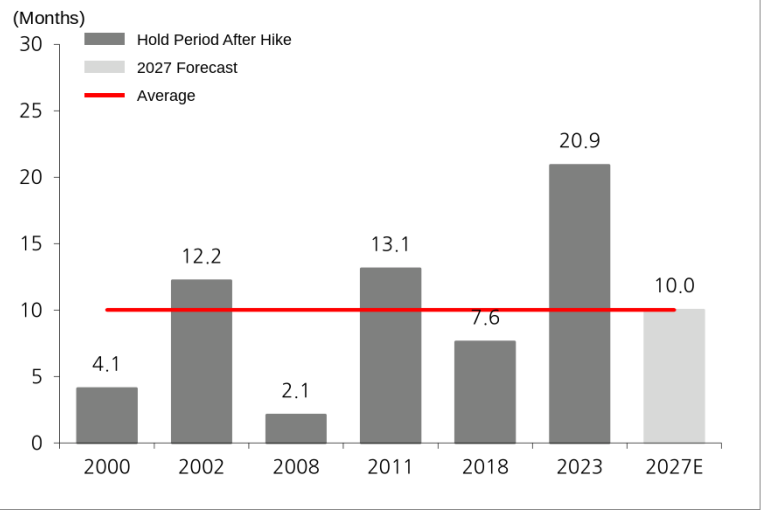
- The Bank of Korea's terminal policy rate is forecast at 3.50%, with additional hikes in November 2026 and February 2027 followed by a hold through year-end
- Considering the lingering effect of oil prices and the re-estimation of potential growth and the neutral rate, a terminal rate of 3.50% is reasonable
  - If the contribution of oil prices to inflation turns negative after 1Q, disinflationary pressure may strengthen. But that is purely due to the base effect
  - We do not believe semiconductors structurally lift potential growth, but the yardstick for monetary policy is the Bank of Korea's estimate
- Looking at expected inflation, growth, financial stability, the external environment and so on, there is no reason to cut anywhere
  - Expected inflation is becoming entrenched in the mid-to-high 2% range, the output gap is positive through 2027, apartment prices keep rising, and the Korea-US policy rate gap is inverted, among other factors

Financial Imbalances Not Improving



Note: Household loans are month-end balances of depository institutions (through July 2026)
Source: Bank of Korea, KB Kookmin Bank, Hanwha Research Center

Average of 10 Months from Last Hike to First Cut

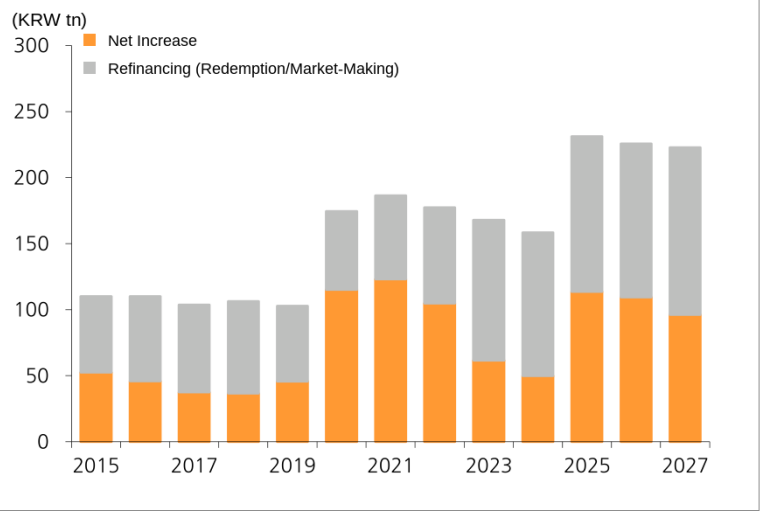


Note: From the date of the last hike in a tightening cycle to the date of the first cut. X-axis is the year of the last hike (Oct 2000, May 2002, Aug 2008, Jun 2011, Nov 2018, Jan 2023)
Source: Bank of Korea, Hanwha Research Center

[Korea] Supply-Demand: Unfavorable Fiscal Policy, Supply Burden Expected to Persist

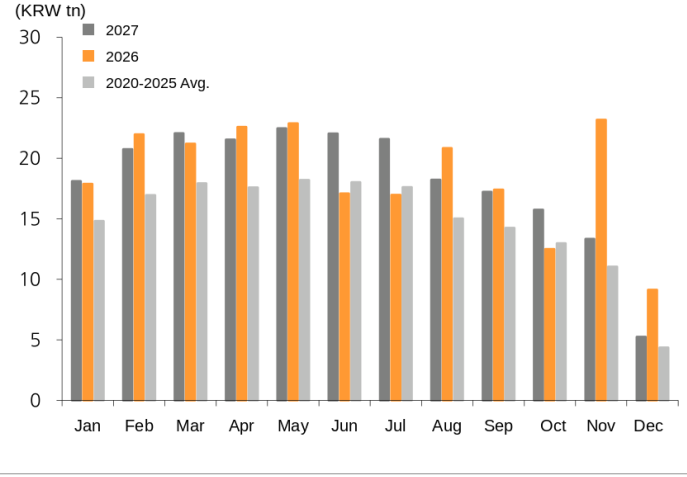
- The 2027 KTB issuance plan is KRW 222.8tn. It is KRW 2.9tn lower than the previous year's plan, but given strong tax revenue, an active issuance stance has been maintained
  - The net increase fell to KRW 96.2tn (-KRW 13.2tn), but remains above the 2020s average (KRW 82.6tn on an original budget basis)
- Current fiscal policy is built on the assumption that the semiconductor cycle will last for a considerable period. If the semiconductor cycle slows, an increased burden on the KTB market is unavoidable
  - The average duration of semiconductor expansion cycles since 2001 is 29-30 months. The current cycle is in its 43rd month (October 2026)
- On a monthly basis, supply of around KRW 18tn is expected. This is similar to 2026 (KRW 18.6tn per month) and more than KRW 3tn above the 2020-2025 average (KRW 14.9tn per month)
  - Applying the 2020-2025 average shares gives a monthly average of KRW 18.2tn and KRW 127.0tn in 1H (58.2% of the annual total). 2Q is the year's largest at KRW 66.1tn

Korea Treasury Bond Issuance Plan by Year



Note: Final plan incl. supp. budgets; 2027 is govt. proposal (redemption 110.7tn, market-making 15.9tn); 2025 per 2nd supp. budget
Source: Ministry of Finance and Economy, Ministry of Planning and Budget, National Assembly Budget Office, Hanwha Research Center

Expected Monthly KTB Issuance in 2027

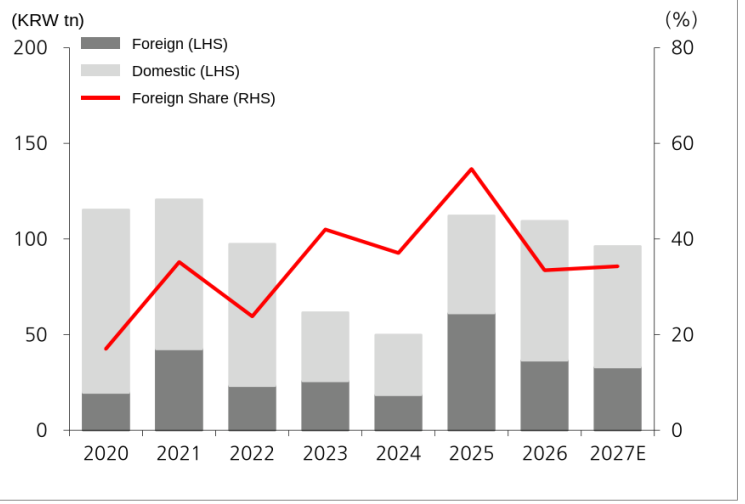


Note 1: Govt. proposal total issuance 222.8tn × monthly share. Historical avg. share = avg. issuance share of 2020-2025 plans (incl. supplementary budgets)
Note 2: Jan-Sep 2026 are preliminary actuals, October is the plan, and Nov-Dec is the remaining volume split by the 2020-2025 average share
Source: Ministry of Finance and Economy, Ministry of Planning and Budget, Hanwha Research Center

[Korea] Supply-Demand: Not Even WGBI

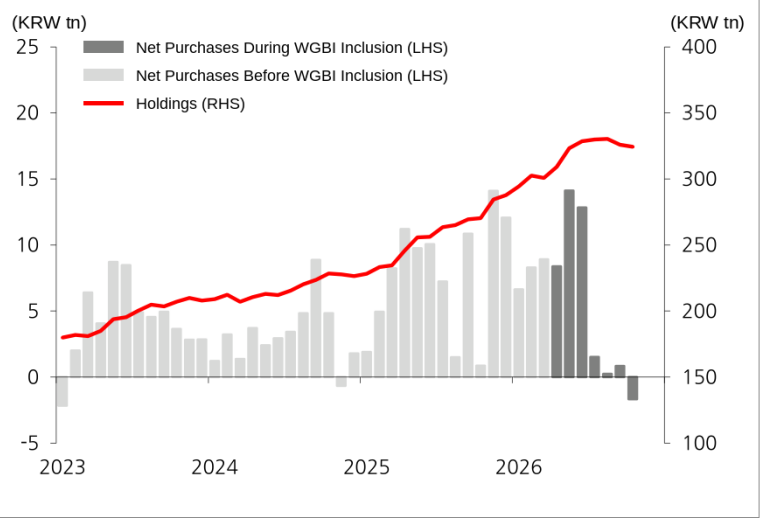
- With WGBI inclusion complete (though its actual effect was also limited), there will be no additional buying from index inclusion in 2027
  - Foreign average monthly net KTB purchases were KRW 8.4tn from January 2025 to June 2026 → KRW 0.2tn since July. Buying momentum was exhausted before inclusion ended
- Foreigners absorbed about 33% (KRW 36.6tn) of the 2026 net KTB increase. In 2027, domestic investors bear the burden of filling the shrinking foreign share
  - Assuming the foreign absorption share at the 2020-2025 average (34%), 2027 absorption is estimated at KRW 33.0tn for foreigners and KRW 63.2tn for domestic investors
  - The foreign share falls KRW 3.6tn from 2026. Even though the net increase falls KRW 13.2tn, the domestic share declines by only KRW 9.6tn, from KRW 72.8tn to KRW 63.2tn
- 2027 is a year without a clear demand-strengthening catalyst

Foreign Share of Net KTB Increase by Year



Note 1: Foreign = change in foreign KTB holdings; domestic = net increase minus foreign. 2020-2025 net increase is actual
Note 2: 2026 is the planned net increase and foreign change through Oct 12; 2027E applies the 2020-2025 average foreign absorption share (34%) to the government proposal net increase
Source: Ministry of Finance and Economy, Yonhap Infomax, Hanwha Research Center

Foreign Net KTB Purchases and Holdings



Note: OTC KTB basis. Net purchases are monthly totals; holdings are month-end (Oct is Oct 12). WGBI inclusion period is April-November 2026
Source: Yonhap Infomax, Hanwha Research Center

III

Credit

Tight & Steady

Si-hwa Han · shhan1229@hanwha.com · 3772-7737

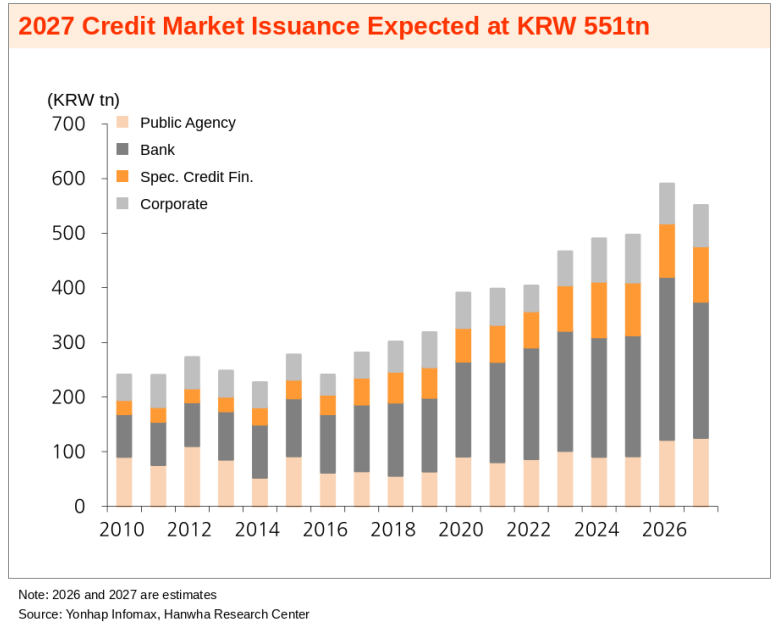
Tight & Steady

- 2027 credit market issuance is forecast at KRW 551tn. Issuance falls from 2026 on reduced bank bond issuance
- Bank bonds KRW 250tn > public agency bonds KRW 125tn > specialized credit finance bonds KRW 101tn > corporate bonds KRW 75tn
- Bank bond net issuance shrinks. Public agency bonds are driven by LH, specialized credit finance bonds by refinancing, and a resumption of corporate bond issuance is expected
- Credit spreads narrow only to a limited extent. Linked to the KTB 3Y's high-in-1H, low-in-2H path, they record a peak in 1H and then narrow
  - Buying is expected to broaden into lower grades, where the high-yield merit stands out

[Table 4] Credit Spreads Narrow to a Limited Extent

Item (bp)	Public Agency AAA 3Y	Bank AAA 3Y	Specialized Credit Finance AA+ 3Y	Specialized Credit Finance AA- 3Y	Corporate AA- 3Y	Corporate A+ 3Y
1Q26	29	30	48	61	55	94
2Q26	31	31	55	70	64	102
3Q26	34	34	58	73	68	106
4Q26(E)	36	38	59	75	69	106
1Q27(E)	37	39	60	76	70	108
2Q27(E)	36	38	59	74	68	105
3Q27(E)	35	37	56	71	66	101
4Q27(E)	34	35	55	69	63	98
2026(E)	33	33	55	70	64	102
2027(E)	36	37	58	73	67	103

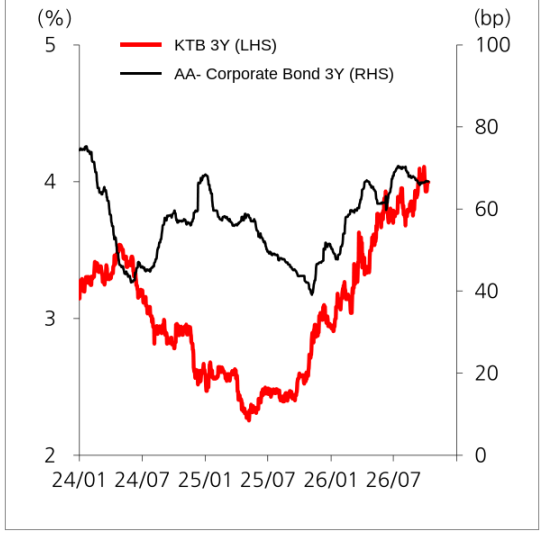
Source: Yonhap Infomax, Hanwha Research Center



Credit Spreads Are Low Considering the KTB Level

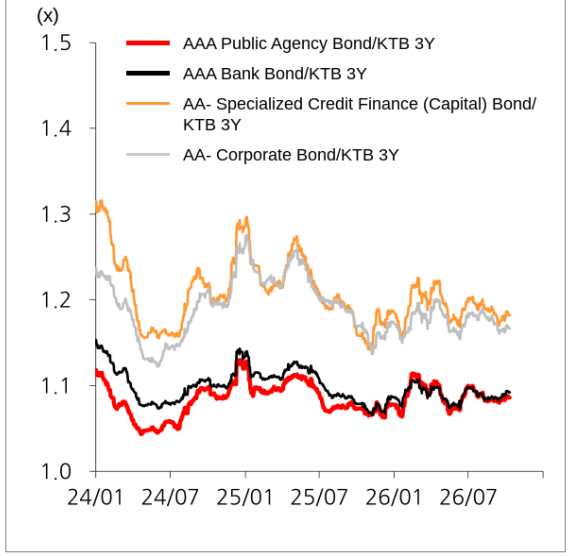
- With this year's surge in KTB yields, credit yields rose by a similar proportion, and 2026 credit spreads widened by 11-18bp
- The yield ratio (credit/KTB yield) is below its post-2010 average in all sectors
- Even compared with the average yield ratio during periods when yields were 3.5% or higher, all sectors are below average. Considering the KTB level, credit spreads are low
- Bank bonds hold an advantage over public agency bonds (+2.4bp) and capital bonds over corporate bonds (+6.1bp), but the gap has narrowed compared with early 2024

Credit Spreads Widen Slightly as KTB Yields Surge



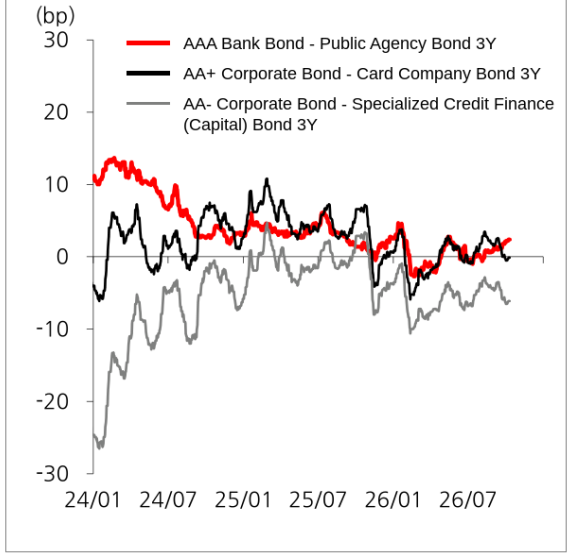
Source: Yonhap Infomax, Hanwha Research Center

Yield Ratio (Credit/KTB Yield) Trend



Source: Yonhap Infomax, Hanwha Research Center

Relative Value Trend by Sector



Source: Yonhap Infomax, Hanwha Research Center

[Public Agency Bonds] KRW 50tn of Refinancing Demand in 1H

- 2027 public agency bond maturities are around KRW 90tn
- 1Q KRW 27tn > 2Q KRW 23tn ≒ 3Q KRW 23tn > 4Q KRW 17tn, so refinancing demand is concentrated in 1H
- By institution, maturities are concentrated in MBS at KRW 17.7tn and KEPCO at KRW 16.7tn
- Korea Land & Housing Corporation (LH), whose issuance is expected to increase in 2027, has maturities of only about KRW 3.0tn, so the supply burden is expected to grow

[Table 5] 2027 Public Agency Bond Maturities of KRW 90tn; Refinancing Demand Concentrated in 1H, with KRW 27tn in 1Q and KRW 23tn in 2Q (Unit: KRW bn)

Item (KRW bn)	2027	1Q	2Q	3Q	4Q	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Total	90,066	26,996	23,417	23,057	16,596	8,087	9,153	9,755	6,691	7,885	8,841	8,352	7,359	7,346	5,449	6,761	4,386
Central Public Enterprises	69,823	21,011	18,511	18,106	12,195	5,311	6,775	8,925	5,763	5,950	6,798	6,730	5,825	5,551	4,151	4,900	3,145
KEPCO	16,700	5,180	3,350	5,160	3,010	1,470	1,760	1,950	980	1,040	1,330	1,840	1,840	1,480	1,380	1,100	530
Korea Housing Finance Corporation	6,930	1,500	2,190	1,830	1,410	620	160	720	690	810	690	750	820	260	300	770	340
Small and Medium Business Corporation	5,420	1,750	2,040	1,010	620	280	620	850	510	800	730	380	220	410	0	540	80
KOGAS	4,010	800	1,340	1,230	640	0	170	630	770	450	120	680	220	330	210	290	140
LH	3,015	1,369	316	773	557	535	250	584	20	279	17	5	137	631	58	391	109
Korea Expressway Corporation	2,950	1,600	230	590	530	210	630	760	60	100	70	70	170	350	0	340	190
Local Public Enterprises	12,854	3,675	3,057	3,371	2,751	1,767	1,518	390	878	1,235	944	1,222	884	1,265	839	871	1,042
Gyeonggi Housing & Urban Development Corp.	4,298	1,677	929	841	850	919	643	115	345	415	170	320	100	421	281	240	330
Seoul Housing & Urban Development Corp.	2,355	420	743	682	510	220	200	0	160	399	184	262	200	220	120	0	390
MBS	17,764	6,132	5,624	3,870	2,138	2,006	2,455	1,671	1,553	1,681	2,390	1,402	1,438	1,030	553	649	936
Federations, etc.	7,390	2,310	1,850	1,580	1,650	1,010	860	440	50	700	1,100	400	650	530	460	990	200

Source: Yonhap Infomax, Hanwha Research Center

[Public Agency Bonds] Public Institution Debt Ratio Shifts to a Rising Path

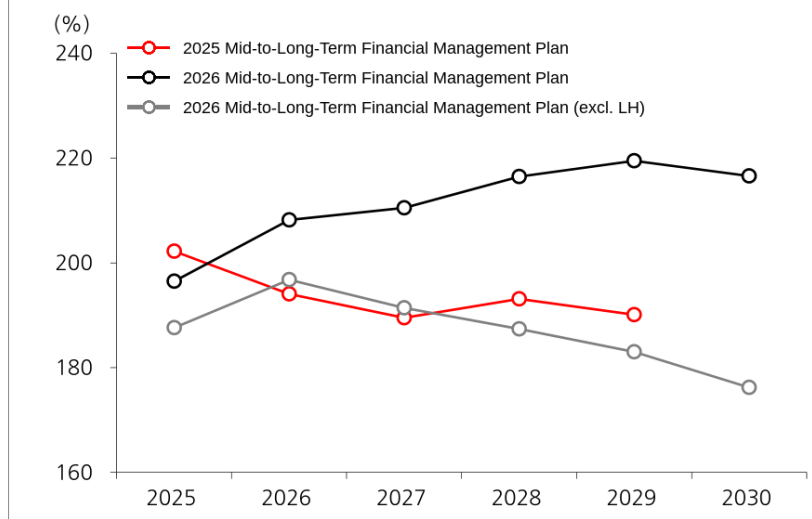
- The government's expansionary fiscal stance continues. 2027 budget proposal total expenditure is KRW 820.9tn (+12.8%), and public housing budget is KRW 30tn (+41.5%)
- The 2026-2030 Public Institutions Mid-to-Long-Term Financial Management Plan puts weight on expanding housing supply through debt. Key investment areas are residential welfare and energy
  - Assets of the 37 public institutions subject to the mid-to-long-term financial management plan are projected to reach KRW 1,457.9tn in 2030, an increase of KRW 371.4tn over five years
  - Debt in 2030 is KRW 997.4tn, an increase of KRW 277.4tn over five years. It is presented as growing 6.7% per year on average
- The debt ratio rises from 208.2% in 2026 to 216.6% in 2030. The plan submitted last year presented a path falling 12%p to 190% in 2029

2026-2030 Public Institutions Mid-to-Long-Term Financial Management Plan



Source: National Assembly of Korea, Hanwha Research Center

Public Institution Debt Ratio Forecast Trend

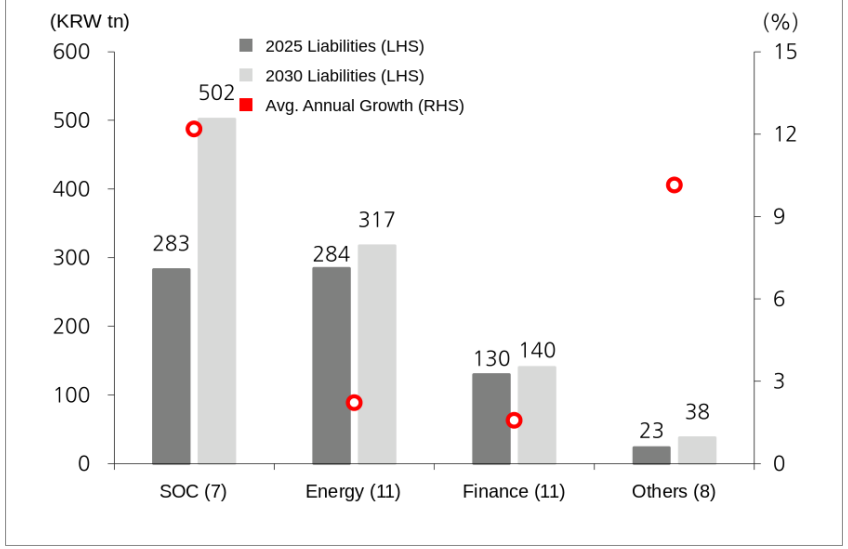


Source: National Assembly of Korea, Hanwha Research Center

[Public Agency Bonds] Debt Increase Concentrated in the SOC Sector

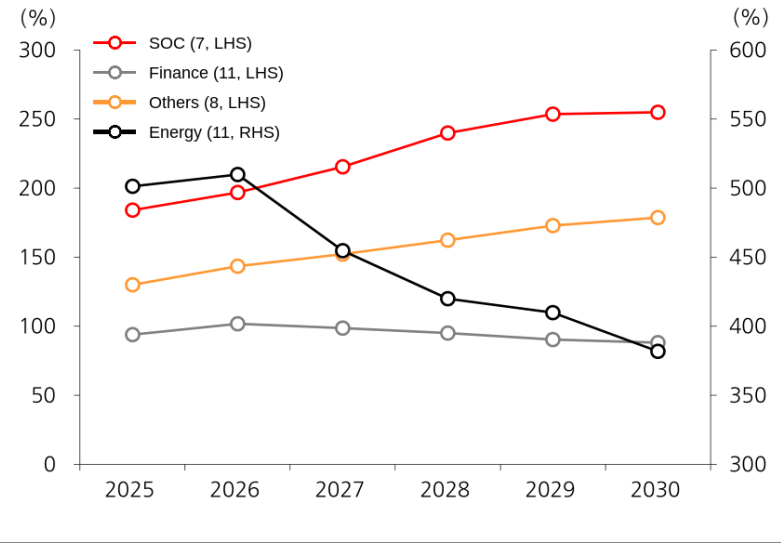
- Because of LH's expanded housing construction starts and earlier project timing, the increase in public institution debt is concentrated in the SOC sector
  - SOC debt in 2030 is KRW 502.2tn (+KRW 219.5tn), 79.1% of the total increase. Average annual growth rate is 12.2%
  - Debt of the energy and financial sectors grows only 2.2% and 1.6% per year on average, and their debt ratios are set to fall to 381.7% and 88%, respectively
- Excluding LH, 2030 debt is KRW 624.6tn (+2.7% per year on average), and the debt ratio improves from 187.6% to 176.2%
- 92% (KRW 177.8tn) of the increase in financial debt is in the SOC sector. Public institutions' market-based borrowing is also expected to increase centered on SOC

Debt Increase Concentrated in SOC Sector



Source: National Assembly of Korea, Hanwha Research Center

SOC Debt Ratio Rises; Energy and Finance Decline

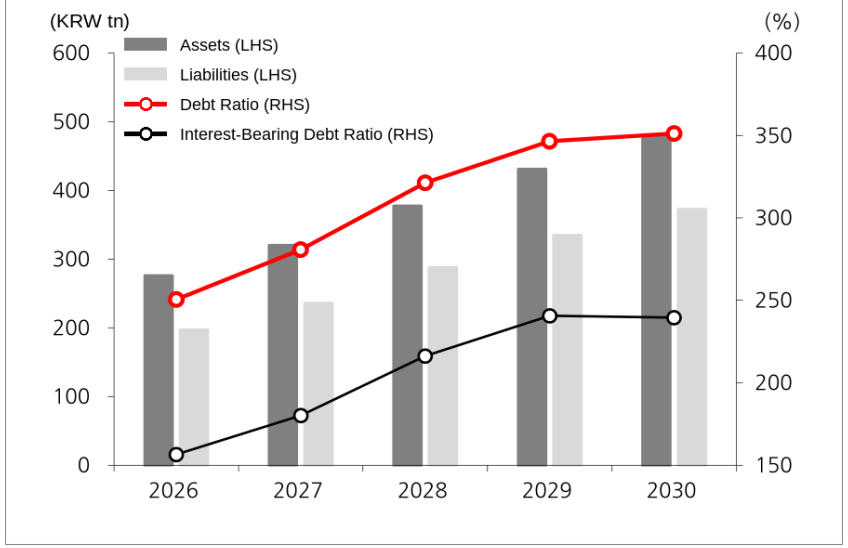


Source: National Assembly of Korea, Hanwha Research Center

[Public Agency Bonds] LH: Profitability Expected to Decline from Public-Led Policy

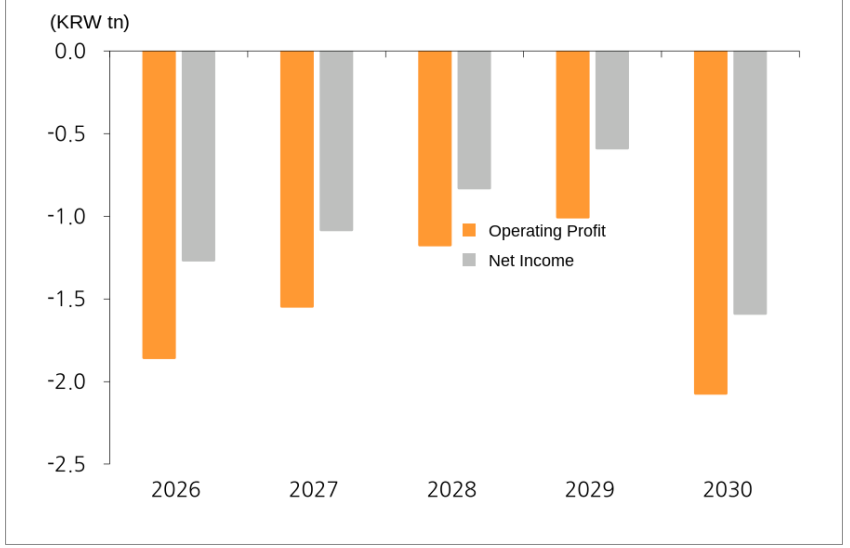
- LH's investment scale is surging due to the public-led housing policy. In 2030, assets are KRW 479tn and debt is KRW 373tn, so assets and debt both surge at the same time
  - Average annual project cost required through 2030 is KRW 62.6tn, KRW 10.6tn more than the previous year's plan
  - Financial debt is projected at KRW 322.5tn, up KRW 182.4tn over five years. The debt ratio worsens from 230.8% in 2026 to 351.2% in 2030
- Profitability is expected to decline as the recovery timing is deferred under the public transition. Net loss narrows for three years, then worsens again to KRW 1.6tn in 2030
  - This is because land sales decrease with the expansion of LH's direct implementation, and recovery of sale and rental revenue is deferred until after completion

LH Asset and Liability Forecast Trend



Source: National Assembly of Korea, Hanwha Research Center

LH Operating Profit and Net Income Forecast Trend

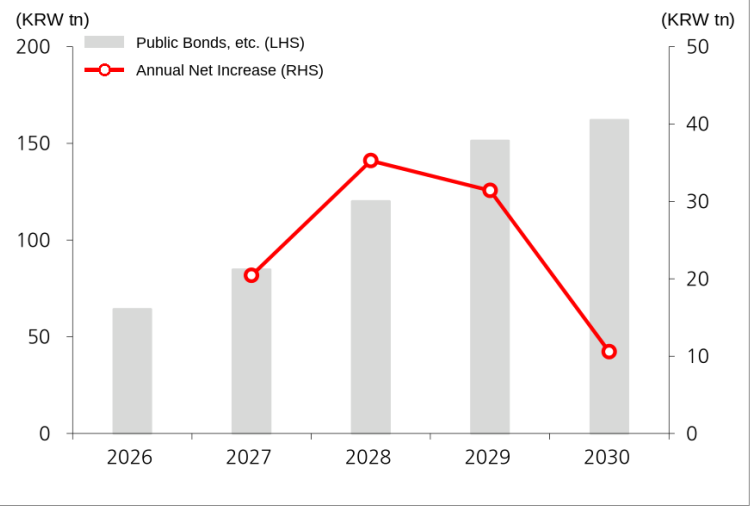


Source: National Assembly of Korea, Hanwha Research Center

[Public Agency Bonds] LH, Not KEPCO

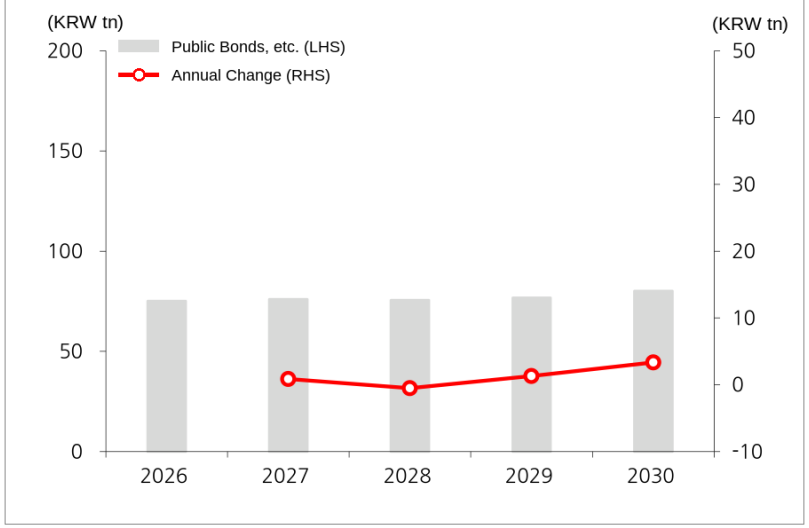
- The LH public bond balance, including short-term borrowings such as CP, is projected at KRW 162.0tn in 2030, up KRW 97.7tn from 2026
  - The annual net increase expands from KRW 20.4tn in 2027 to KRW 35.3tn in 2028. Supply burden is concentrated in 2028 and 2029
  - From 2028, public housing orders are to expand to over KRW 30tn per year. This reflects project costs increasing gradually to KRW 73.1tn in 2028 and KRW 76.1tn in 2029
- Going forward, public bond supply and demand is expected to be driven by LH rather than KEPCO, the former main supplier. KEPCO's debt increases only slightly, and its debt ratio is also expected to fall
  - LH debt increases KRW 199.1tn over five years. This is 72% of the total increase for public institutions

LH Public Bonds: Annual Net Increase



Note: Includes short-term borrowings such as CP
Source: National Assembly of Korea, Hanwha Research Center

KEPCO Public Bonds: Annual Change

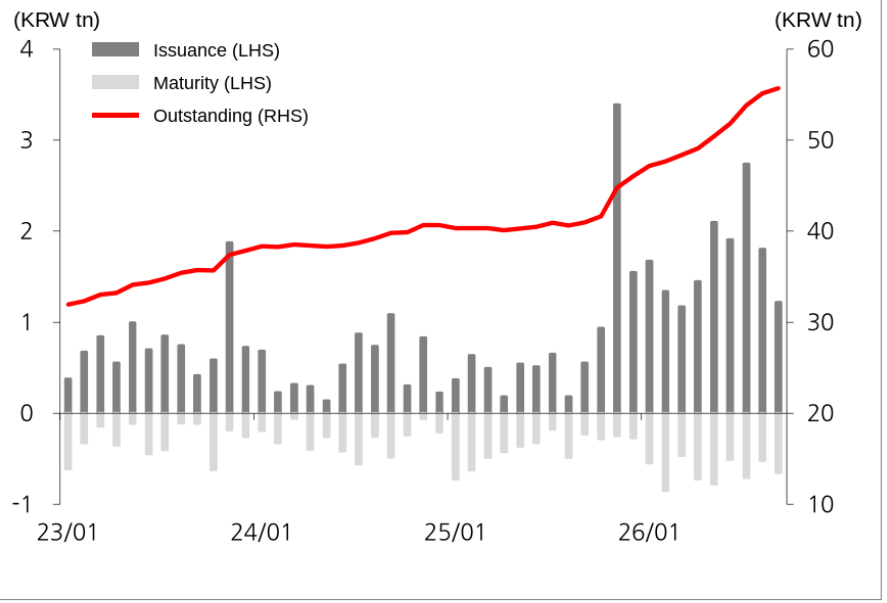


Source: National Assembly of Korea, Hanwha Research Center

[Public Agency Bonds] LH: Crowding-Out Effect Will Be Limited

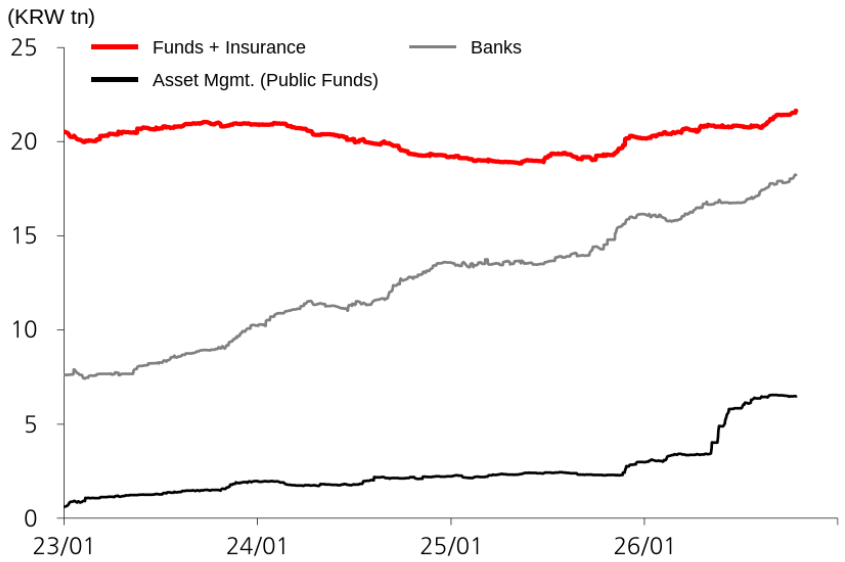
- The crowding-out effect on the public bond market, as during the 2022 surge in KEPCO bond issuance, is expected to be limited
  - Because LH is an institution subject to mandatory government loss compensation and a 0% bank risk weight applies, bank demand is solid
- LH public bond balance was KRW 55.2tn in September, up KRW 15.0tn from end-2024
  - Over the same period, banks' LH holdings increased KRW 4.5tn. Banks effectively absorb 30% of LH net issuance

LH Public Bond Balance Up KRW 14.5tn from End-2024



Source: Yonhap Infomax, Hanwha Research Center

LH Balance by Investor Type



Source: Yonhap Infomax, Hanwha Research Center

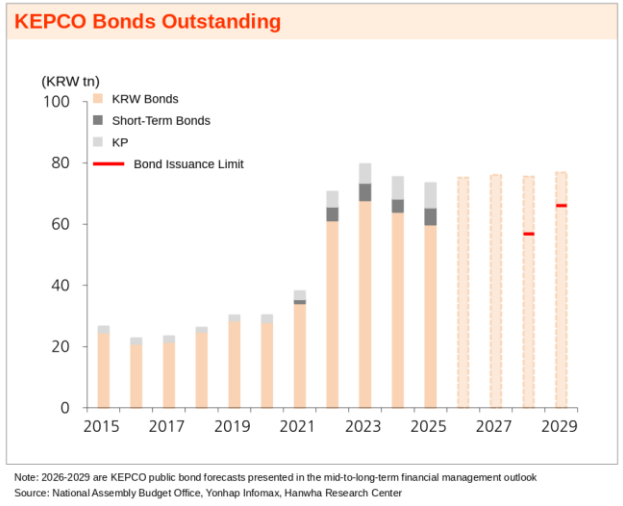
[Public Agency Bonds] KEPCO: Bond Balance Persistently Exceeds the Statutory Limit

- From 2028, KEPCO's bond issuance limit is normalized from the existing 5 times to 2 times the sum of capital stock and reserves
- According to the National Assembly Budget Office, KEPCO's bond issuance multiple is expected to persistently exceed the statutory limit even after the special provision sunsets
  - The bond issuance multiple exceeds 2x at 2.3x in 2028 and 2.1x in 2029. It is a structure that is not resolved by one-off redemptions at the sunset point
  - Based on the mid-to-long-term financial management plan, the end-2027 bond balance of KRW 76.0tn is estimated to exceed the 2x limit of KRW 56.8tn by KRW 19.2tn
  - To comply with the statutory limit in 2028, bond redemption of KRW 19.2tn per year or an increase of KRW 9.6tn in the sum of capital stock and reserves is needed

[Table 6] KEPCO Bond Issuance Multiple Persistently Exceeds the Statutory Limit Even After the Special Provision Sunsets

Item (KRW bn)	2025 (Actual)	2026	2027	2028	2029
Capital Stock	3,210	3,210	3,210	3,210	3,210
Reserves (Forecast)	21,121	22,183	25,208	29,832	33,525
Capital Stock + Reserves	24,331	25,393	28,418	33,042	36,735
Issuance Limit		121,653	126,964	56,835	66,083
Outstanding Issuance		75,188	76,049	75,546	76,832
Bond Issuance Multiple Forecast (x)	3.0	3.0	2.7	2.3	2.1
Meets 2x Statutory Limit	×	×	×	×	×

Note: Separate basis
Source: National Assembly Budget Office, Hanwha Research Center



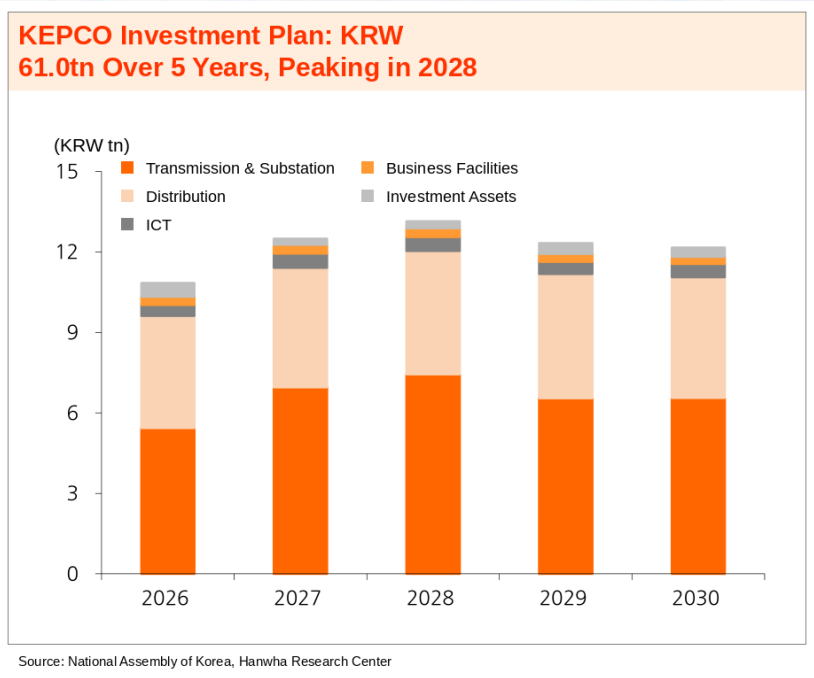
[Public Agency Bonds] KEPCO: Earnings Improvement Alone Will Not Achieve Compliance

- Options for complying with the statutory limit are (1) bond redemption, (2) increasing capital stock and reserves, (3) deferring the special provision sunset
- To bring the end-2027 bond balance of KRW 76.0tn within the 2x limit, reserves must be increased by KRW 13.7tn
  - Cumulative 2026-27 net income in the mid-to-long-term financial management plan is KRW 6.8tn. Even assuming all profits are retained (no dividends), there is a shortfall of KRW 6.9tn
  - It is difficult to comply with the limit through reserve increases from earnings improvement alone. A combination of reduced dividends, paid and free capital increases, expanded subsidiary dividends and bond redemption is needed
- The investment burden also limits capacity for bond redemption. Investment for 2026-30 is expected at KRW 61.0tn, or KRW 12.2tn per year on average
- KEPCO has stated that it will reach the 2x bond issuance multiple by end-2030 and plans to pursue discussions with the government on amending the KEPCO Act. The likelihood of extending the special provision is high

[Table 7] Reserve Increase Required to Comply with the Statutory Limit at End-2027

Item (KRW bn)	End-2025	End-2026	End-2027
Net Income		1,725	5,042
Cumulative Net Income (A)		1,725	6,767
Capital Stock	3,210	3,210	3,210
Reserves	21,121	22,846	27,888
Capital Stock + Reserves (B)	24,331	26,056	31,097
Bond Balance (C)		75,188	76,049
Required Capital Stock + Reserves (D = C ÷ 2)		37,594	38,024
Reserve Shortfall (C-B)		11,538	6,927

Note 1: Reserves are the end-2025 reserves (retained earnings after deducting the year-end dividend + other reserves) plus cumulative net income (A)
Note 2: Assumes full retention of net income
Note 3: The 2x limit applies after the special provision sunsets (2027.12.31). End-2026 falls within the 5x special provision period and is a reference figure
Source: National Assembly Budget Office, National Assembly of Korea, Hanwha Research Center



[Bank Bonds] Shorter Issuance Maturities Increase Refinancing Demand

- 2027 bank bond refinancing demand is KRW 218tn. It exceeds KRW 193tn in 2024 and KRW 195tn in 2025. 1H KRW 125tn > 2H KRW 93tn
  - Reflecting higher interest rate volatility, bank bond issuance maturities shortened in 2026. The share of issuance with maturities of one year or less in January-September rose YoY
- Specialized banks are at around KRW 156tn. With the expansion of policy finance, it stays at a high level above 2024 (KRW 131tn) and 2025 (KRW 134tn), following 2026
- Commercial banks are at KRW 55tn. Hana KRW 16.7tn > Shinhan KRW 12.8tn ≒ Woori KRW 12.6tn > KB Kookmin KRW 9.1tn. Maturities are concentrated in March at KRW 7.6tn and July at KRW 8.3tn

[Table 8] 2027 Bank Bond Maturities of KRW 218tn: Specialized Banks KRW 156tn, Commercial Banks KRW 55tn (Unit: KRW bn)

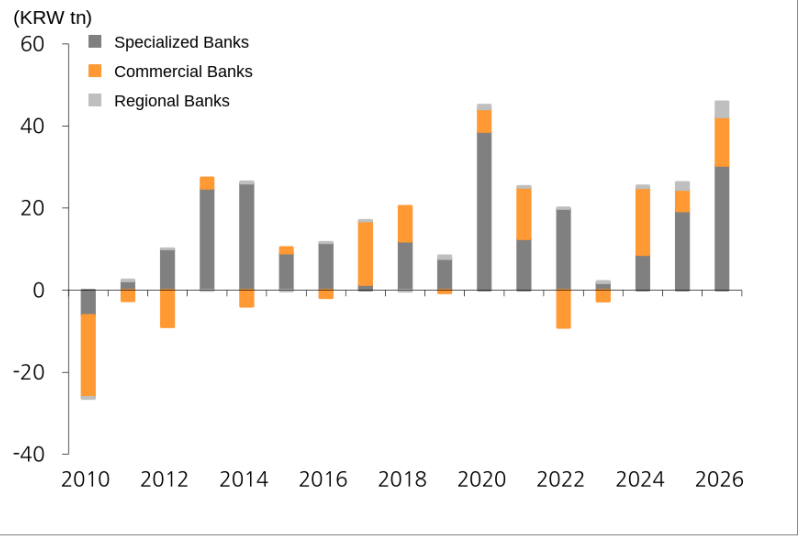
Item (KRW bn)	2027	1Q	2Q	3Q	4Q	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Total	218,247	59,640	65,384	55,022	38,201	14,580	18,000	27,060	22,344	21,970	21,070	23,681	16,011	15,330	15,470	14,041	8,690
Specialized Banks	155,647	43,740	46,504	36,492	28,911	12,410	13,280	18,050	15,564	16,580	14,360	14,951	10,301	11,240	9,570	11,701	7,640
KDB	58,467	15,890	15,714	13,862	13,001	4,840	5,130	5,920	5,134	5,640	4,940	5,131	4,741	3,990	4,620	4,331	4,050
IBK	58,870	17,990	19,170	11,100	10,610	4,340	5,700	7,950	6,380	7,500	5,290	5,480	2,700	2,920	2,720	5,150	2,740
KEXIM	18,470	5,330	4,680	4,900	3,560	1,920	1,610	1,800	1,620	1,340	1,720	2,100	1,410	1,390	1,470	1,240	850
NH Nonghyup	14,510	3,110	4,830	5,110	1,460	590	690	1,830	1,430	1,620	1,780	2,080	920	2,110	480	980	0
Suhyup	5,330	1,420	2,110	1,520	280	720	150	550	1,000	480	630	160	530	830	280	0	0
Commercial Banks	54,860	12,970	16,570	16,620	8,700	1,790	3,550	7,630	5,810	4,890	5,870	8,310	4,980	3,330	5,850	1,850	1,000
Hana	16,730	4,060	3,940	6,510	2,220	210	1,150	2,700	1,800	1,190	950	3,650	1,600	1,260	1,840	380	0
Shinhan	12,800	2,520	4,080	3,740	2,460	960	450	1,110	1,010	1,240	1,830	1,400	1,660	680	1,460	800	200
Woori	12,610	2,290	4,390	4,100	1,830	0	510	1,780	1,860	1,100	1,430	1,990	1,130	980	680	450	700
KB Kookmin	9,080	3,560	3,330	1,360	830	520	1,340	1,700	670	1,160	1,500	960	240	160	510	220	100
iM Bank	2,230	450	530	420	830	100	100	250	170	200	160	310	0	110	830	0	0
SC	1,410	90	300	490	530	0	0	90	300	0	0	0	350	140	530	0	0
Regional Banks	7,740	2,930	2,310	1,910	590	380	1,170	1,380	970	500	840	420	730	760	50	490	50
Busan	4,100	1,440	840	1,310	510	220	470	750	360	130	350	340	730	240	50	460	0
Gwangju	1,190	580	430	180	0	160	260	160	180	200	50	80	0	100	0	0	0
Gyeongnam	1,380	730	330	290	30	0	330	400	120	0	210	0	0	290	0	30	0
Jeju	280	0	230	0	50	0	0	0	100	100	30	0	0	0	0	0	50
Jeonbuk	790	180	480	130	0	0	110	70	210	70	200	0	0	130	0	0	0

Source: Yonhap Infomax, Hanwha Research Center

[Bank Bonds] Issuance Market Led by Specialized Banks

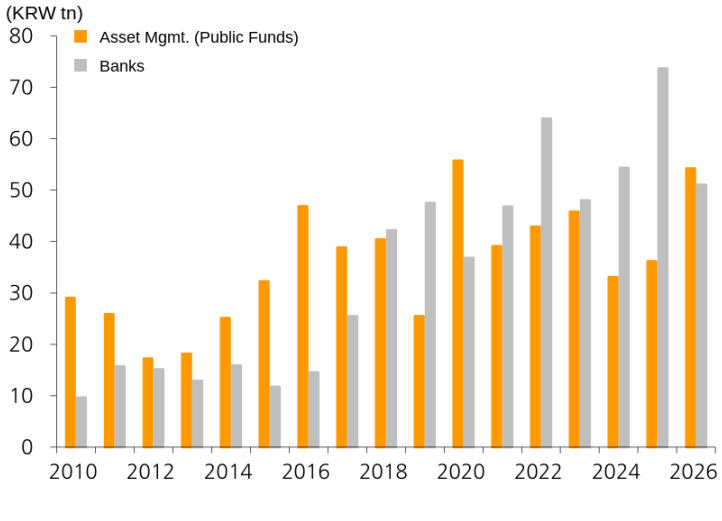
- Specialized bank bond issuance in 2026 is KRW 160.2tn as of October, 73.5% of total bank bond issuance
- With expanded productive finance and large-scale annual maturity refinancing of KRW 173tn, net issuance is KRW 29.6tn
- This year, government-guaranteed bond issuance, such as KDB advanced strategic industry fund bonds and KEXIM supply chain stabilization bonds, has also increased
- Specialized bank bonds are highly liquid assets, so steady demand from banks flows in and absorbs the increase in issuance. This year in particular, investment trust net purchases exceeded bank demand

Annual Bank Bond Net Issuance Trend



Note: 2026 is cumulative through October
Source: Yonhap Infomax, Hanwha Research Center

Annual Net Purchases of Specialized Bank Bonds by Banks and Investment Trusts

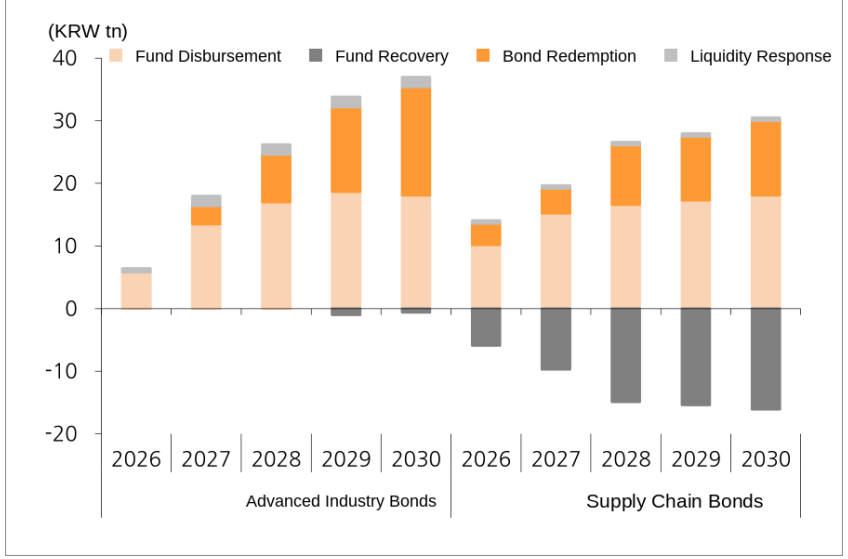


Source: Yonhap Infomax, Hanwha Research Center

[Bank Bonds] Government-Guaranteed Bond Issuance for Policy Finance Continues

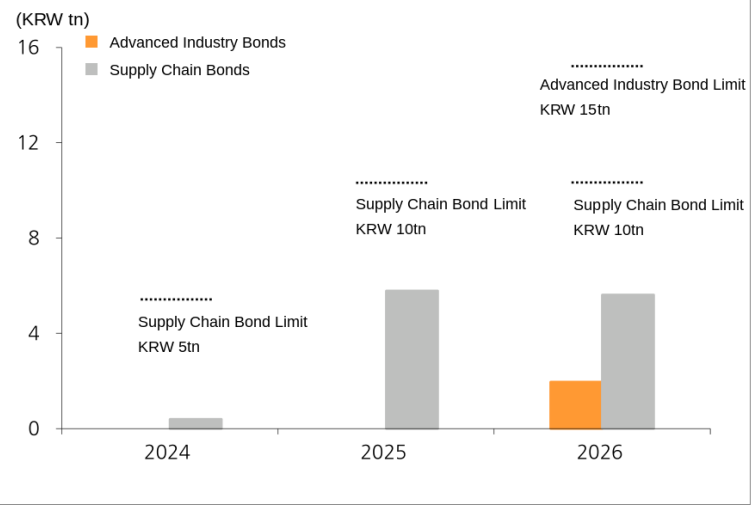
- The 2027 advanced industry bond issuance limit is set at KRW 18tn: fund disbursement KRW 13.5tn + refinancing KRW 2.9tn + liquidity response KRW 1.6tn
- The supply chain bond limit is KRW 10tn, the same as this year: fund disbursement KRW 15.2tn + refinancing KRW 4tn + liquidity response KRW 0.5tn
  - Fund recovery is KRW 9.7tn, more than half of disbursements
- 2026 issuance results are KRW 1.96tn of advanced industry bonds and KRW 5.6tn of supply chain bonds. Actual 2027 issuance will also fall short of the limit
- Government-guaranteed KP (Korea Paper) supply is also increasing. After the first KP (USD 500mn) of supply chain bonds in 2026, policy funds' foreign currency funding is expected to expand
  - KP issuance is highly likely, such as the Korea-US Strategic Investment Corporation bonds with a government guarantee limit of USD 7bn and KEXIM strategic export finance fund bonds of KRW 1.2tn

Estimated Bond Issuance Limits by Fund



Source: Bill Information System, Hanwha Research Center

Specialized Banks: Cumulative Government-Guaranteed Bond Issuance

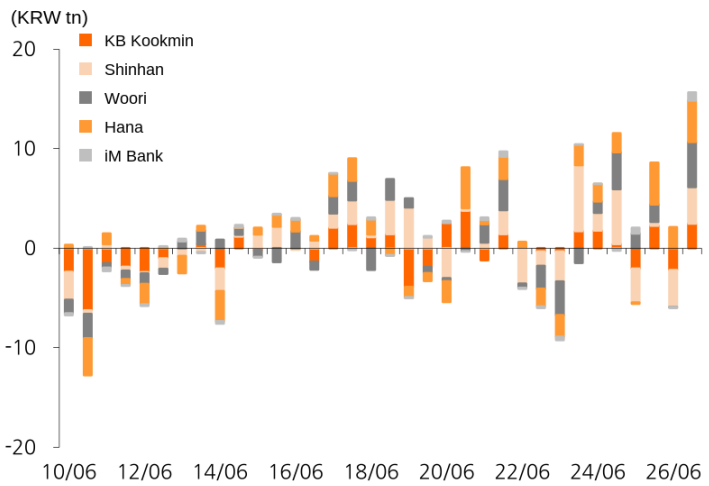


Note: 2026 is cumulative through October
Source: Yonhap Infomax, Hanwha Research Center

[Bank Bonds] Temporary Increase in Commercial Bank Issuance in 2H26

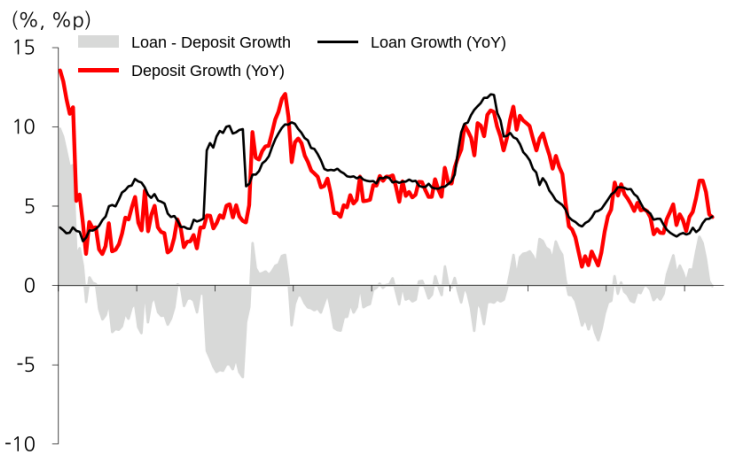
- Commercial bank bonds turned from net redemption in 1H 2026 to net issuance. This is the result of 4Q refinancing volume plus demand for managing regulatory ratios due to corporate deposit maturities
- Through 1H, incentives for bank bond issuance were limited. With loan growth held to the 3% range by household loan regulation, deposits were solid due to corporate fund inflows from semiconductors and other sectors
- Deposit growth has exceeded loan growth for 14 consecutive months. However, the gap narrowed from 3.1%p in May to 0.3%p in August and then inverted to -0.04%p in September
- With commercial bank supply burden limited, bank bonds are strong relative to public agency bonds. The spread gap narrowed from an average of 8.0bp in 2023-25 to 0.3bp in 1H 2026
  - The spread gap of bank bonds versus public agency bonds inverted in February-April, and is widening again to 2bp in October with increased issuance in 2H

Commercial Bank Net Issuance Rises in 2H26



Note: 2H26 is cumulative through October
Source: Yonhap Infomax, Hanwha Research Center

Deposit Growth Exceeds Loan Growth for 14 Consecutive Months

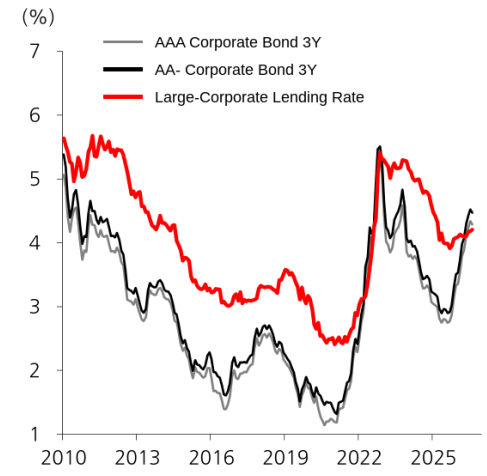


Source: Bank of Korea, Hanwha Research Center

[Bank Bonds] Corporate Funding Returns to Corporate Bonds

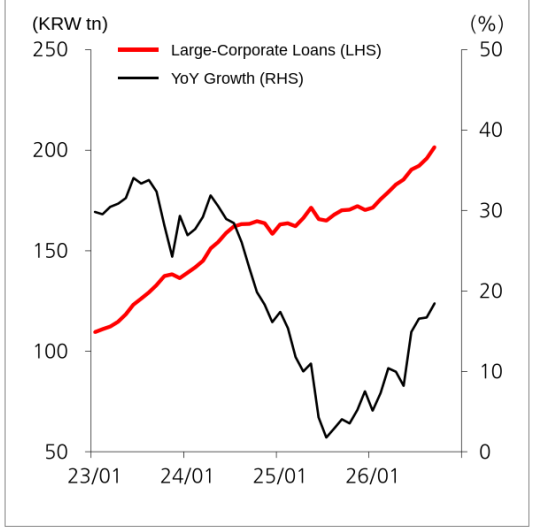
- As the policy rate hiking cycle began, market yields surged and corporate bond yields inverted below bank lending rates for the first time in 42 months
- Corporate funding shifted to bank loans, so corporate bond issuance was sluggish. Since the beginning of the year, net redemptions of KRW 1.0tn in high-grade and KRW 5.2tn in lower-grade corporate bonds have been recorded
- Large-corporate loan balances at the Big 5 banks were KRW 201tn in September, up 18.4% YoY, while SME loans increased only 2.8%
  - Large-corporate loans surged temporarily this year. The share of large corporations in the monthly increase in corporate loan balances keeps rising: 40% in July → 59% in August → 72% in September
- In 2027, with interest rate volatility easing, companies that postponed issuance are expected to return to the corporate bond market. With loan demand slowing, the incentive for bank bond issuance weakens

Corporate Bond Yields Invert Below Bank Lending Rates



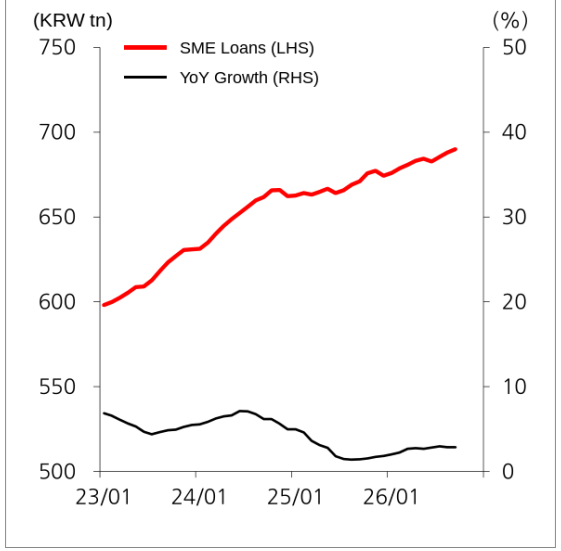
Note: Large-corporate lending rate is based on new loans
Source: Yonhap Infomax, Hanwha Research Center

Big-5 Banks: Large-Corporate Loan Balance Trend



Source: Yonhap Infomax, Hanwha Research Center

Big-5 Banks: SME Loan Balance Trend



Source: Yonhap Infomax, Hanwha Research Center

[Bank Bonds] Introduction of the Systemic Risk Buffer

- Under the Financial Services Commission's August 13 measures, the household debt aggregate target was relaxed to 3% (originally 1.5%), but capital regulation on mortgages was tightened
- In January 2027, the types of high-risk mortgages are expanded and risk weights are raised to 2-4 times (as an example)
- In 2H 2027, the Systemic Risk Buffer (SSyRB) is introduced. Each bank must additionally accumulate capital of 0-1%p depending on the share of mortgage loans in risk-weighted assets
  - The household debt-to-GDP ratio at end-2025 was 88.6%, above the 80% regulatory trigger presented as an example
- Banks are expected to respond by adjusting mortgage lending rather than raising capital. With loan growth constrained, the incentive for bank bond issuance weakens

[Table 9] Strengthened Capital Regulation on Mortgages

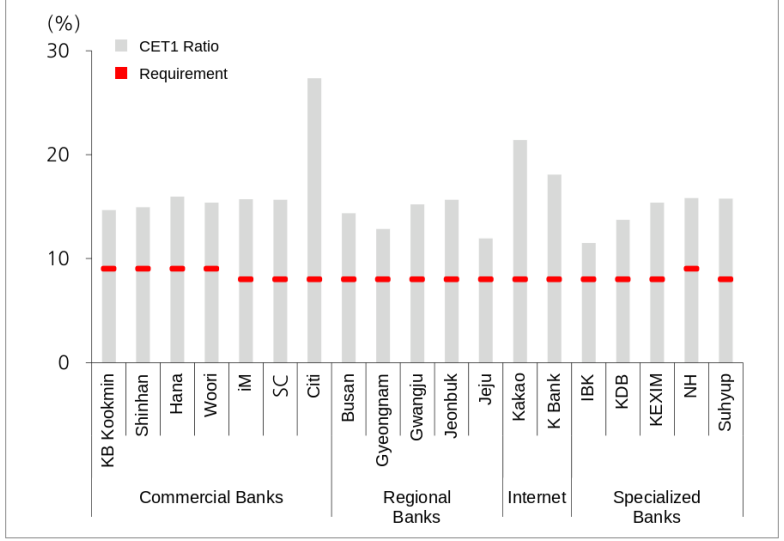
Item	Key Details
Background	Household debt aggregate growth target relaxed to 3% (originally 1.5%), while curbing speculative housing loans through jeonse loans, DSR and capital regulation
	Current regulation focuses on the borrower level (LTV, DSR), and there is no additional capital that banks build to respond to household debt risk
	Adding bank-level capital regulation raises the capital required when originating mortgages, curbing the incentive to supply mortgages
Raising Risk Weights on High-Risk Mortgages (January 2027)	Expand the types of high-risk mortgages (new and existing) and raise risk weights
	Current ①: interest-only/grace-period installment repayment, 3 or more loans, LTV above 60% (average RW about 2x)
	Current ②: principal repayment ratio below 10% when extending maturity/grace period (average RW about 2.5x)
	Improved ③: high-amount/high-DSR mortgages (e.g., RW 2-4x)
	Improved ④: high-priced housing/high-LTV mortgages (e.g., RW 2-4x)
	- Raise risk weights to 2-4x (example), applied not only to new loans but also to existing loan balances → bank RWA increases
	- Detailed imposition criteria to be finalized within 4Q 2026 after consultation with banks and analysis of the capital ratio impact
Introduction of Household-Sector Systemic Risk Buffer (SSyRB) (2H 2027)	Systemic Risk Buffer (SSyRB: Sectoral Systemic Risk Buffer): set when the household debt-to-GDP ratio exceeds a certain level (e.g., 80%), with additional bank-level capital accumulation within an upper limit
	- End-2025 household debt-to-GDP ratio of 88.6% exceeds 80%
	- Additional capital accumulation of 0-1%p per bank. Applied in tiers according to the share of residential mortgages within RWA. The growth rate of mortgages, etc. is to be reviewed
	- 4Q 2026 imposition criteria finalized → January 2027 supervisory regulation amended, 6-month pilot operation → 2H accumulation obligation imposed

Source: Financial Services Commission, Hanwha Research Center

[Bank Bonds] Shareholder Return Targets Also Constrained

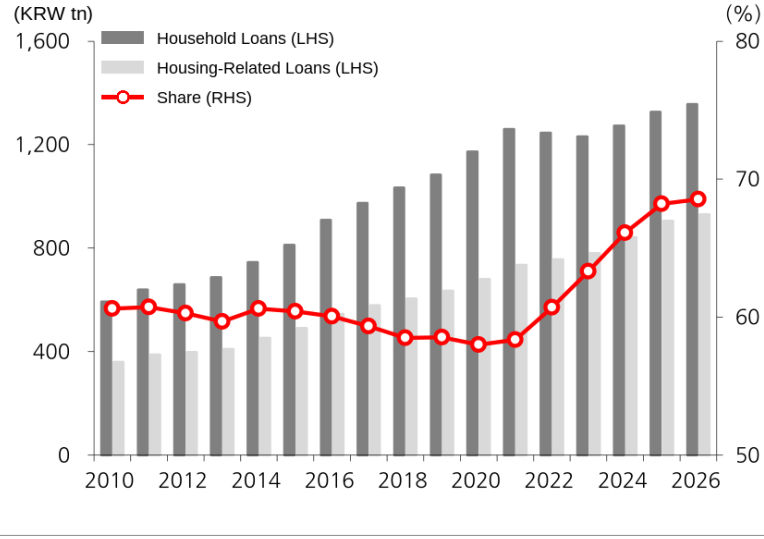
- The Big 5 banks' CET1 ratio is 14.7-16.0%. Even adding the SSyRB cap of 1%p, they have a cushion of at least 4.7%p over the regulatory requirement
- However, the four major financial holding companies are at 13.2-13.7%, close to the 13% shareholder return threshold
  - Capacity for shareholder returns shrinks immediately when risk-weighted assets increase. This is a burden on bank loan expansion
- Banks are expected to respond by reducing high-risk mortgage lending rather than raising capital. This is a factor slowing household loan growth

Domestic Banks' CET1 Ratio



Note: CET1 regulatory requirement. KB Kookmin, Shinhan, Hana, Woori and NH are at 9% reflecting the 1%p D-SIB surcharge; other banks at 8%
Source: FISIS, Hanwha Research Center

Mortgage Share of Household Loans Rising



Note: 2026 is as of end-July
Source: Bank of Korea, Hanwha Research Center

[Specialized Credit Finance Bonds] Largest-Ever Refinancing in a Rate-Hiking Period

- 2027 specialized credit finance bond refinancing demand is KRW 92.3tn: card company bonds KRW 31.7tn, capital bonds KRW 60.6tn
- Maturities of both card company bonds and capital bonds are record highs. They exceed 2024 (KRW 28.7tn, KRW 55.1tn)
- Three-year bonds issued in large volumes during the 2024 rate-decline period are maturing. 2027 specialized credit finance bond issuance is expected to increase from previous years

[Table 10] 2027 Maturities: Card Company Bonds KRW 31.7tn, Specialized Credit Finance Company (Capital) Bonds KRW 60.6tn (Unit: KRW bn)

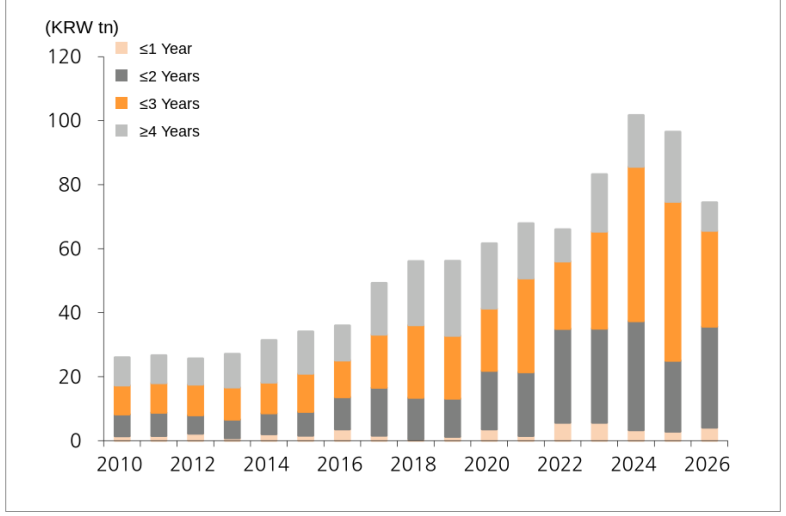
Item (KRW bn)	2027	1Q	2Q	3Q	4Q	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Specialized Credit Finance Bonds	92,340	21,454	24,946	23,020	22,920	6,921	6,449	8,084	8,272	8,089	8,585	8,035	7,308	7,677	7,291	7,740	7,890
Card Company Bonds	31,695	7,469	8,755	8,320	7,150	2,489	2,060	2,920	2,940	2,765	3,050	3,190	2,630	2,500	1,930	2,430	2,790
Shinhan	5,040	1,180	1,550	1,240	1,070	430	290	460	350	670	530	460	380	400	120	430	520
KB Kookmin	4,880	1,200	1,220	1,230	1,230	400	400	400	420	400	400	410	400	420	420	410	400
Hyundai	4,145	909	965	1,320	950	339	310	260	540	165	260	330	660	330	270	290	390
Samsung	3,930	760	1,210	1,180	780	320	0	440	400	430	380	420	380	380	180	220	380
BC	990	180	140	400	270	0	150	30	40	80	20	140	100	160	60	80	130
Woori	3,660	790	1,140	1,020	710	250	250	290	390	270	480	580	160	280	230	290	190
Hana	2,790	380	950	660	800	120	80	180	260	260	430	350	200	110	80	320	400
Lotte	6,260	2,070	1,580	1,270	1,340	630	580	860	540	490	550	500	350	420	570	390	380
Specialized Credit Finance Company (Capital) Bonds	60,646	13,985	16,191	14,700	15,770	4,432	4,389	5,164	5,332	5,324	5,535	4,845	4,678	5,177	5,361	5,310	5,100
Hyundai	4,420	660	1,500	930	1,330	160	290	210	240	690	570	360	140	430	200	500	630
KB	6,050	1,430	1,540	1,520	1,560	480	450	500	490	480	570	540	460	520	550	530	480
Hana	5,770	1,380	1,470	1,440	1,480	440	440	500	500	480	490	470	470	500	470	500	510
JB Woori	4,460	1,110	1,140	1,100	1,110	370	360	380	380	380	380	380	360	360	350	380	380
Woori Financial	4,170	1,080	1,050	1,020	1,020	370	370	340	330	350	370	340	350	330	340	340	340
KDB	3,380	540	930	860	1,050	160	130	250	310	300	320	290	280	290	380	370	300
IBK	3,330	790	920	1,020	600	220	200	370	280	330	310	340	360	320	160	210	230
BNK	2,970	710	760	760	740	230	250	230	260	250	250	250	260	250	240	240	260
Hyundai Commercial	2,930	990	610	500	830	280	260	450	230	180	200	130	180	190	350	360	120
NH Nonghyup	2,830	750	620	710	750	230	260	260	220	160	240	230	230	250	240	260	250
Shinhan	2,760	540	590	800	830	170	150	220	220	240	130	300	260	240	300	240	290
iM	1,950	590	490	450	420	200	180	210	150	160	180	130	140	180	160	140	120
Mirae Asset	1,380	340	290	380	370	130	70	140	120	70	100	100	130	150	140	110	120
A	13,308	2,800	4,028	2,880	3,600	930	900	970	1,536	1,162	1,330	880	990	1,010	1,435	1,100	1,065
BBB+	299	114	60	55	71	17	35	62	26	24	10	0	25	30	46	25	0
Unrated	192	22	88	72	10	5	2	15	17	36	35	5	10	57	0	5	5

Source: Yonhap Infomax, Hanwha Research Center

[Specialized Credit Finance Bonds] Shorter Issuance Maturities Recur in a Rising-Rate Period

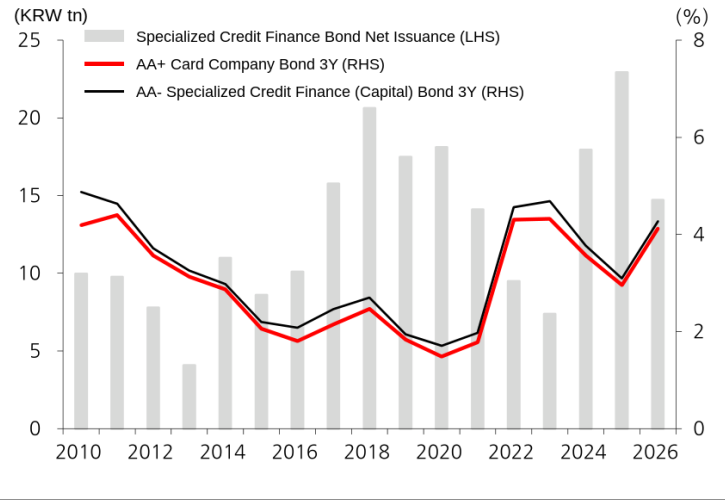
- During the 2022-23 rising-rate period, the share of issuance with maturities of two years or less increased to 52.9% and 42.2%
  - Refinancing of short-term bonds increased in 2024, so maturities surged to KRW 83.8tn. After yields turned lower in 2024, issuance normalized centered on 3-year bonds
- In 2026, the KTB 3Y yield rose an average of 102bp for the year. The share of issuance with maturities of two years or less is 47.9%, so shorter issuance maturities have recurred
- After the end of the policy rate hiking cycle is confirmed, issuance is expected to normalize centered on 3-year bonds, as in 2024

Specialized Credit Finance Bond Maturities Shortening



Note: 2026 is cumulative through October
Source: Yonhap Infomax, Hanwha Research Center

Net Issuance Shrinks in Rate-Hike Periods

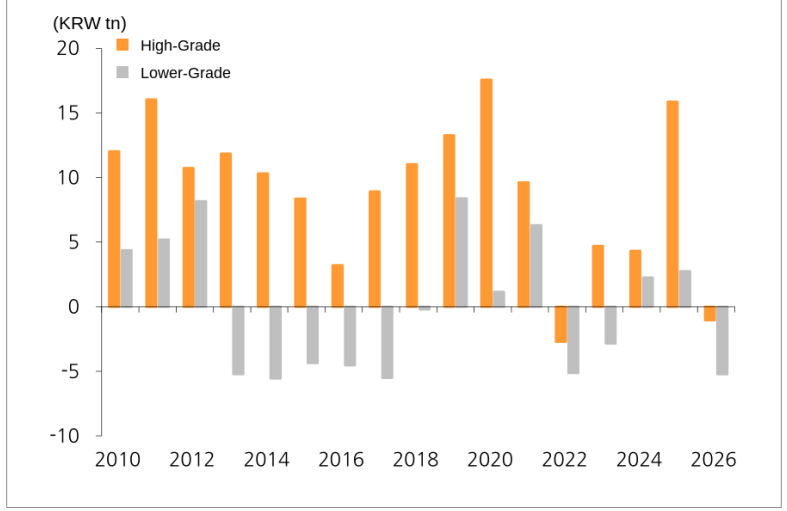


Note: 2026 is cumulative through October
Source: Yonhap Infomax, Hanwha Research Center

[Corporate Bonds] Net Redemptions Continue, Lower-Grade Investor Sentiment Deteriorates

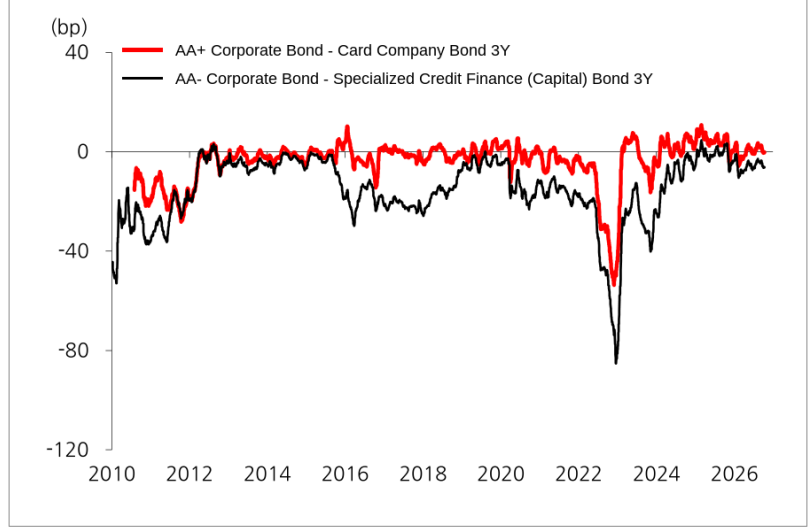
- With the issuance market shrinking under the high-rate stance, corporate bonds have continued to see net redemptions through the year. Since the beginning of the year, net redemptions of KRW 1.0tn in high-grade and KRW 5.2tn in lower-grade bonds have been recorded
- With lower-grade investor sentiment worsening due to the JR Global REIT and Joongang Group defaults, BBB-rated demand forecasts have seen unsold portions
- With investment trust net purchase volume down 75% YoY, the AA- corporate bond 3Y spread widened 17bp from the start of the year
- Corporate bonds are relatively strong versus specialized credit finance bonds amid sluggish issuance
  - However, on an AA- basis, relative value still exceeds the long-term average since 2010 of -14bp. It has only partly resolved the weakness of 2025

Corporate Bond Issuance Market Weak Amid High Rates in 2026



Note: 2026 is cumulative through October
Source: Yonhap Infomax, Hanwha Research Center

Corporate Bond Relative Value vs. Specialized Credit Finance Bonds

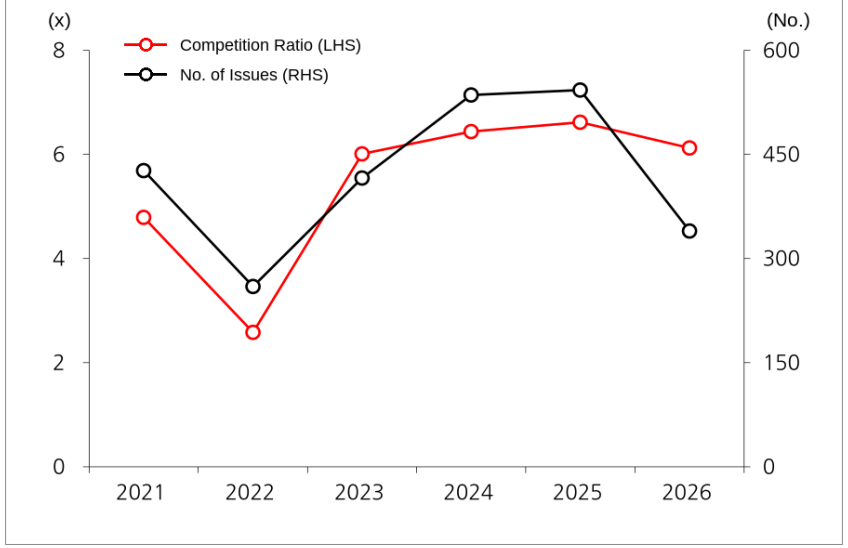


Source: Yonhap Infomax, Hanwha Research Center

[Corporate Bonds] Demand Forecasts Solid, Issuance Expected to Concentrate in Early 2027

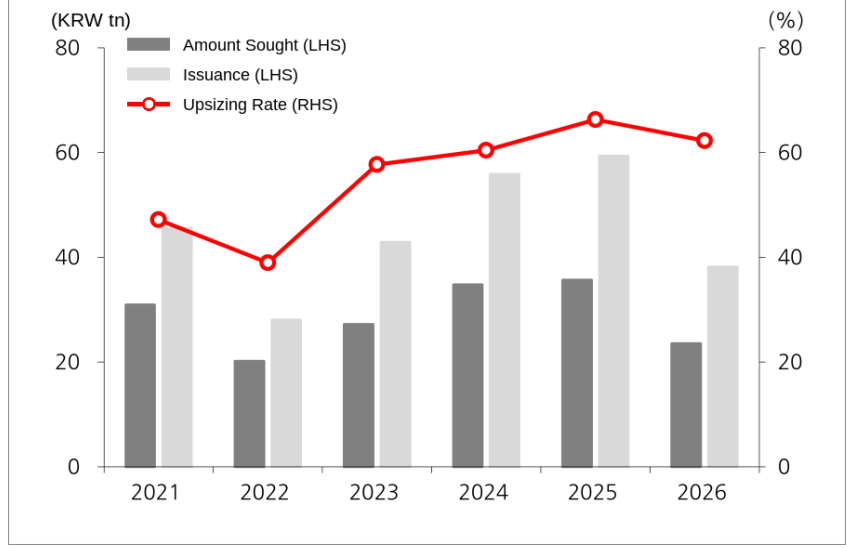
- Although corporate bond issuance was sluggish, the demand forecast competition ratio of 6.1x and upsizing rate of 62% held at the 2023-25 average level
- However, the share of issuance priced below the market average yield was 60%, down from the recent three-year average of 71%, so issuers' pricing power has weakened
- With the policy rate hike, short-term funding rates also rose. The increased short-term borrowing is also expected to act as a refinancing burden in 2027
- Issuance deferred this year is expected to resume in 2027. In particular, issuance is expected to concentrate in 1Q to take advantage of the early-year effect

Demand Forecast Competition Ratio Stays at 3-Year Average



Source: Dart, Hanwha Research Center

Upsizing Rate Also Stays at Average



Source: Dart, Hanwha Research Center

[Corporate Bonds] Tighter Securities Firm Liquidity Regulation

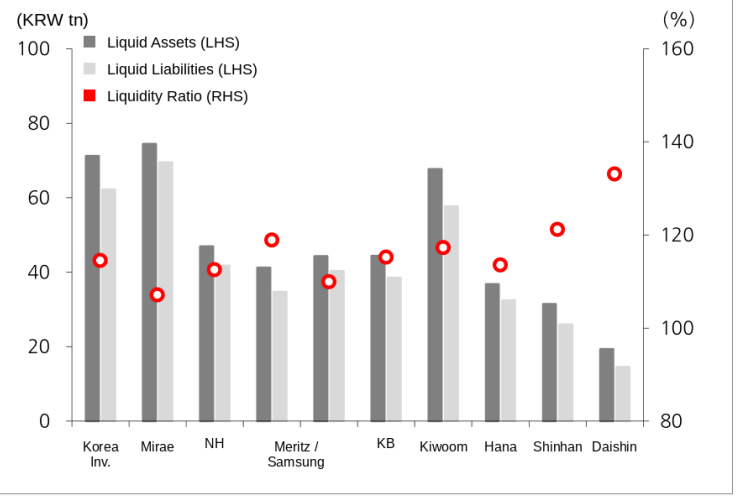
- With tighter securities firm liquidity regulation, securities bond issuance is expected to increase and issuance maturities to lengthen
- From 2027, the securities firms subject to liquidity management expand to 49 domestic firms, and the new adjusted liquidity ratio is introduced
  - Korea Ratings forecasts that the new adjusted liquidity ratios of the 19 securities firms in its analysis will be 106% for 1 month and 104% for 3 months, each down 12%p from the current level
- Securities firms will manage their liquidity ratios by increasing liquid assets or by reducing liquid liabilities through issuing long-term CP or corporate bonds

[Table 11] Introduction of the New Adjusted Liquidity Ratio for Securities Firms

Item	Key Details
Background	During the 2022 Legoland incident, securities firms faced funding difficulties despite liquidity ratios above 100%
	Because discount rates were not applied and contingent liabilities were not reflected, it was difficult to gauge crisis response capacity
Implementation Schedule	Regulation change notice (5.21-6.30) → system development → takes effect 2027.1.1
Expansion of Regulated Firms (January 2027)	Current: 10 comprehensive investment business operators, 13 issuers of derivative-linked securities
	Improved: expanded to all 49 securities firms (excluding 12 foreign branches)
	- The obligation to maintain 1-month and 3-month liquidity ratios of 100% or more is unchanged
Liquid Assets: Discount Rates Applied	0%: government bonds, public bonds, special bonds, bank bonds, AAA bonds, physical government bond ETFs
	7%: AA-grade bonds / 10%: A-grade and below bonds
	15%: stocks, foreign currency securities, open-end funds, ETFs / 30%: synthetic ETFs
	- Assets sold under RP and assets provided as securities lending collateral are deducted in full without exception
	- Fund liquidation period: open-end funds use the redemption period, closed-end funds use the remaining maturity
Liquid Liabilities: Contingent Liability Add-On	Balance of refinancing issued securities × MAX (add-on rate, past 1-year average execution rate)
	- Add-on rate: A1 16%, A2 and below 60% (based on the 4Q 2022 execution rate)
	The full balance of loan and investment commitments is added to 1-month and 3-month liquid liabilities
	Outflow rates differ for secured liabilities (RP sold 0-100%, securities lending 15-100%)

Source: Financial Services Commission, Hanwha Research Center

Securities Firms' Current Liquidity Ratios

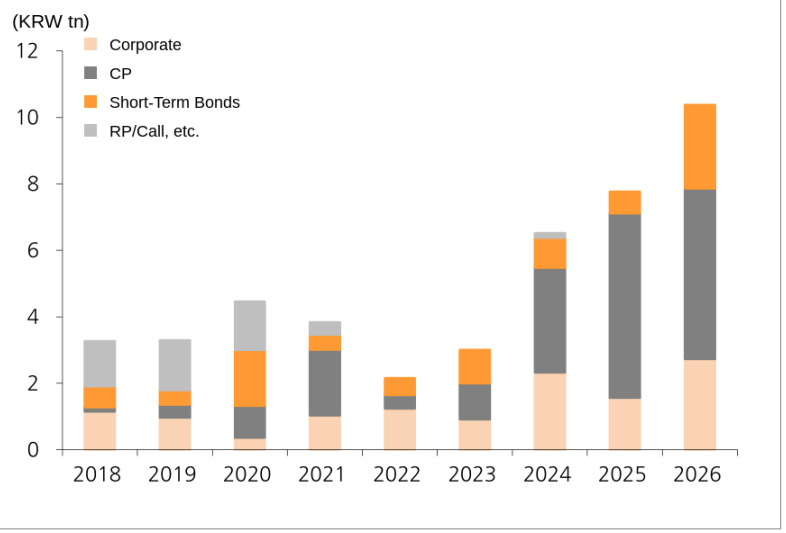


Note: As of end-2Q26
Source: FISIS, Hanwha Research Center

[Corporate Bonds] Securities Bond Issuance Maturities Lengthen

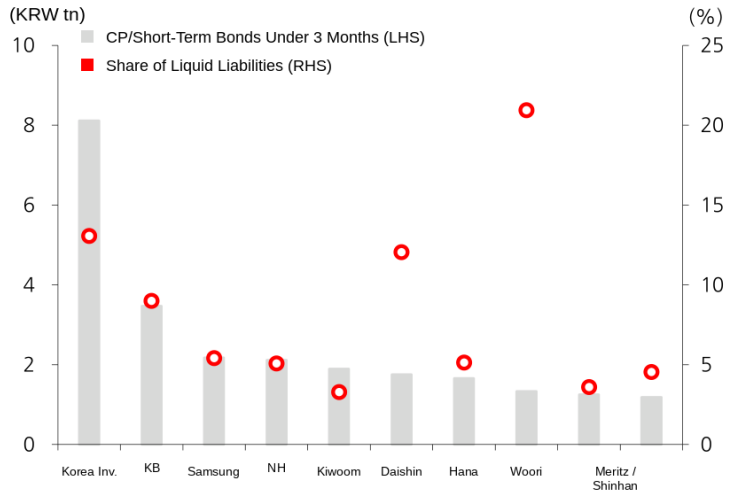
- Securities firms are understood to be already lengthening funding maturities. Cumulative securities bond issuance in 2026 is KRW 11tn, already exceeding last year's full-year issuance
- In September alone, five firms including NH, Hana and Daishin issued KRW 1.4tn to redeem CP and short-term bonds. Issuance by small and mid-sized firms such as DB, Hanyang and LS also continued
- Most securities firms issue securities bonds to refinance CP/short-term bonds. This year, 72% of total issuance was used to refinance short-term bonds
- Short-term bonds with remaining maturities under 3 months are concentrated in large securities firms with equity capital of KRW 4tn or more. The top 10 prime securities firms hold 66% of the total balance
- Going forward, securities bonds will be issued mainly by AA-rated prime securities firms

Refinancing by Purpose



Note: 2026 is cumulative through October
Source: Yonhap Infomax, Hanwha Research Center

Share of CP/Short-Term Bond Refinancing in Securities Bond Proceeds



Source: SEIBro, FISIS, Hanwha Research Center

IV

Equities

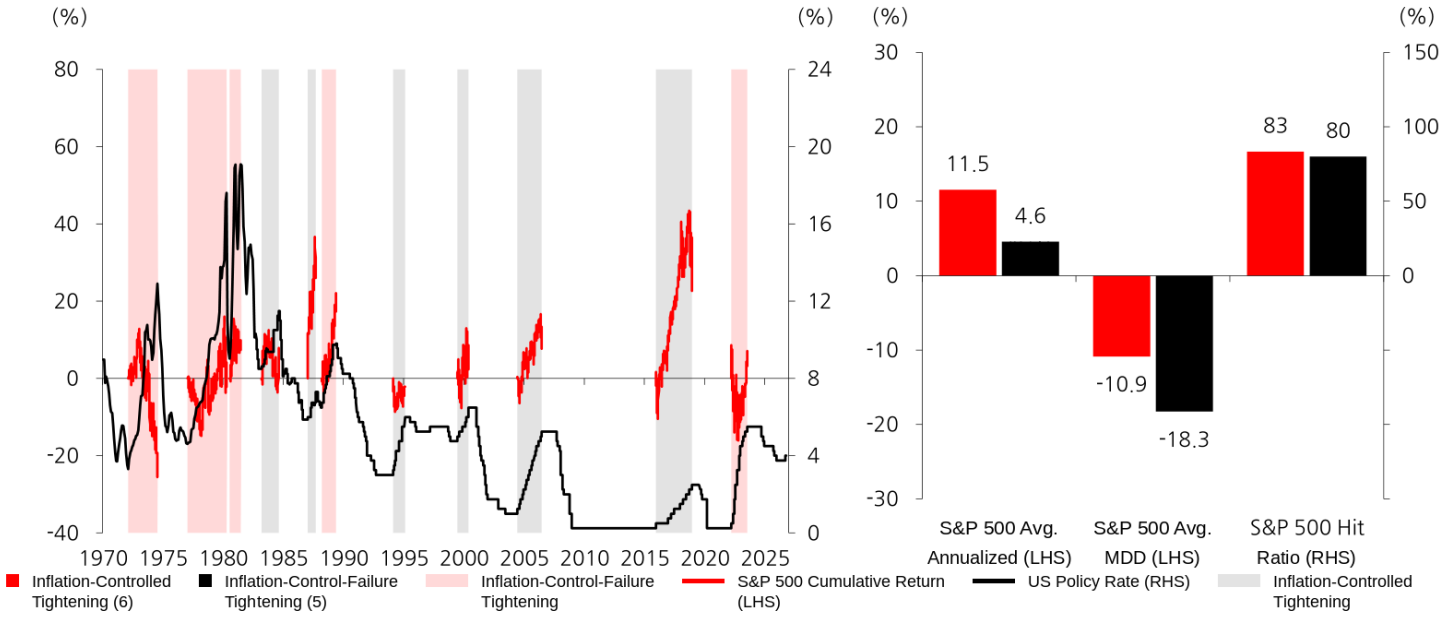
A KOSPI Re-rating Formula Easier Than You Think

Jang-kwon Ha · research.ha@hanwha.com · 3772-7717

Good Tightening, Bad Tightening

- Average return in tightening cases where inflation was successfully controlled (annualized): 11.5% / Hit ratio: 83% (5/6)
- Average return in tightening cases where inflation control failed (annualized): 4.6% / Hit ratio: 80% (4/5)
- Equities can rise even during Fed rate-hiking periods, but the size of the gain tends to differ depending on whether inflation control succeeds

Oil Shocks 1 and 2 and the 2022 Tightening Phase Are Representative Cases of Difficulty Controlling Inflation

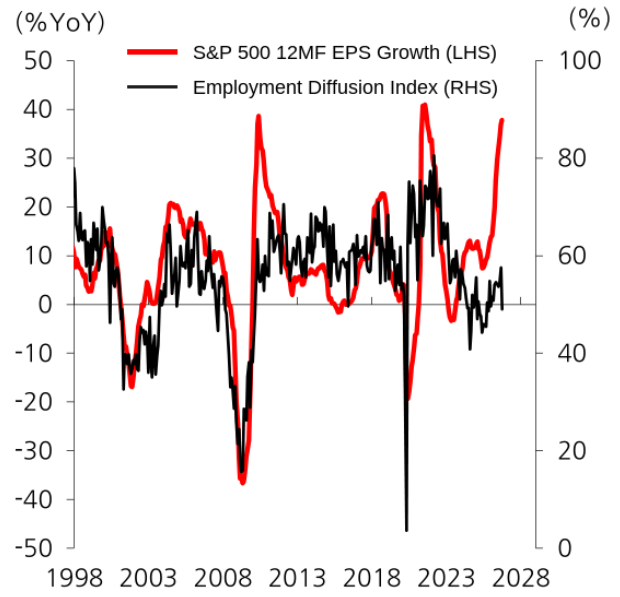


Note: Shaded periods run from the first policy rate hike to the last hike; returns in the right chart are also calculated over those periods
Source: Refinitiv, Hanwha Research Center

A Macro Environment Where Gentle Music Keeps Playing. Keep Dancing

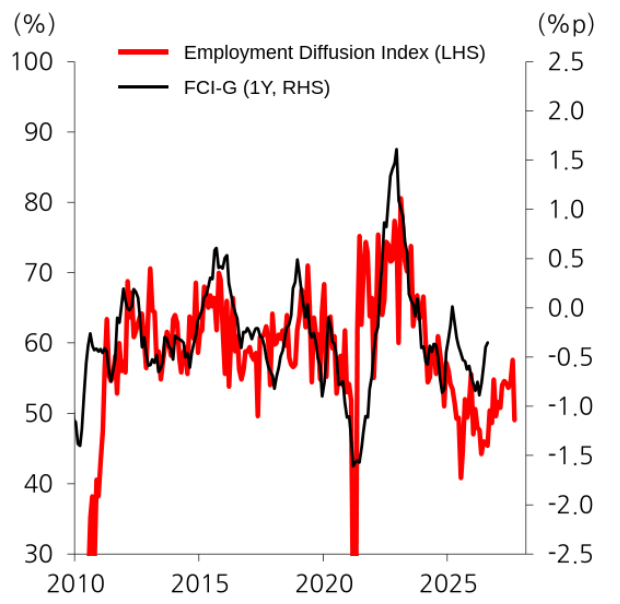
- Normally, corporate earnings growth, inflation and employment tend to move together with the economic cycle
- However, in the current phase, the divergence between employment and corporate earnings is at its maximum
- While overall employment has not changed much, layoff pressure is building, centered on high-wage industries

Divergence Between Earnings Change and Employment Maximized



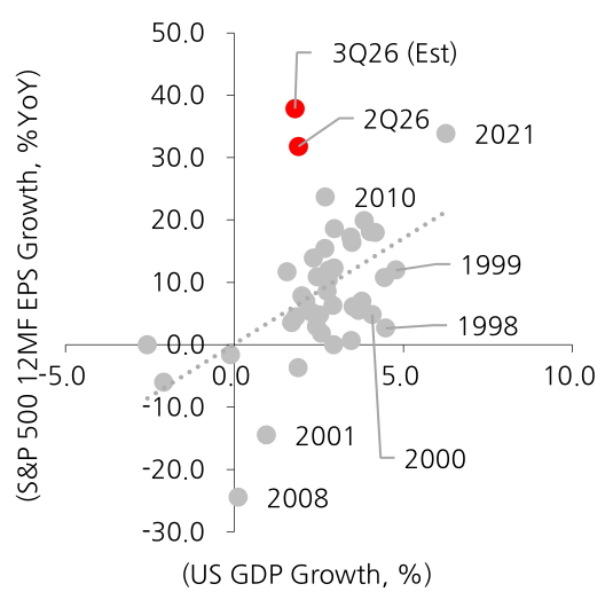
Source: Refinitiv, Hanwha Research Center

Current Employment Remains Lukewarm



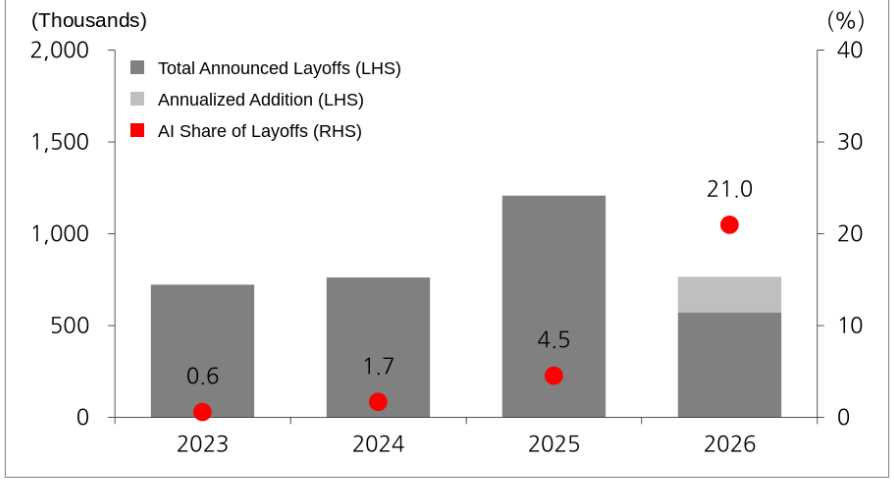
Source: Refinitiv, Hanwha Research Center

AI Investment Leads Growth



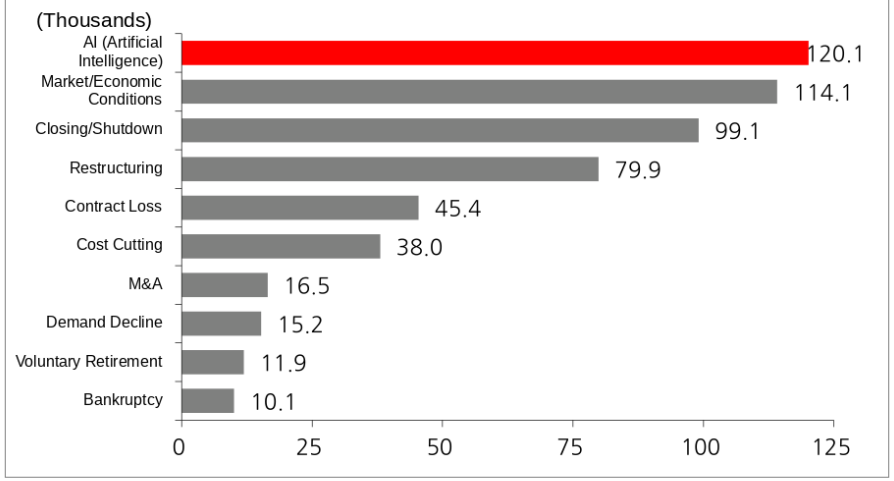
Source: Refinitiv, Hanwha Research Center

Explosive Rise in AI Share of Layoffs



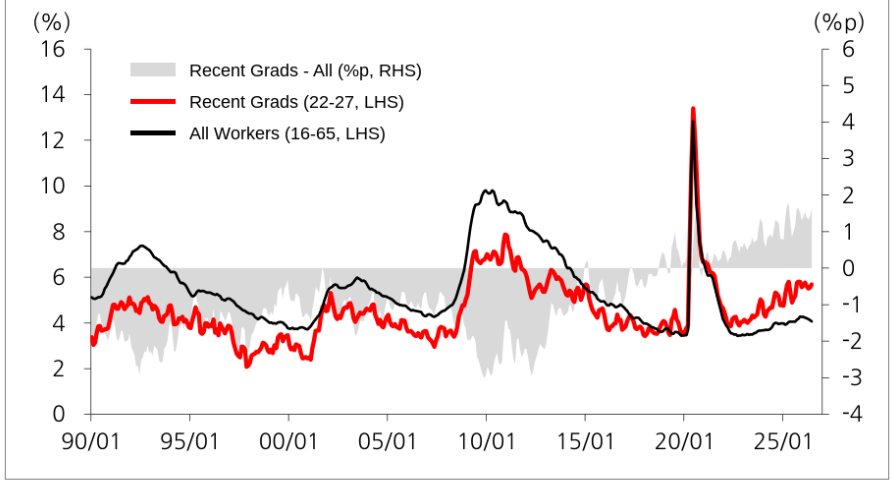
Source: Challenger, Gray & Christmas, Hanwha Research Center

AI Is Now the No. 1 Reason for Layoffs



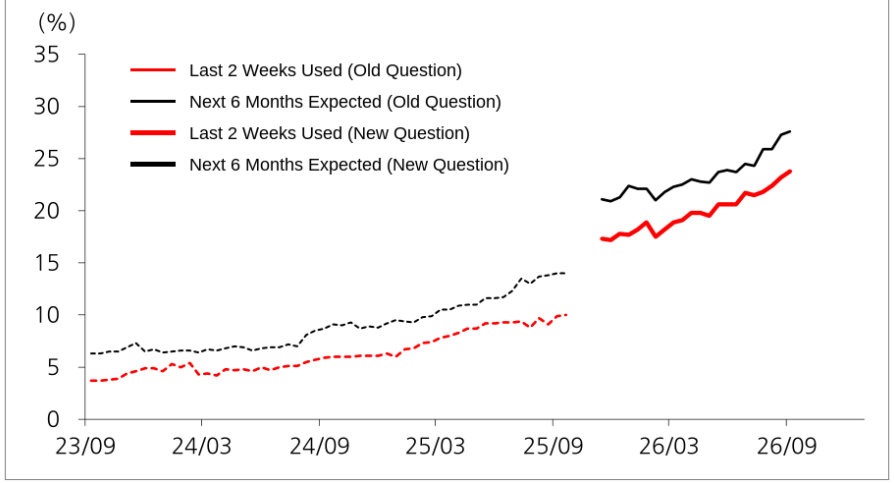
Source: Challenger, Gray & Christmas, Hanwha Research Center

Weak Unemployment, Led by College Graduates



Source: U.S. Bureau of Labor Statistics, Hanwha Research Center

US Companies Accelerate AI Adoption

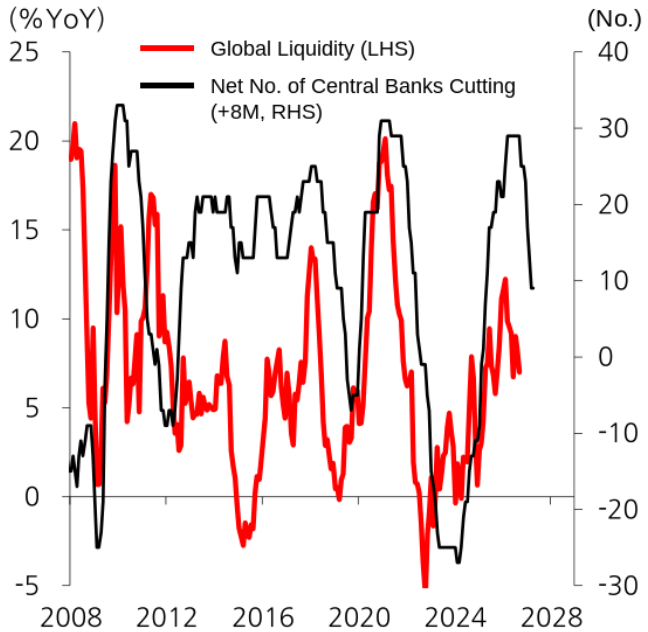


Note: Gap from the Oct-Nov 2025 shutdown
Source: U.S. Census Bureau, Hanwha Research Center

Macro Gravity Felt by Equities Is About to Pass Its Peak

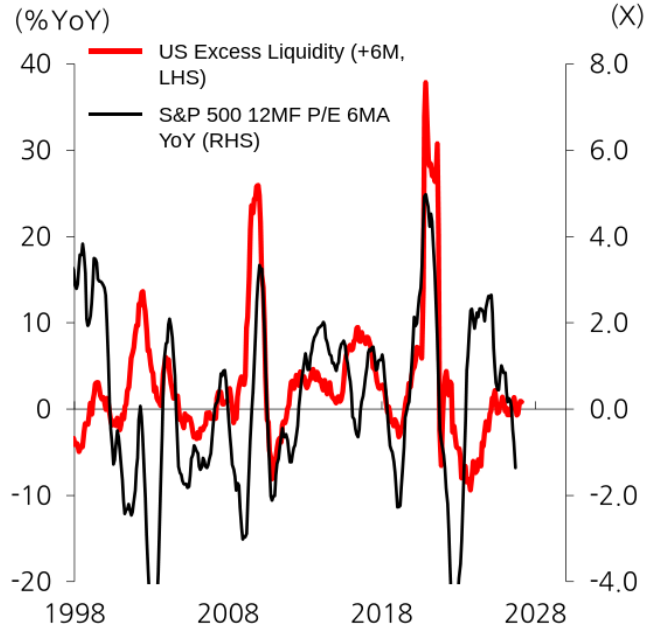
- High interest rates and sluggish liquidity growth lasting through 1Q next year are factors limiting multiple expansion in US equities
- With the Misery Index peaking out, we judge that the macro burden felt by equities is passing its peak

Liquidity Growth Expected to Slow Gradually for Now



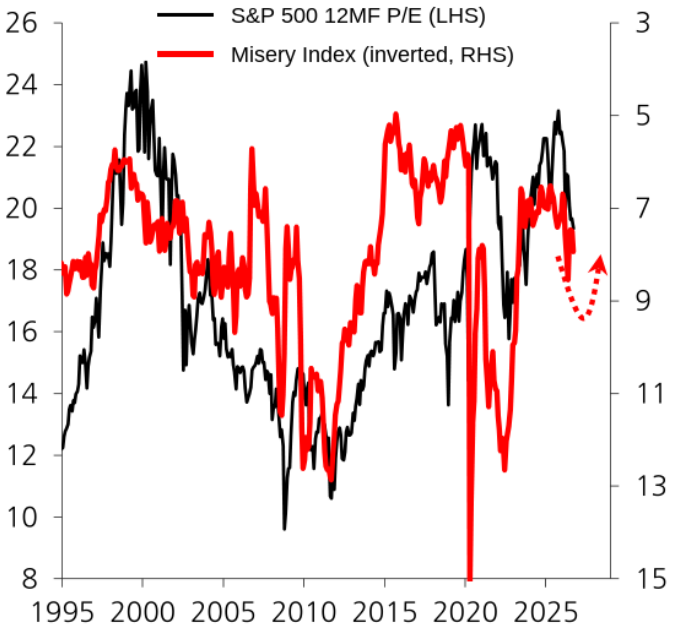
Source: Refinitiv, Hanwha Research Center

Limited Momentum for Multiple Expansion for Now



Source: Refinitiv, Hanwha Research Center

Misery Index Peak-Out

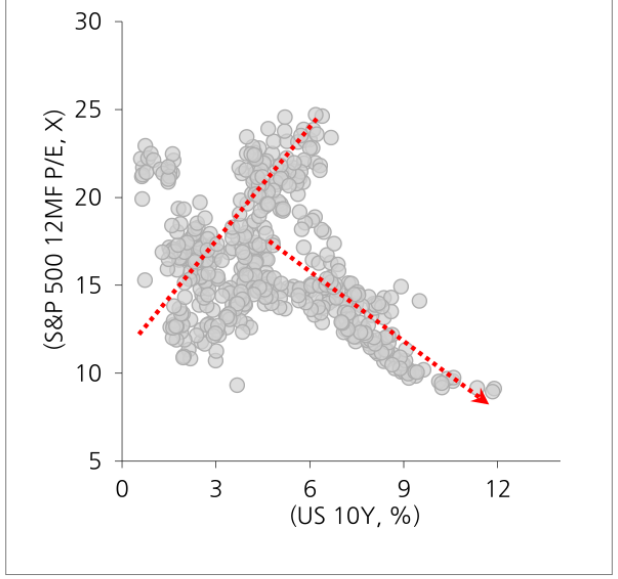


Source: Refinitiv, Hanwha Research Center

The "Bad Is Good" Market Will Be Extended

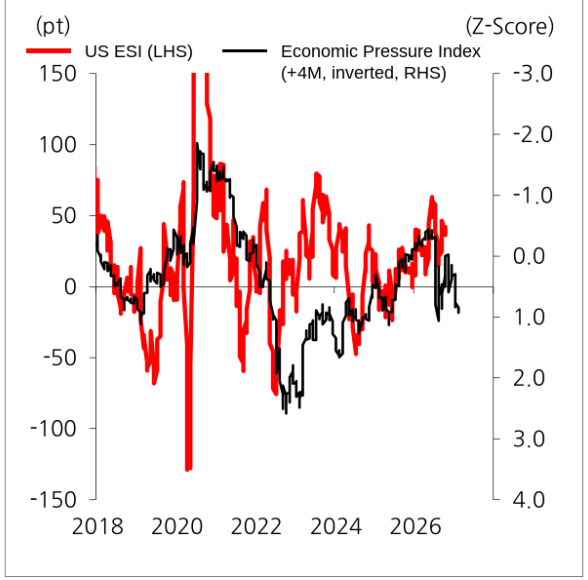
- Unless accompanied by a trend increase in inflation, a "Bad Is Good" market is expected to continue in terms of the economy
- We judge that Korean equity valuation is not greatly affected by the 10-year KTB yield. The critical point for the US 10-year Treasury yield that could trigger risk-off in US equities is around 6%
- The accumulated burden of high interest rates and high oil prices may act as downward pressure on the economy with a lag, potentially lowering economic expectations in the short term

Risk-Aversion Threshold Is 6%



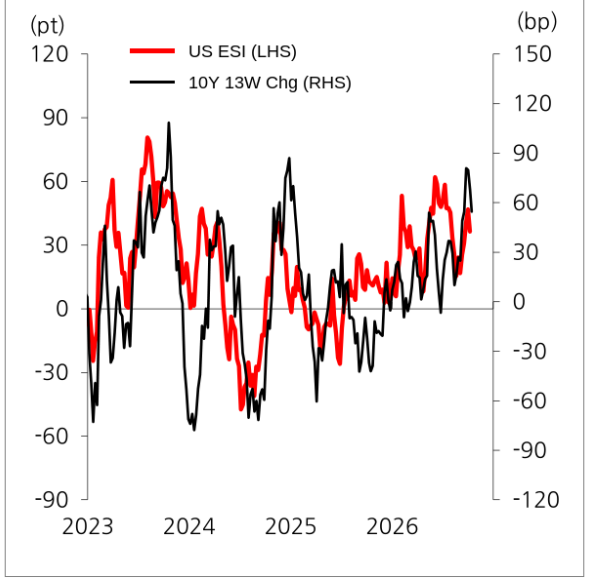
Source: Refinitiv, Hanwha Research Center

Accumulating Burden of High Rates and High Oil Prices



Source: Refinitiv, Hanwha Research Center

Economic Expectations May Be Lowered



Source: Refinitiv, Hanwha Research Center

The Trump Sandbag: Time to Get Lighter

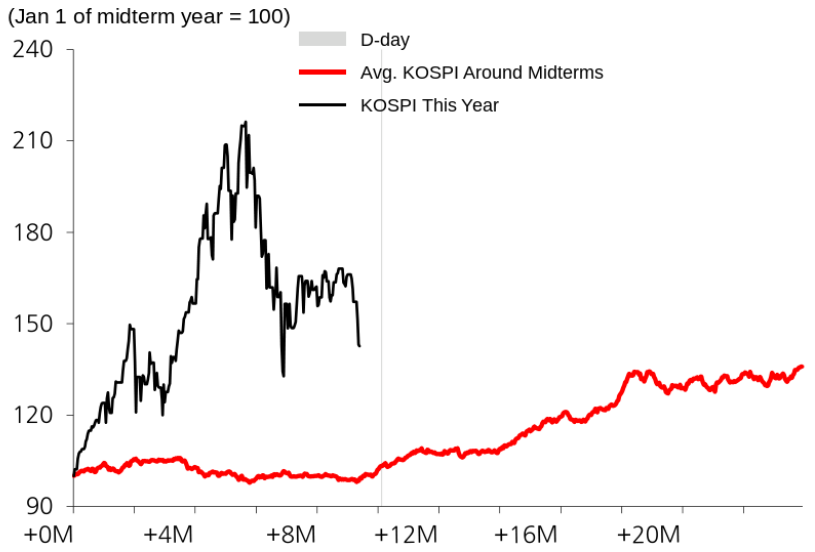
- US equities are the most efficient in terms of earnings. Even so, in phases of rising political uncertainty inefficiency is maximized and the tendency to follow seasonality strengthens
- Korean equities are also confirmed to follow this seasonal pattern
- Midterm-related uncertainty is usually concentrated in Q2-Q3 of the election year and tends to pass its peak around late September to early October, about one month before the election

Midterm Uncertainty Concentrates in Q2-Q3 of the Election Year; Eases from One Month Before the Election



Source: Bloomberg, Hanwha Research Center

Average KOSPI Trend Around Midterm Elections

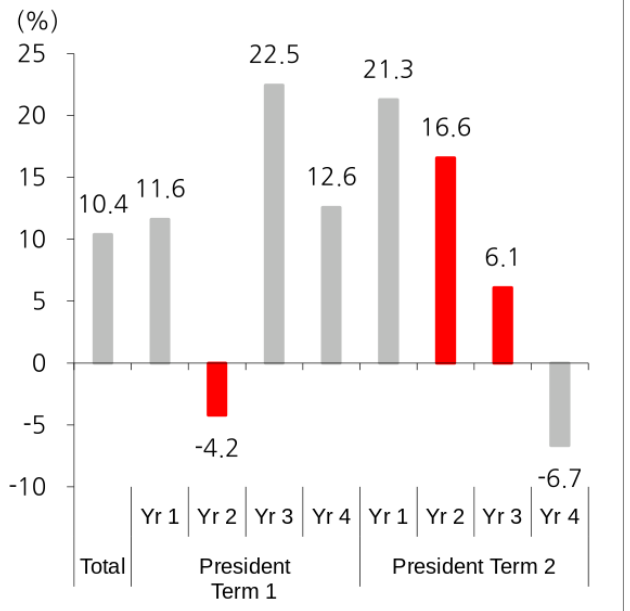


Source: Bloomberg, Hanwha Research Center

The Trump Sandbag: Time to Get Lighter

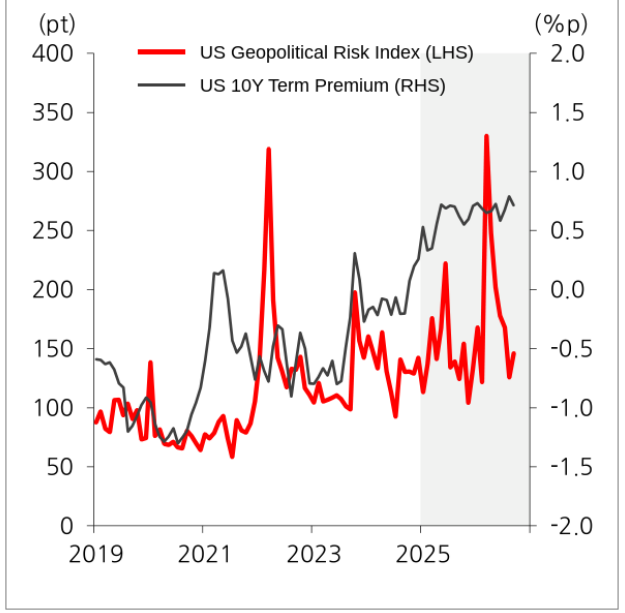
- Second-term presidential midterms posted above-average returns. As the third year begins, equities tend to reflect lame-duck concerns
- However, this needs to be interpreted the opposite way in Trump's second term. Since the start of his second term, trade and geopolitical uncertainty has continued to stand out, including tariff conflicts and issues involving Venezuela, Greenland and Iran
- The trend in the US 10-year Treasury term premium and the narrowing of the US equity valuation premium support this

S&P 500 Returns by Presidential Term



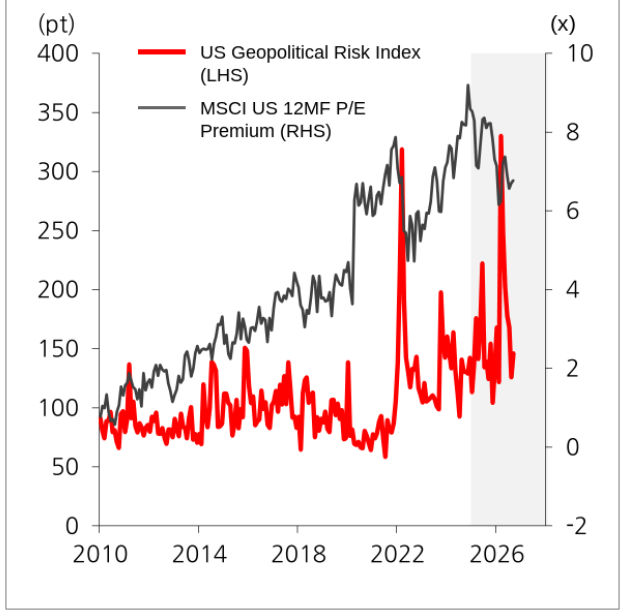
Source: Refinitiv, Hanwha Research Center

Term Premium Rises as Trump's Second Term Begins



Source: Refinitiv, Hanwha Research Center

US Valuation Premium Also Starts to Shrink

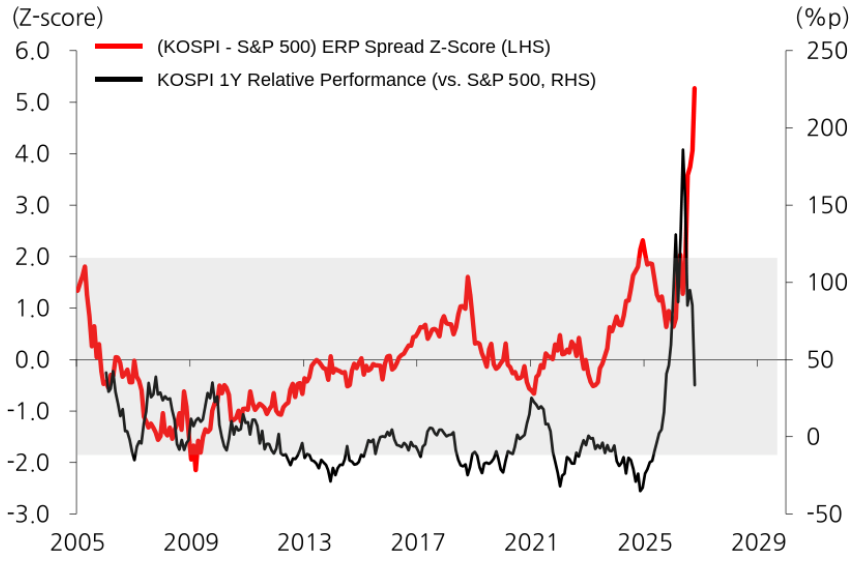


Source: Refinitiv, Hanwha Research Center

Korea's Appeal Versus the US at the +5SD Level

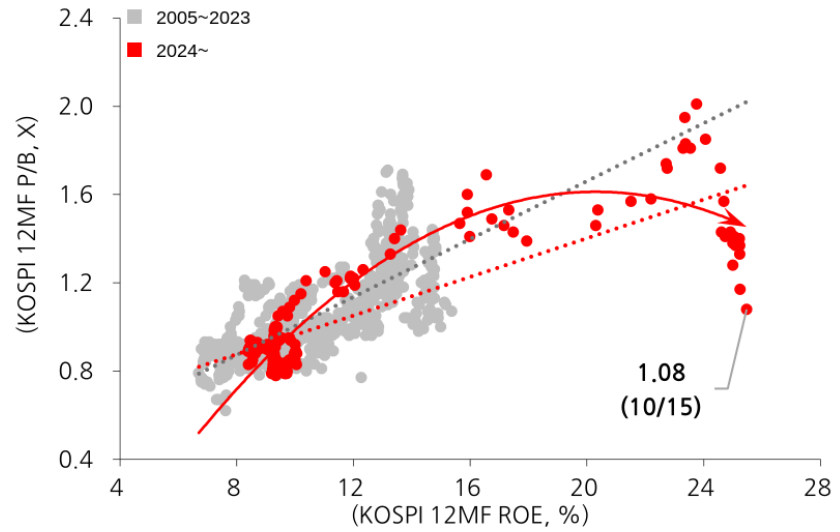
- On an ERP spread basis, Korean equities' relative appeal versus the US is very high, at +5 standard deviations from the average
- The gap between the Philadelphia Semiconductor Index and the KOSPI has also widened significantly
- The value-up premium that had formed from 2H 2025 on the back of value-up policy also turned from a premium to a discount after the sharp fall in June

Recent Plunge Prices in Lack of Growth Sustainability + ROE Decline



Source: Refinitiv, Hanwha Research Center

Taiwan Case: Re-Rating Begins in Earnest After ROE Confirms Short-Term Bottom

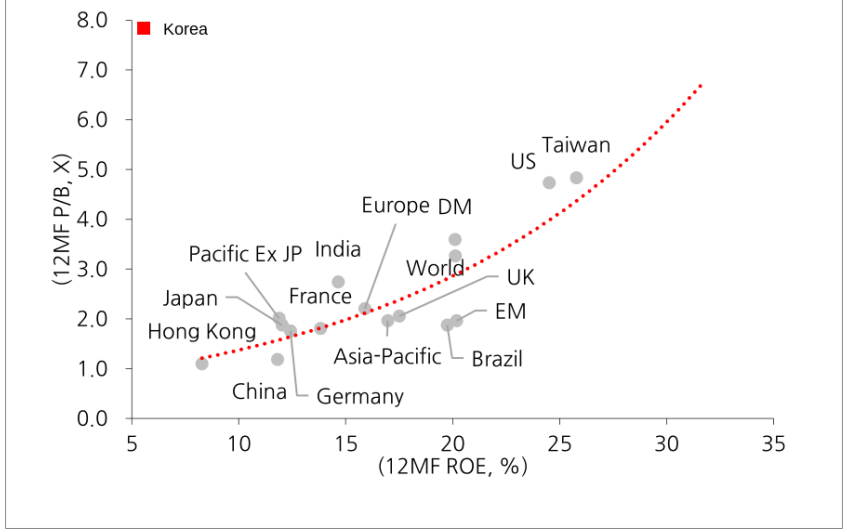


Source: Refinitiv, Hanwha Research Center

No Confidence in Growth. A P/E Re-rating Is Premature

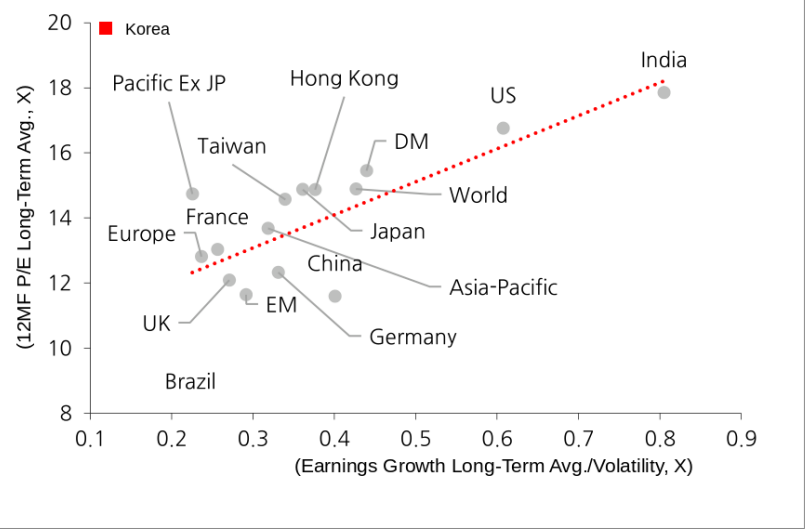
- On the MSCI country 12MF ROE vs. PBR matrix, Korea's PBR is near the bottom despite its high ROE
- For P/E valuation, confidence in the sustainability of growth is important. A low assessment of growth quality acts as a factor in the P/E discount
- Comparing Korea, Japan and Taiwan, Korea's ROE rose to the highest level among the three, helped by the improvement in the earnings structure of the two memory companies in 1H
- However, the ROE implied in market prices is less than half of the expected ROE
- For Korea to re-rate, the sustainability of the elevated ROE level needs to be proven

MSCI Country 12MF ROE vs. PBR Matrix



Source: Refinitiv, Hanwha Research Center

Korea: A Market Lacking Confidence in Growth Sustainability

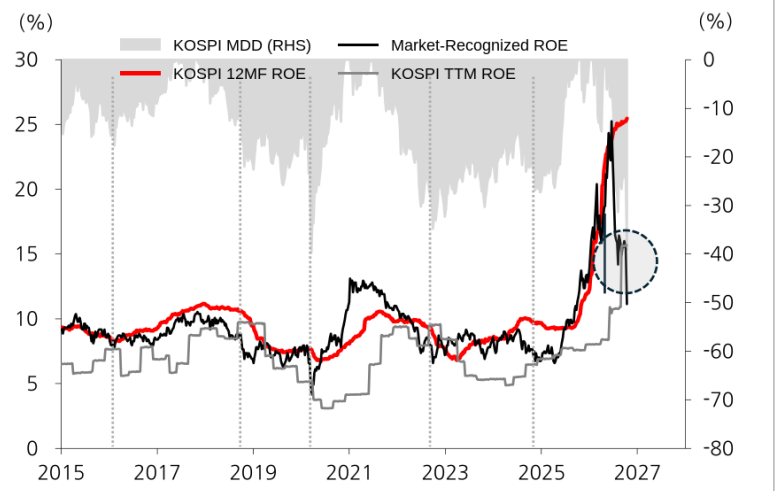


Source: Refinitiv, Hanwha Research Center

Re-rating Expected to Begin in Earnest After ROE Stabilization Is Confirmed

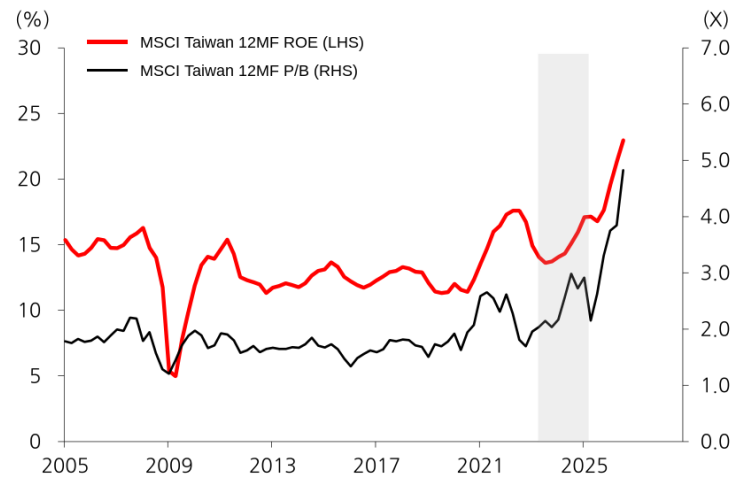
- Expectations for sustained growth were absent from the outset, and the recent plunge excessively priced in concerns that future ROE would not only fail to improve but fall short of even the current TTM ROE level
- The point at which the market-recognized ROE level began to fall below TTM ROE usually coincided with the point of maximum index drawdown. Considering this, we judge that even the 6,000pt level is already a deep-value zone, and recommend an active buying response at current index levels
- Even if it does not extend to a narrative of sustained growth like Taiwan's TSMC, expectations for a structural improvement in the ROE level after the LTA and the 2027 HBM price negotiations remain valid. In Taiwan's case, a short-term bottom was confirmed after ROE hit a new high, and the valuation re-rating began in earnest

Recent Plunge Prices in Lack of Growth Sustainability + ROE Decline



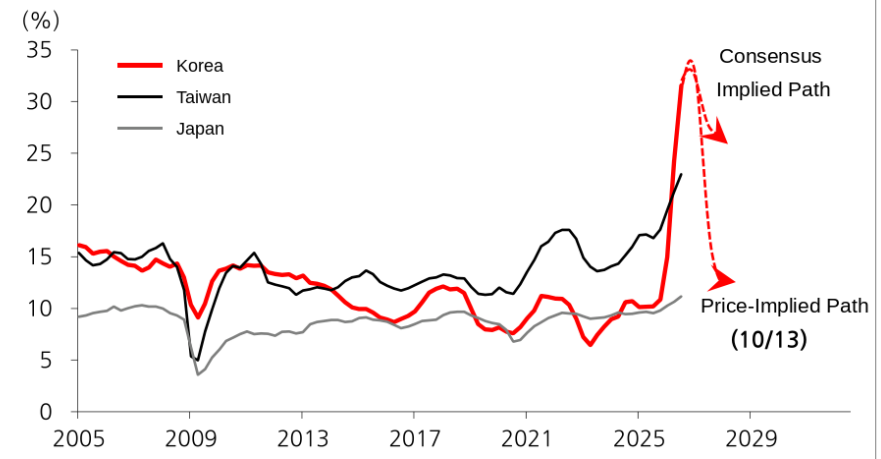
Source: WISEfn, Hanwha Research Center

Taiwan Case: Re-Rating Begins in Earnest After ROE Confirms Short-Term Bottom



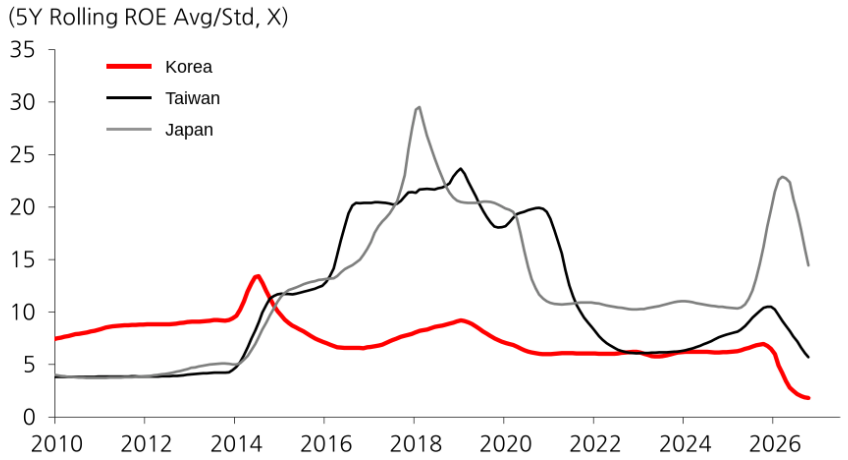
Source: Refinitiv, Hanwha Research Center

MSCI Country 12MF ROE: Korea Surges; Sustainability Must Be Proven



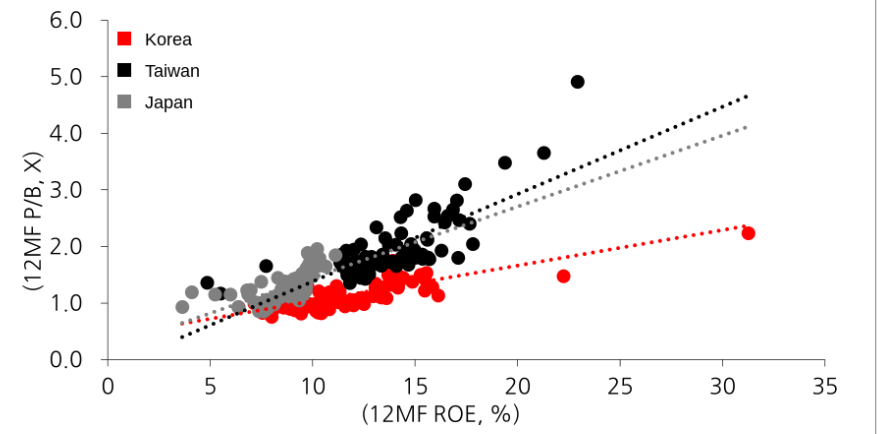
Source: Refinitiv, Hanwha Research Center

Korea Has the Lowest ROE Quality. Taiwan Offers Level, Japan Offers Low Volatility



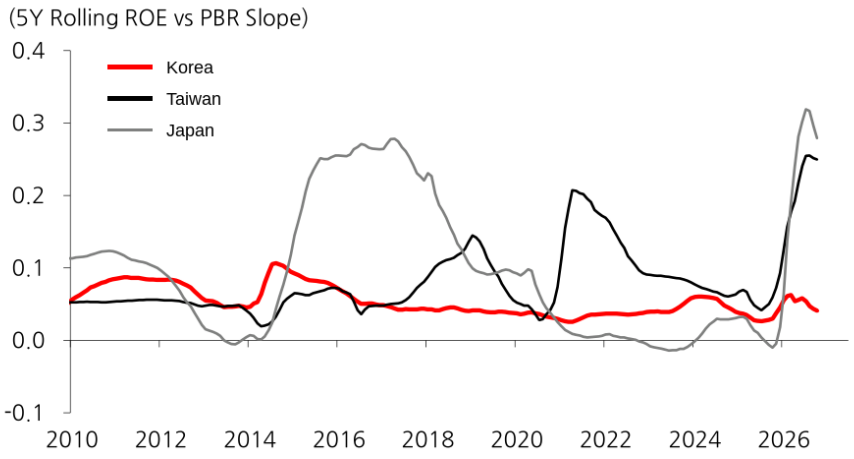
Source: Refinitiv, Hanwha Research Center

MSCI Country 12MF ROE vs. P/B Matrix



Source: Refinitiv, Hanwha Research Center

ROE Quality Translates Directly into PBR Re-Rating

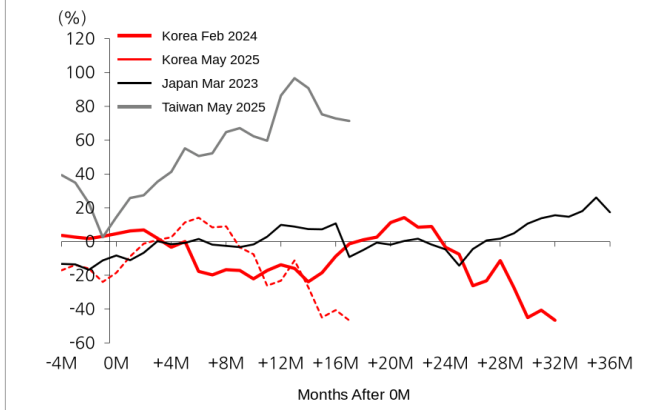


Source: Refinitiv, Hanwha Research Center

Policy Only Assists

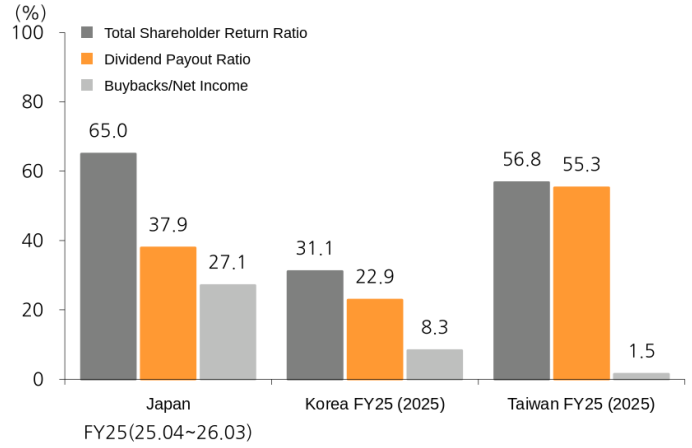
- Taiwan introduced the Power UpTW policy in June 2024, after the AI premium had already been priced into share prices
- Looking at the Japan and Taiwan cases, for a full-scale valuation re-rating, building confidence in a structural improvement in ROE matters more than the value-up policy itself
- Whereas Taiwan priced in growth expectations from AI innovation ahead of the policy, Japan's value-up policy plays the role of supporting the valuation floor during equity market plunges
- Looking at cases that (1) have cyclical characteristics, (2) saw ROE rise sharply in a short period due to industry structural change, but (3) still lack expectations for sustained growth, in most cases the re-rating begins in earnest from the moment the trough starts to rise

Deviation from ROE-PBR Matrix Trend
Line: Korean Market Without Policy Impact



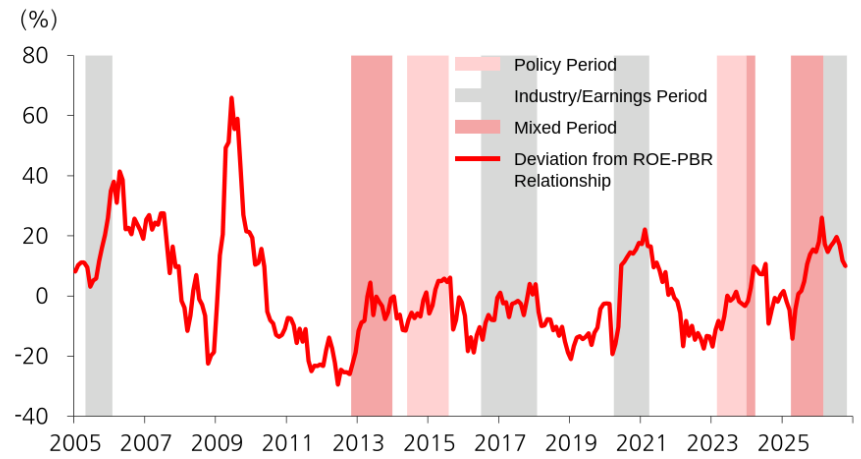
Note: 0M: Japan Mar 2023 TSE request; Korea Feb 2024 Value-up support plan/May 2025 Commercial Act revision; Taiwan May 2025 Power UpTW
Source: Refinitiv, Hanwha Research Center

Value-Up Policy Progress in Korea, Japan and Taiwan



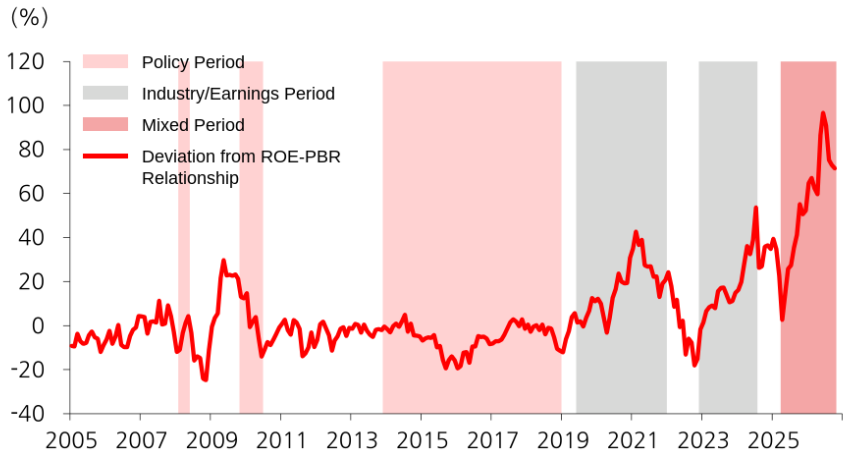
Source: I&N, JPX, KRX, FSC·TWSE, Hanwha Research Center

Japan: Average Level Rose, Recovering After Yen Carry Unwind and Tariff Plunge Following the 2023 Request



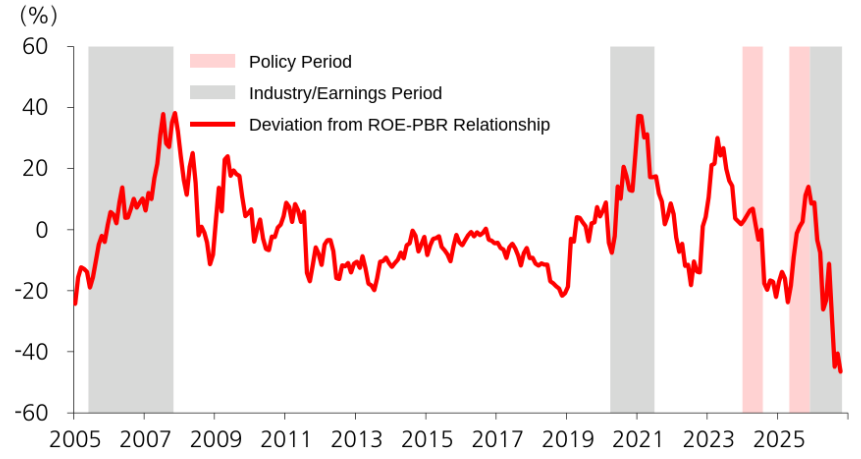
Source: Refinitiv, Hanwha Research Center

Taiwan: Never Re-Rated Through Policy



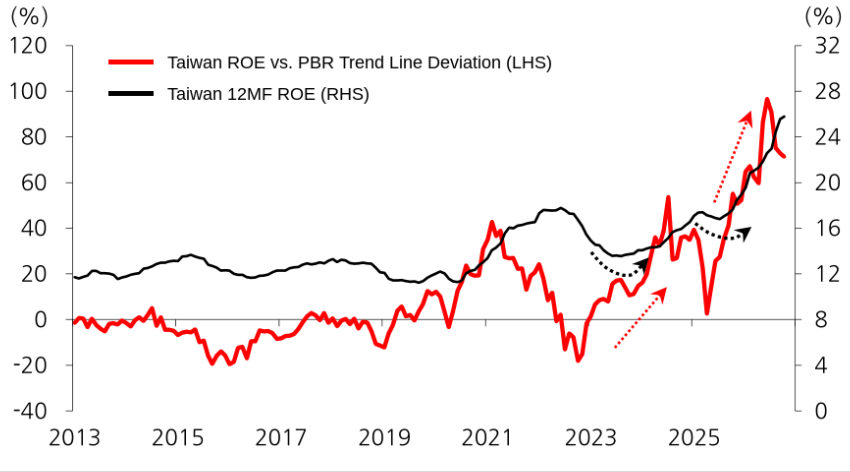
Source: Refinitiv, Hanwha Research Center

Korea: Rose During Commercial Act Revision; Restoring ROE Confidence Is the Key



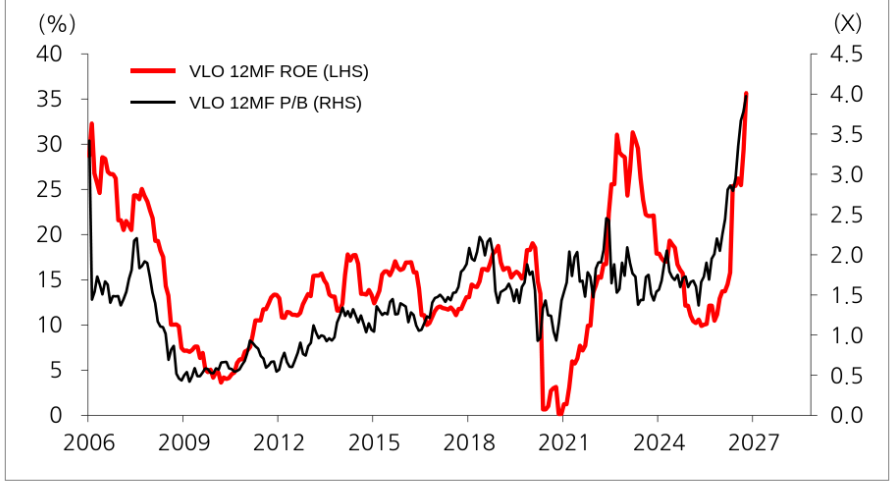
Source: Refinitiv, Hanwha Research Center

Confirming the Bottom Is the Key



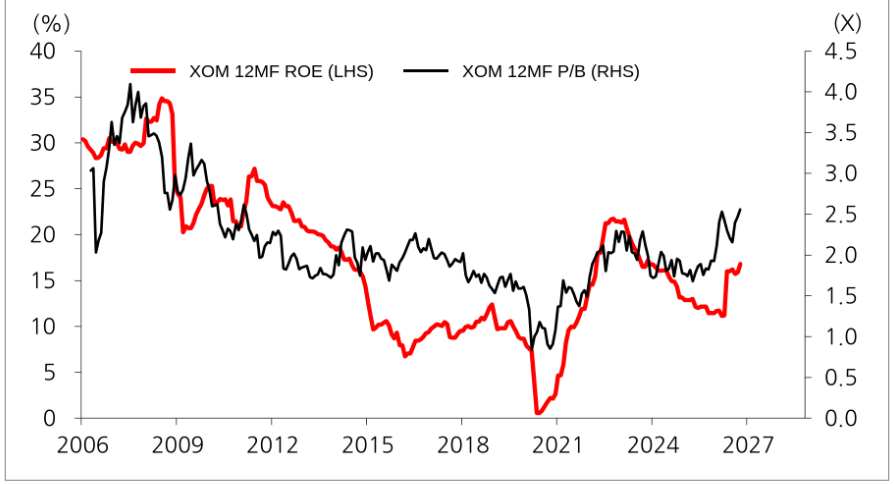
Source: Refinitiv, Hanwha Research Center

Valero Energy (VLO US)



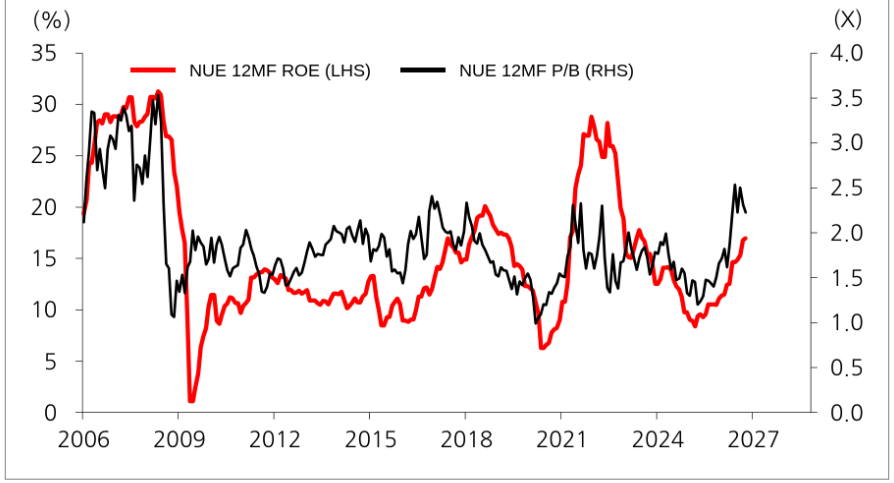
Source: Refinitiv, Hanwha Research Center

ExxonMobil (XOM US)



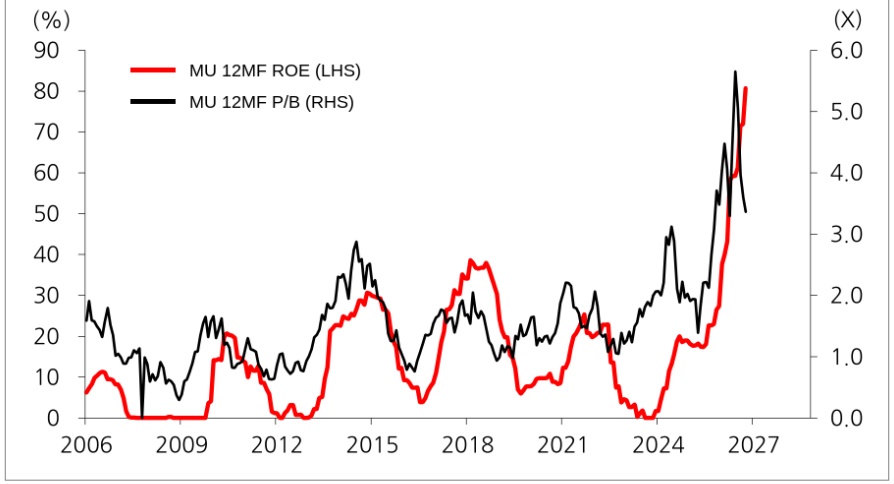
Source: Refinitiv, Hanwha Research Center

Nucor (NUE US)



Source: Refinitiv, Hanwha Research Center

Micron Technology (MU)

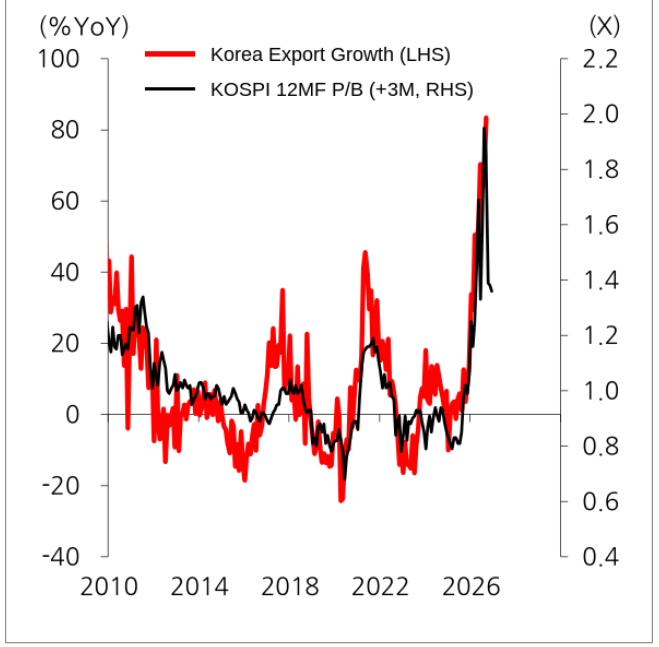


Source: Refinitiv, Hanwha Research Center

KOSPI 12MF P/B Estimation Model

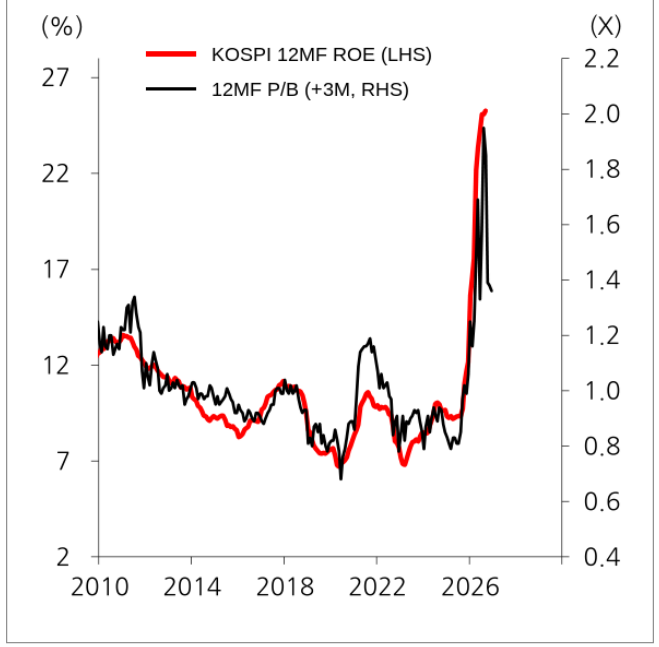
- KOSPI 12MF P/B(t) = f(a(t+3), b(t+3))
- a: Korea export growth rate (determines the direction of 12MF P/B changes)
- b: KOSPI 12MF ROE (determines the reference level of 12MF P/B)

Export Growth Determines Direction of P/B Changes



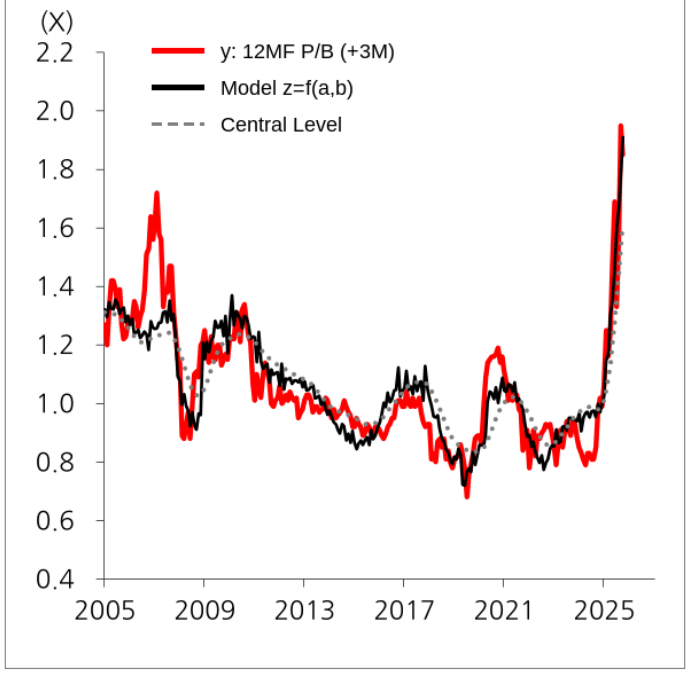
Source: WISEfn, Hanwha Research Center

12MF ROE Determines the Reference Level of P/B



Source: WISEfn, Hanwha Research Center

KOSPI 12MF P/B Estimation Model



Source: WISEfn, Hanwha Research Center

2027 KOSPI Target of 8,500pt

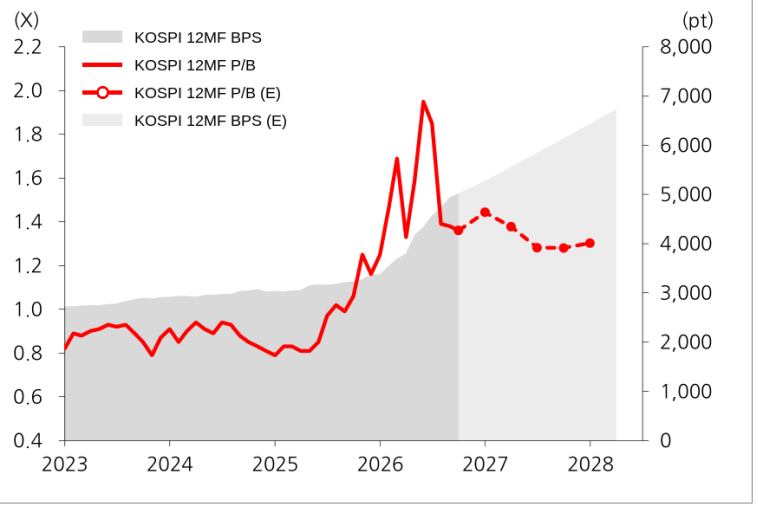
- Although the contribution of export growth gradually declines, expectations for a medium-term rise in the two memory companies' ROE level are reflected in 1H, so valuation is not expected to change much within the year. The rise in book value will drive most of the index's gain
- The market-recognized ROE level is conservatively assumed at 15% at year-end. We assume a large rise in 1H reflecting LTA and price negotiation momentum, followed by a gradual rise
- The KOSPI TTM P/B matrix can estimate the valuation flow according to fundamentals (export growth), but its limitations are clear in the current phase, when equity is rising steeply. A 12MF P/B that can reflect changes in equity needs to be adopted
- In 2027, the KOSPI is expected to be low in 1H and high in 2H, while the 12MF P/B is expected to be high in 1H and low in 2H. Most of the rise will come from rising book value
- We present a 2027 KOSPI band of 6,000-9,300pt and a target of 8,500pt

[Table 12] End-2027 KOSPI Target 12MF P/B of 1.32x

Item (x, pt)	4Q26	1Q27	2Q27	3Q27	4Q27	2027 Full Year
Target 12MF P/B	1.44	1.39	1.29	1.28	1.32	1.32
Target	7,600	7,750	7,550	7,900	8,500	8,500
Upper Bound	8,300	8,500	8,400	8,600	9,300	9,300
Lower Bound	5,800	6,500	6,700	7,000	7,500	6,500

Source: Hanwha Research Center

Most of the Rise Expected to Come from Rising Book Value

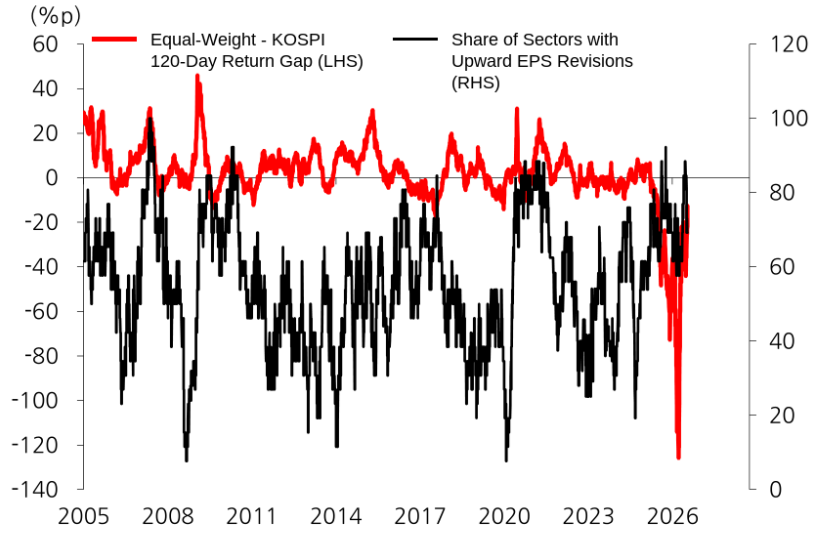


Source: WISEfn, Hanwha Research Center

A Macro Environment Favorable for Broadening Starts in 2Q

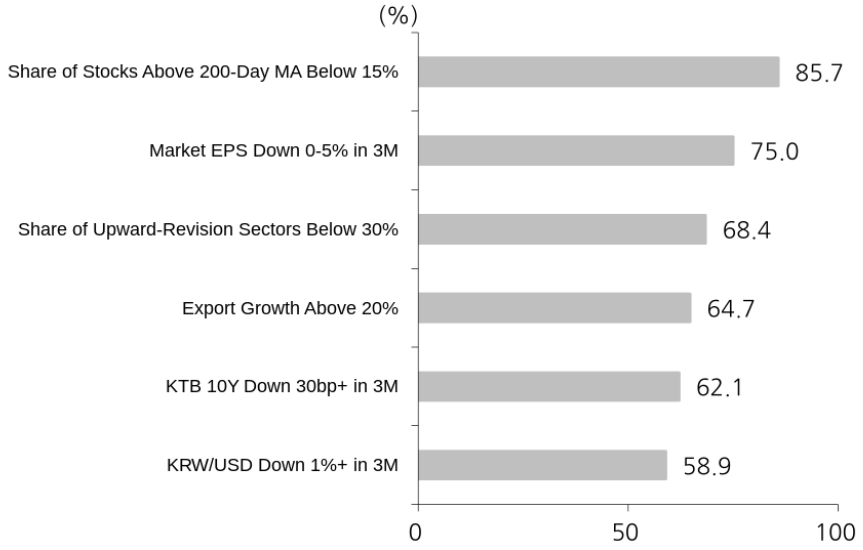
- The two memory companies' relative momentum will stand out as the 2027 HBM price negotiations, the LTA negotiations and the earnings improvement in 1Q next year are confirmed
- We expect the current KRW/USD exchange rate and 10-year KTB yield to persist at current levels through 1Q, then decline gradually from 2Q
- We expect a macro environment favorable for broadening to be created from 2Q

Incumbent Leaders Expected to Stay Ahead Until Long-Term Yields and KRW/USD Begin to Fall in Earnest



Source: WISEfn, Hanwha Research Center

Macro Environment Favorable for Broadening Starts in 2Q

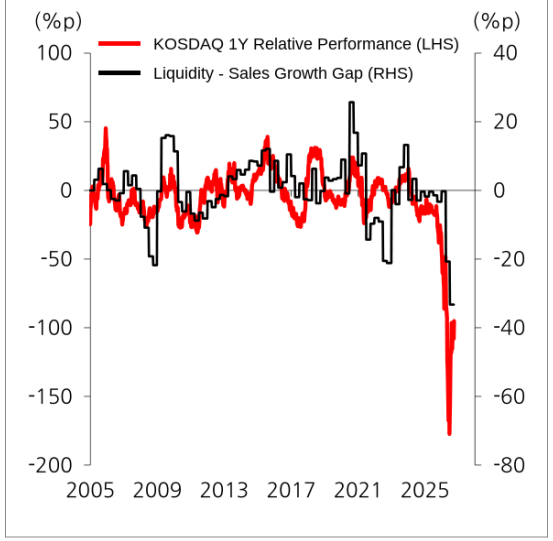


Source: WISEfn, Hanwha Research Center

Determinants of KOSDAQ Relative Performance

- KOSDAQ's relative performance versus the KOSPI tends to improve when liquidity momentum is stronger than changes in the Korean equity market's sales growth rate
- We judge that liquidity volatility will determine relative performance through 1Q next year
- In 1Q, the memory sector, which has the greatest room for a valuation re-rating, is relatively attractive
- KOSDAQ's relative strength is expected to stand out from the phase in which the liquidity growth rate turns upward

Excess Liquidity vs. Fundamentals Determines KOSDAQ Relative Performance



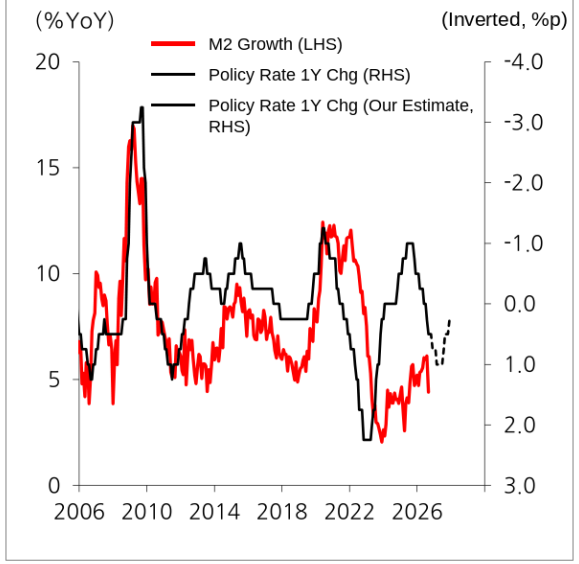
Source: WISEfn, Hanwha Research Center

Korea Equity Market Sales Consensus



Source: WISEfn, Hanwha Research Center

M2 Growth vs. 1Y Change in BOK Policy Rate



Source: WISEfn, Hanwha Research Center

V

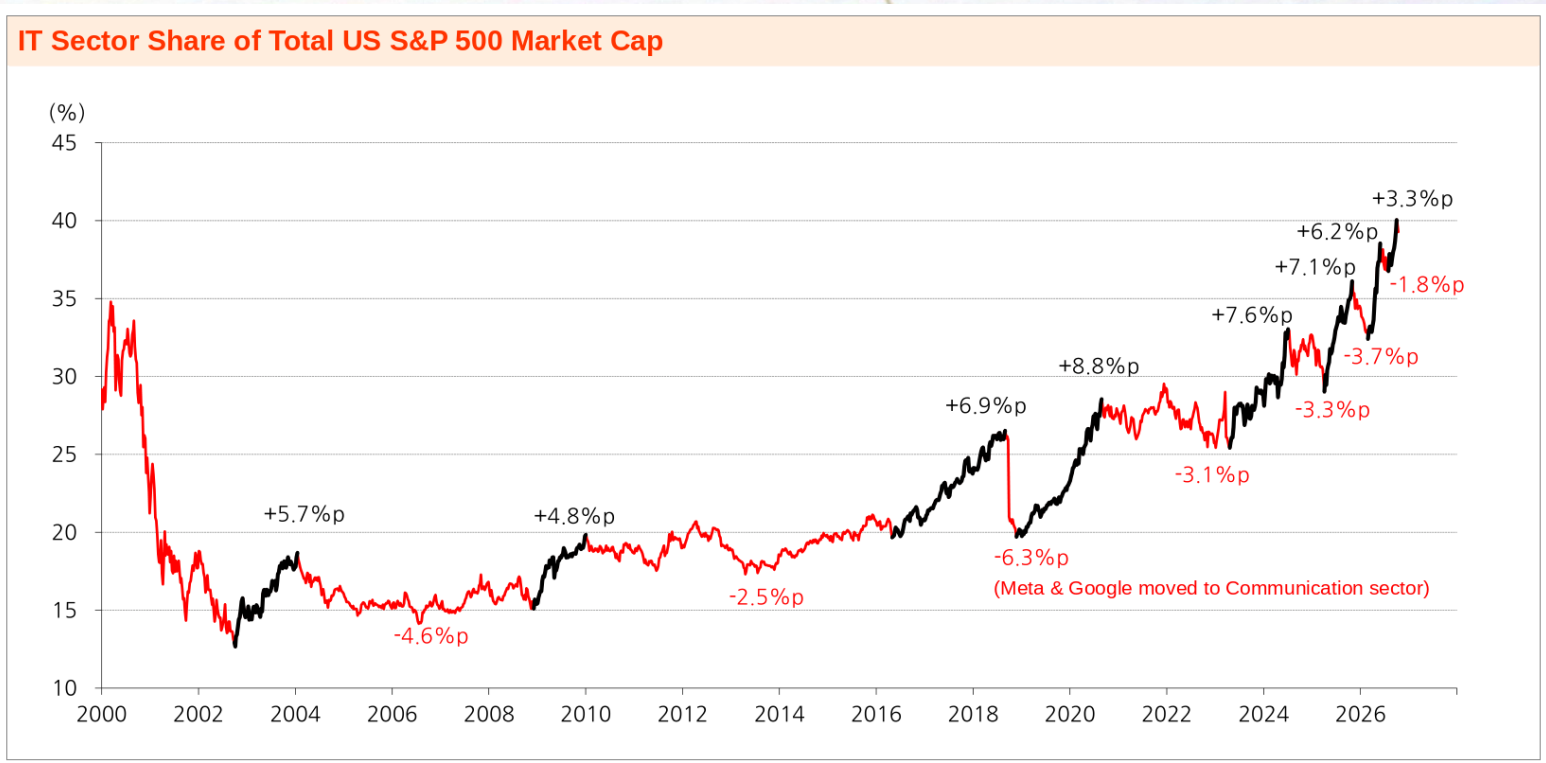
Quant

A Picture Where Bio Becomes the Story

Hyun-kook Ahn · hg.ahn@hanwha.com · 3772-7646

US IT Sector Weight Breaks 40% in Early October

- The IT sector's share of total S&P 500 market capitalization rose to about 35% at the time of the IT bubble and then fell
- The IT sector weight has risen steadily, repeating a pattern of rising 6-7%p and then retracing 2-3%p. It broke 40% in early October, thanks to NVDA, AAPL and MSFT
- Unless new sectors emerge and lower the IT weight as in 2018, excessive concentration becomes a burden every time IT rises

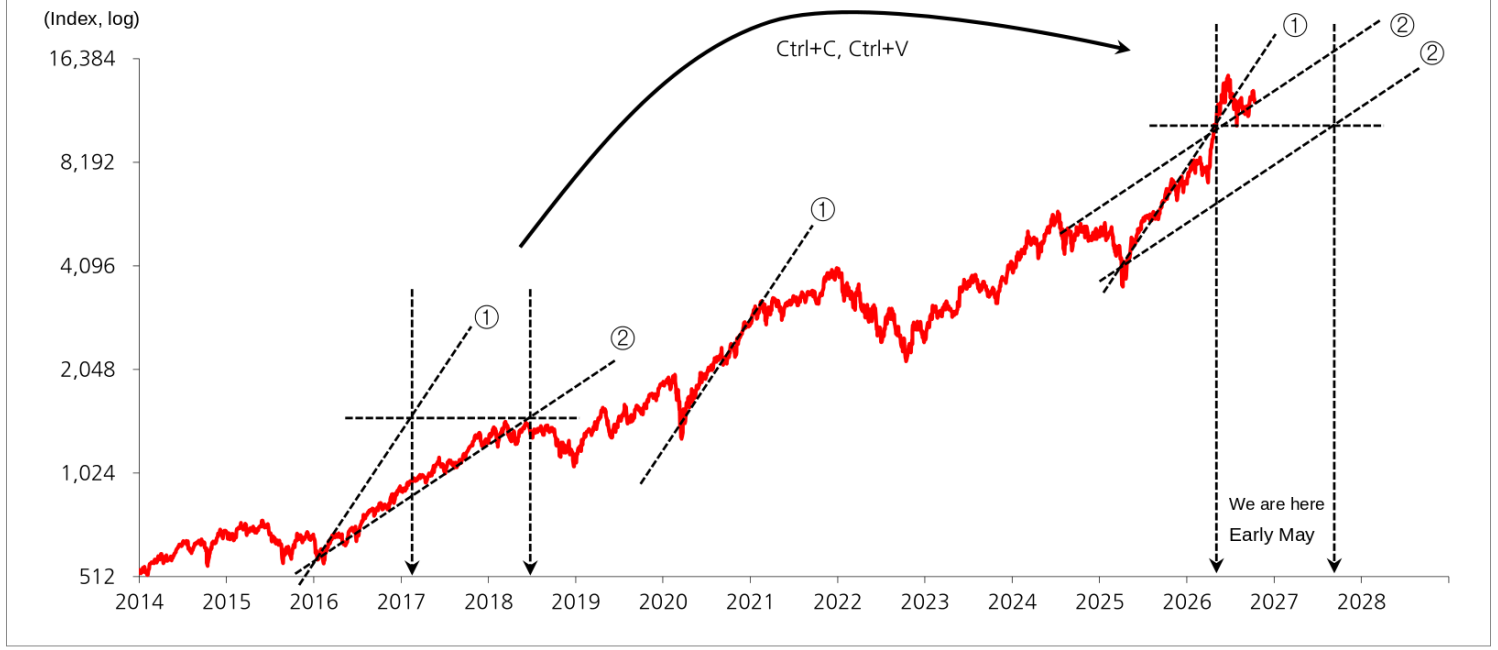


Source: Refinitiv, Hanwha Research Center

Semiconductors with a Lower-Level Slope

- The current semiconductor cycle is a record-setting cycle, with AI semiconductor and legacy semiconductor cycles running at the same time
- The 2016 cycle lasted about two years as a server cycle with the advent of the Fourth Industrial Revolution. The current cycle is more than twice as fast as 2016
- The Philadelphia Semiconductor Index formed a high on June 22 and is undergoing price and time corrections. The pace is now adjusted toward the slope of (2) rather than the slope of (1)

US Philadelphia Semiconductor Index Trend

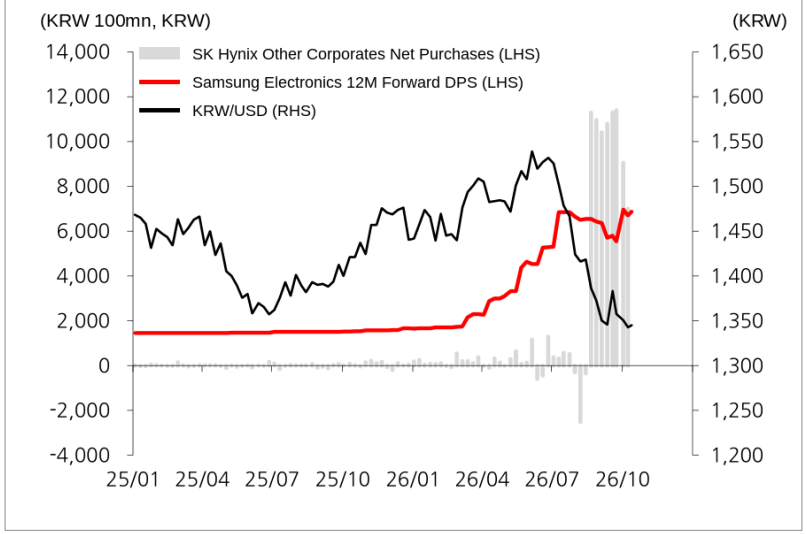


Source: Refinitiv, Hanwha Research Center

Only the Exchange Rate Has Come Down

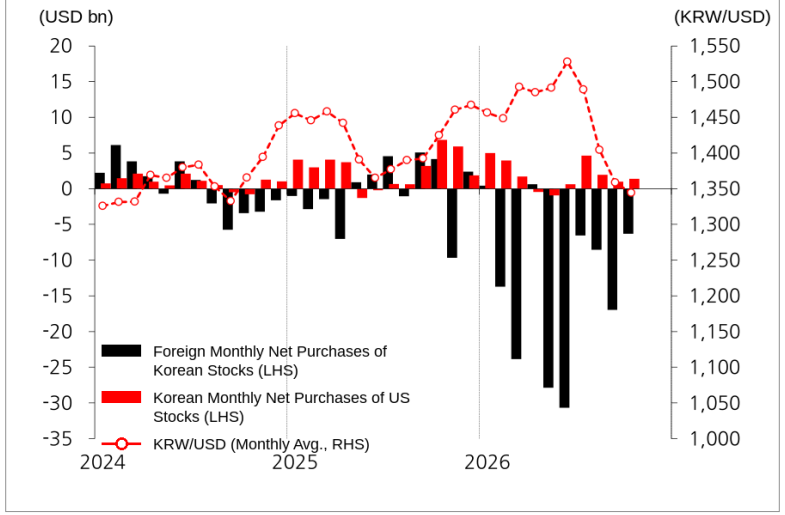
- During periods when the KRW/USD exchange rate moved toward won strength, events occurred such as 1) falling oil prices, 2) upward revisions to Samsung Electronics' dividend estimates, and 3) SK Hynix net purchases by other corporates
- Korean monthly net purchases of US stocks in August and September have not increased significantly, even considering the decline in the exchange rate
- The exchange rate range that major companies presented when giving 2026 guidance was roughly KRW 1,350-1,400. Below KRW 1,350 there is a negative impact
  - Hyundai Motor KRW 1,370, Samsung Biologics KRW 1,400, Kia KRW 1,370, HD Group including HD Hyundai Heavy Industries KRW 1,350, Doosan Bobcat KRW 1,360

Samsung Electronics and SK Hynix Shareholder Returns and KRW/USD



Source: Refinitiv, Hanwha Research Center

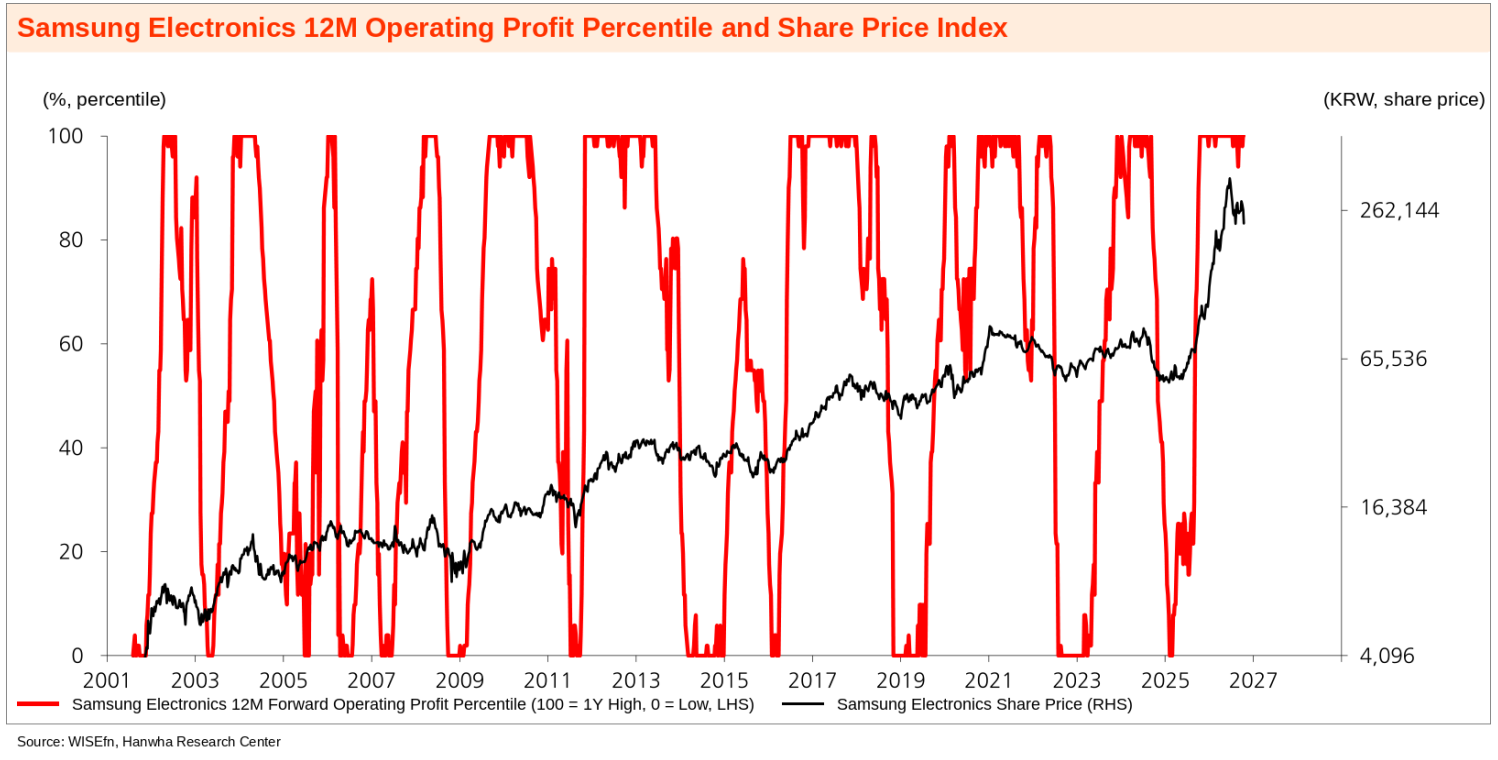
Foreign Net Purchases of Korean Equities and Korean Net Purchases of US Equities



Note: October cumulative through the 12th
Source: SEIBro, Hanwha Research Center

Pace of Samsung Electronics Earnings Upgrades vs. Pace of Won Strength

- Percentile is used for Samsung Electronics' earnings cycle. It expresses where the current estimate stands among the past year's earnings estimates
- Near 100 means the current 12-month forward operating profit estimate is at its highest level in the past year. 0 is the bottom. It is useful for share price troughs/peaks
- Last October, Samsung Electronics rose to the 100th percentile and has stayed there until now. However, in 6 of the last 14 weeks it temporarily dipped below 100% and fluctuated

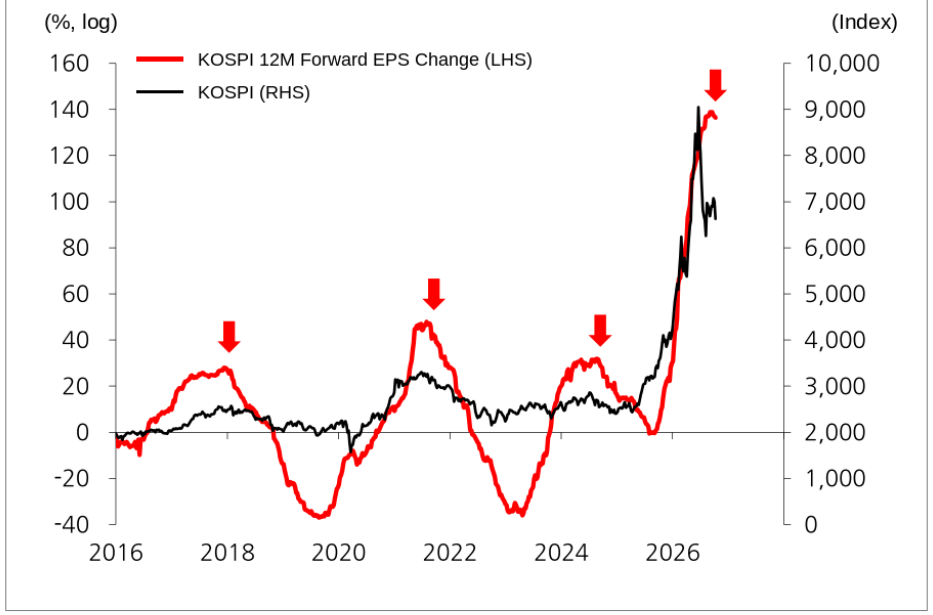


Source: WISEfn, Hanwha Research Center

Korea's Earnings Growth Rate Hit 139% and Is Now 136%

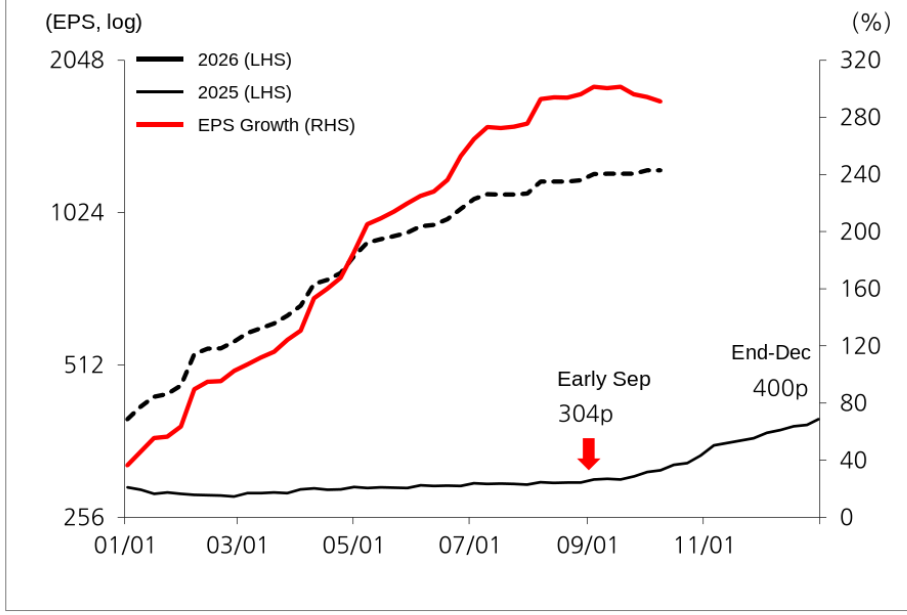
- The KOSPI moves closely with the 12-month forward EPS change rate. So far, the YoY earnings growth rate has supported the rise in share prices
- One point worth considering is that Korea's earnings estimates began to be revised up in earnest from last September, slightly later than the US
- As 2026 earnings upgrades stalled, the YoY growth rate fell slightly from its peak

KOSPI 12M Forward EPS Change and KOSPI



Source: Refinitiv, Hanwha Research Center

Last Year vs. This Year KOSPI 12M Forward EPS

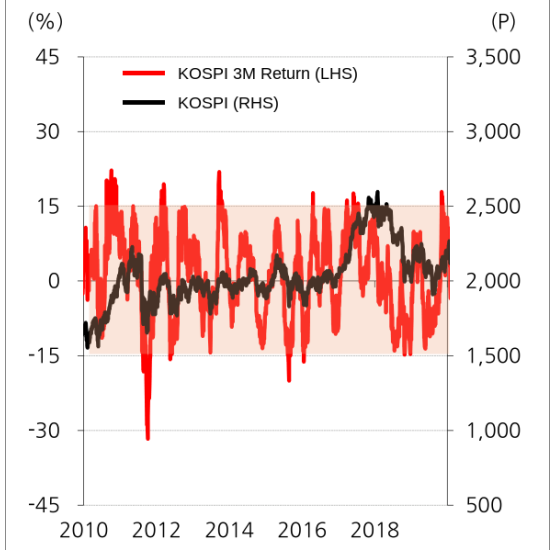


Source: Refinitiv, Hanwha Research Center

The Won Is Strong, So Neutral Despite How It Feels

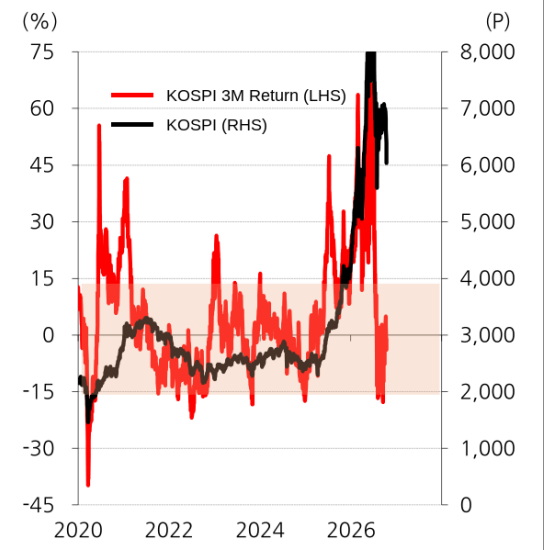
- KOSPI short-term troughs are relatively similar when the exchange rate is considered along with index moves. Historically, a USD-converted KOSPI 3-month return of around -15% has marked a short-term trough
- It broke below -10% from late July and recorded a low of -16.8% on August 6. This is the first signal since the martial law incident at the end of 2024
- The KOSPI recently plunged from around 7,000p to briefly below 6,000p, but accompanied by won strength, the 3-month return is -0.9%, a neutral level

USD-Converted KOSPI 3M Returns, 2010-19



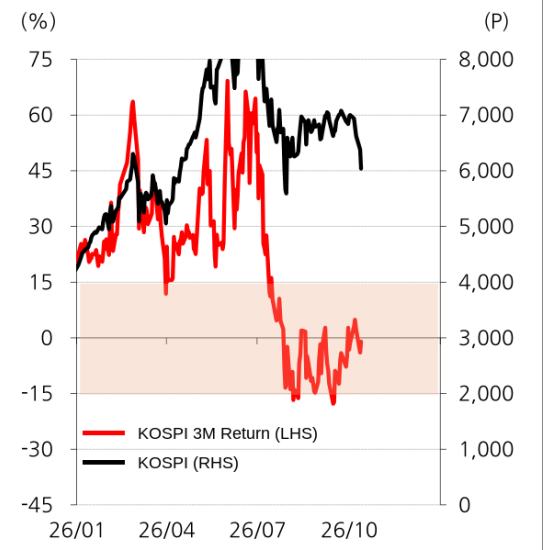
Source: WISEfn, Hanwha Research Center

USD-Converted KOSPI 3M Returns Since 2020



Source: WISEfn, Hanwha Research Center

USD-Converted KOSPI 3M Returns Since 2026

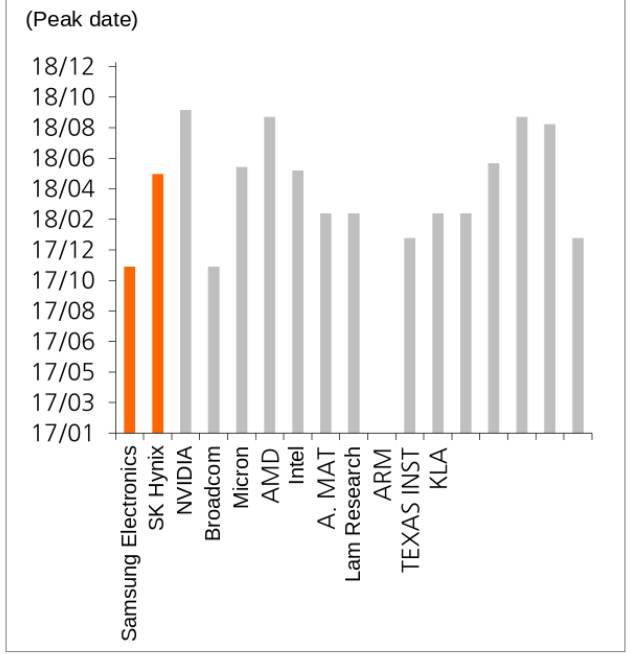


Source: WISEfn, Hanwha Research Center

NVIDIA and AMD Must Not Turn Down

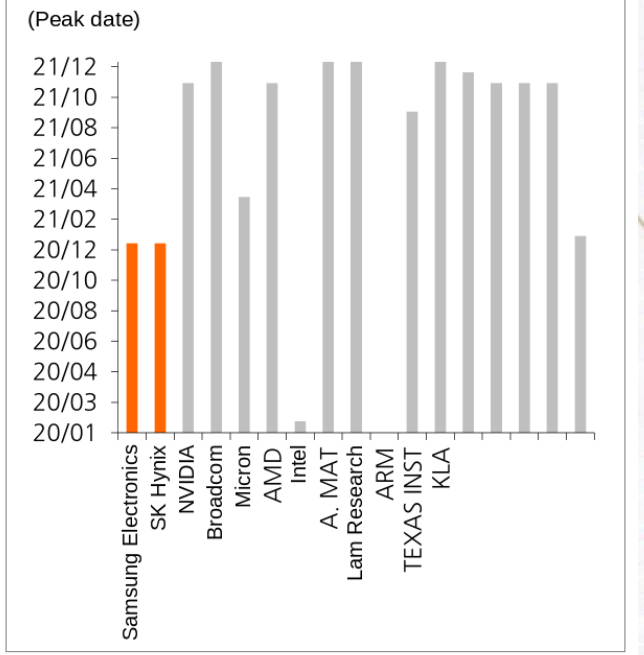
- We analyzed when semiconductor peaks appeared in the periods when semiconductor stocks did well: 2017-2018, 2020-2021 and since 2025
- What the first two periods have in common is that Korea's Samsung Electronics and SK Hynix peaked earliest among semiconductor stocks
- Judging by past time lags alone, NVIDIA and AMD are expected to keep raising their highs

Semiconductor Stock Peak Timing, 2017-2018



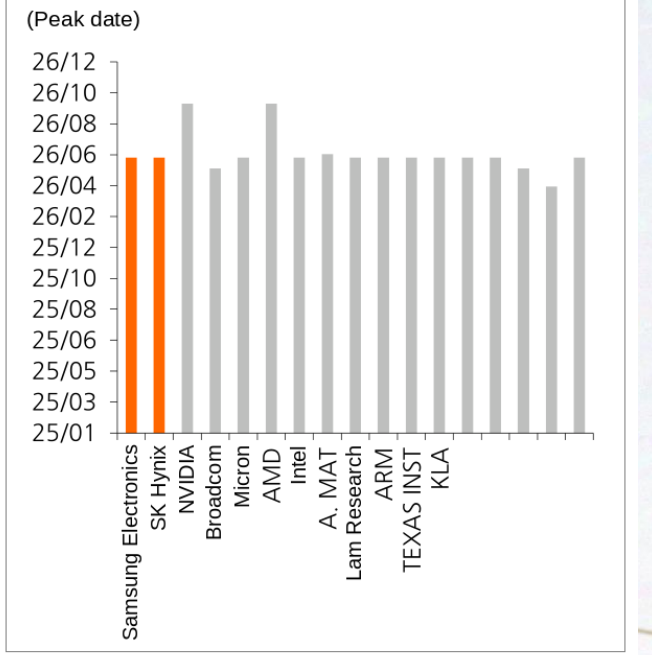
Source: Refinitiv, Hanwha Research Center

Semiconductor Stock Peak Timing, 2020-2021



Source: Refinitiv, Hanwha Research Center

Semiconductor Stock Peak Timing, Since 2025

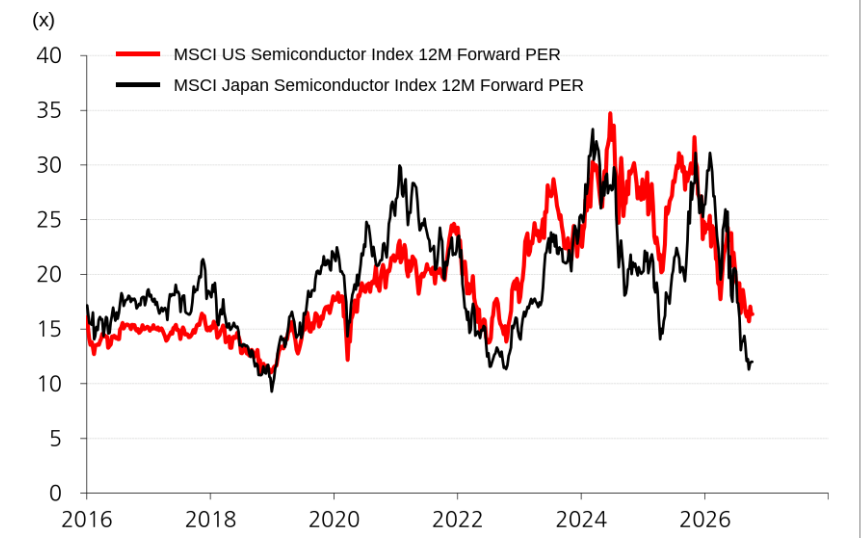


Source: Refinitiv, Hanwha Research Center

Japanese Semiconductors Are Cheaper than US Semiconductors

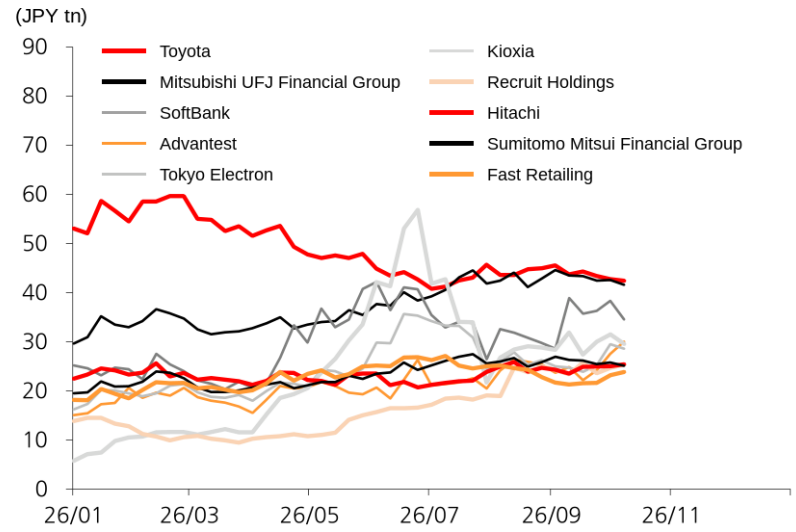
- Over the past 10 years, the 12-month forward PERs of US and Japanese semiconductors have alternated in leading. Before 2020, Japanese semiconductors were stronger
- Since 2020, US semiconductors have been stronger as their relative PER rose, but recently the Japanese semiconductor PER has fallen to 12x, down to its previous trough
- Unlike Korea, Japan does not have an overwhelming market cap weight in any particular company, and is relatively evenly distributed. The order of rebound is NVIDIA, Japanese semiconductors, Korean semiconductors

PER Trend of US and Japan Semiconductor Indices



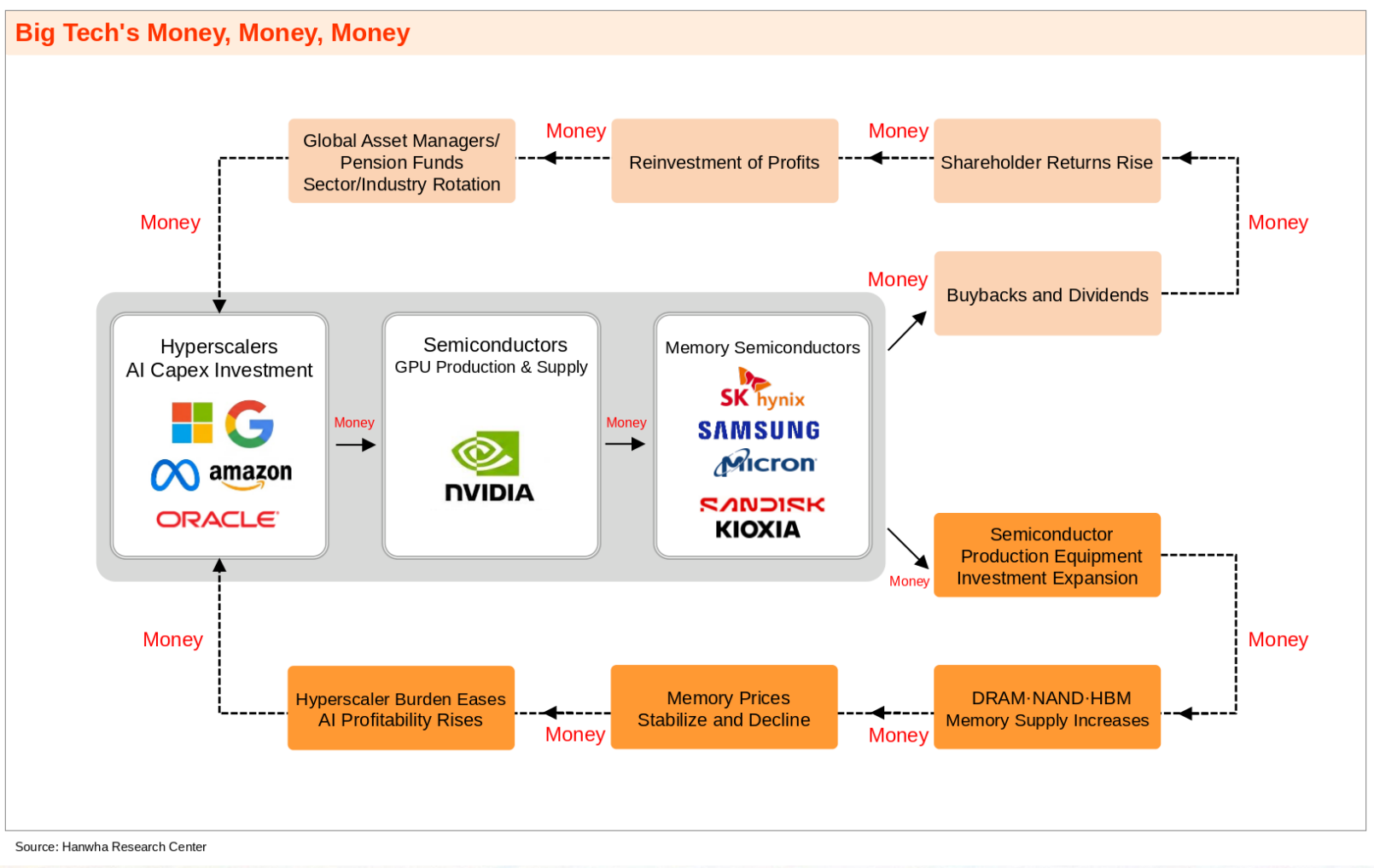
Source: Refinitiv, Hanwha Research Center

Nikkei 225 Top 10 by Market Cap



Source: Refinitiv, Hanwha Research Center

Semiconductor Virtuous Cycle Structure

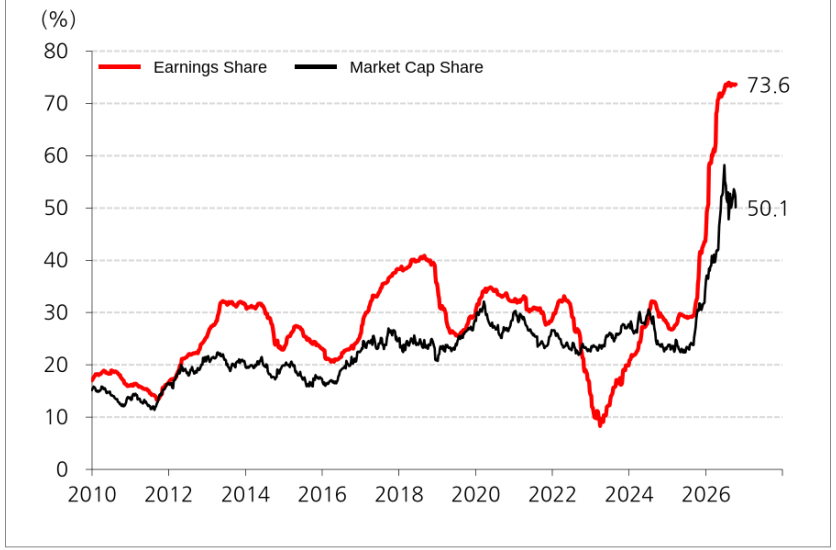


Source: Hanwha Research Center

How High Can It Go

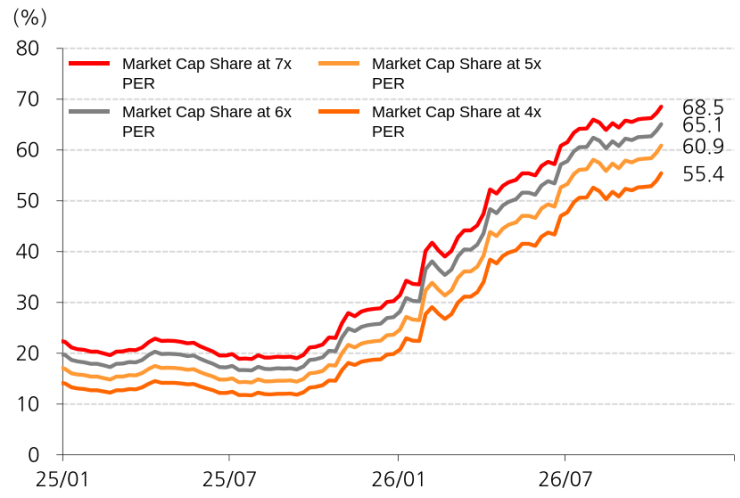
- Samsung Electronics' and SK Hynix's 12-month forward net income accounts for 73.6% of total KOSPI earnings. Their market cap share is about 50%. The peak was 58.3%
- The two stocks' combined 12-month forward PER is currently 3.2x. At 4x, the market cap share would be 55.4%, and at 7x it would rise to 68.5%
- Analysts' current target prices are at a PER of 7-8x. Aramco (government stake of about 90%) in Saudi Arabia's Tadawul index accounts for about 70% of market capitalization

Samsung Electronics and SK Hynix: Share of KOSPI Earnings and Market Cap



Source: WISEfn, Hanwha Research Center

Samsung Electronics and SK Hynix Market Cap Share by PER

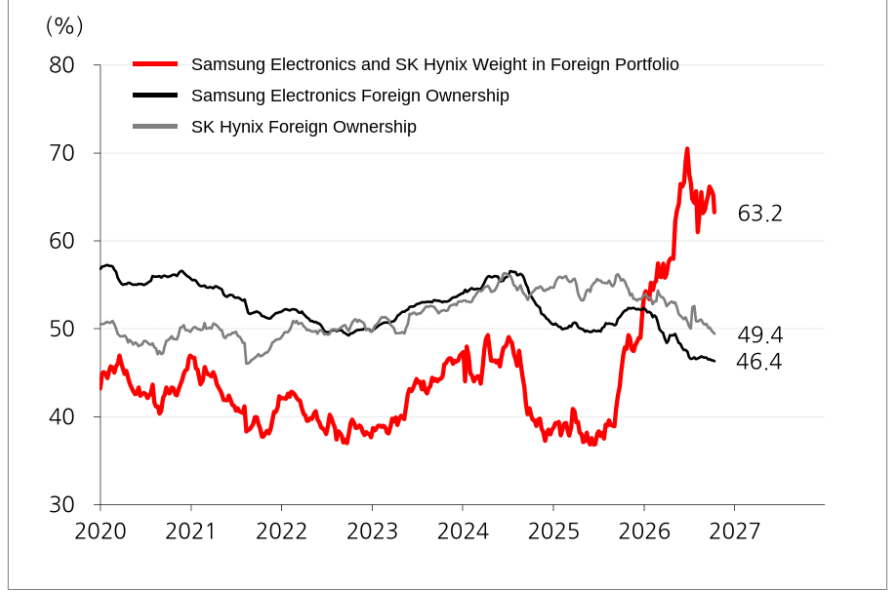


Note: Market cap share by PER calculated assuming all other conditions are equal
Source: WISEfn, Hanwha Research Center

A Weight That Does Not Fall Easily Even When Sold

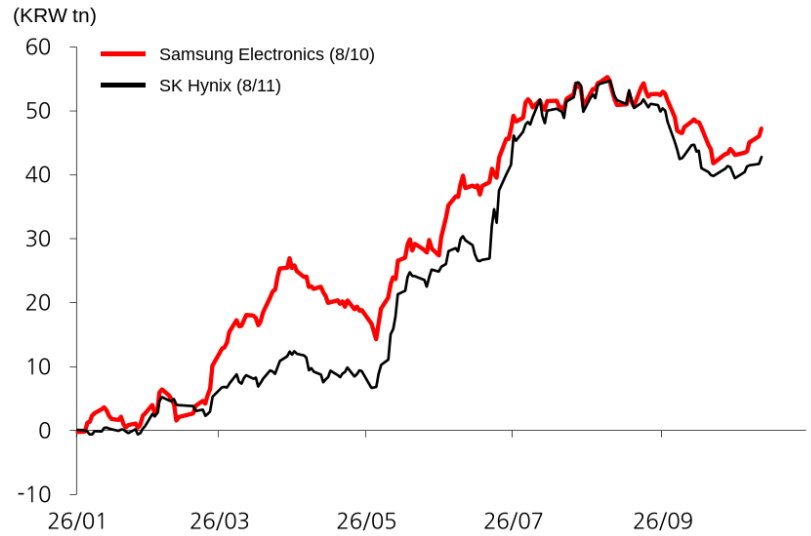
- The foreign ownership of Samsung Electronics and SK Hynix has been trending lower this year
- Within total KOSPI foreign market capitalization (the foreign portfolio), the foreign market cap share of Samsung Electronics and SK Hynix is currently 63.2%
- Even as each stock's foreign ownership falls, its portfolio weight does not fall easily. Individuals' cumulative net purchases peaked in early August

Samsung Electronics and SK Hynix Weight in Foreign Portfolio and Foreign Ownership of Each



Source: WISEfn, Hanwha Research Center

Cumulative Individual Net Purchases of Samsung Electronics and SK Hynix YTD (Peak in Parentheses)



Source: WISEfn, Hanwha Research Center

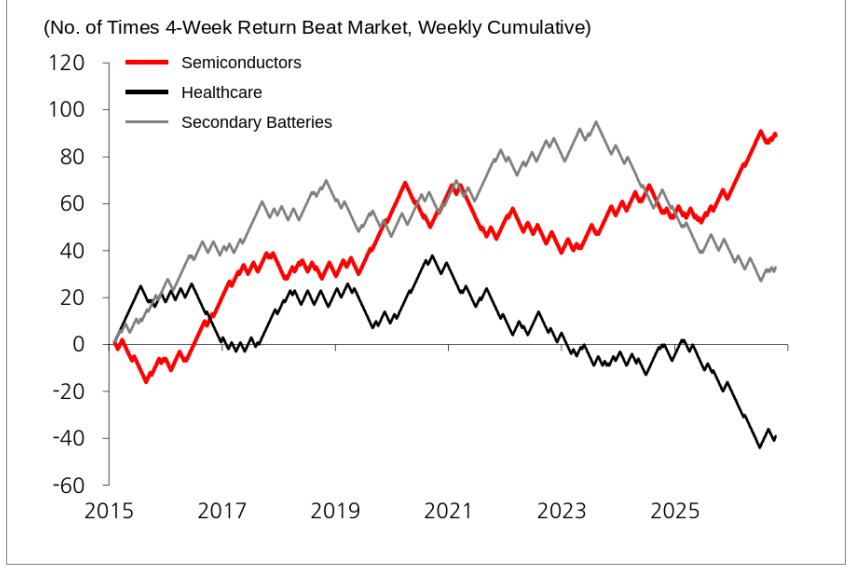
Money Flows from the Top Down (Share Price Change Since August; Stocks That Fell More than Semiconductors Shaded)



Bio Stocks Poised to Benefit

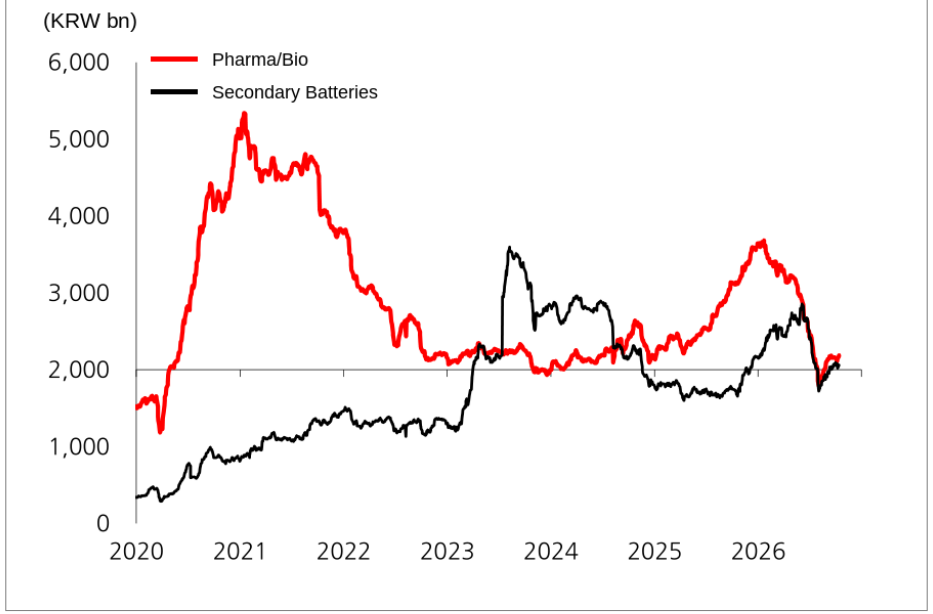
- Korea's No. 1 sector by market capitalization is semiconductors, with healthcare and secondary batteries alternating in second and third place
- Experience shows that when No. 1 pulls ahead of the market, Nos. 2 and 3 post returns below the market. That is, when semiconductors pause, Nos. 2 and 3 come back to life
- The pharma/bio sector's credit balance has fallen to slightly above KRW 2tn, the lowest level since 2020

Number of Times Semiconductors, Healthcare and Secondary Batteries Beat the Market over the Past Month (Cumulative)



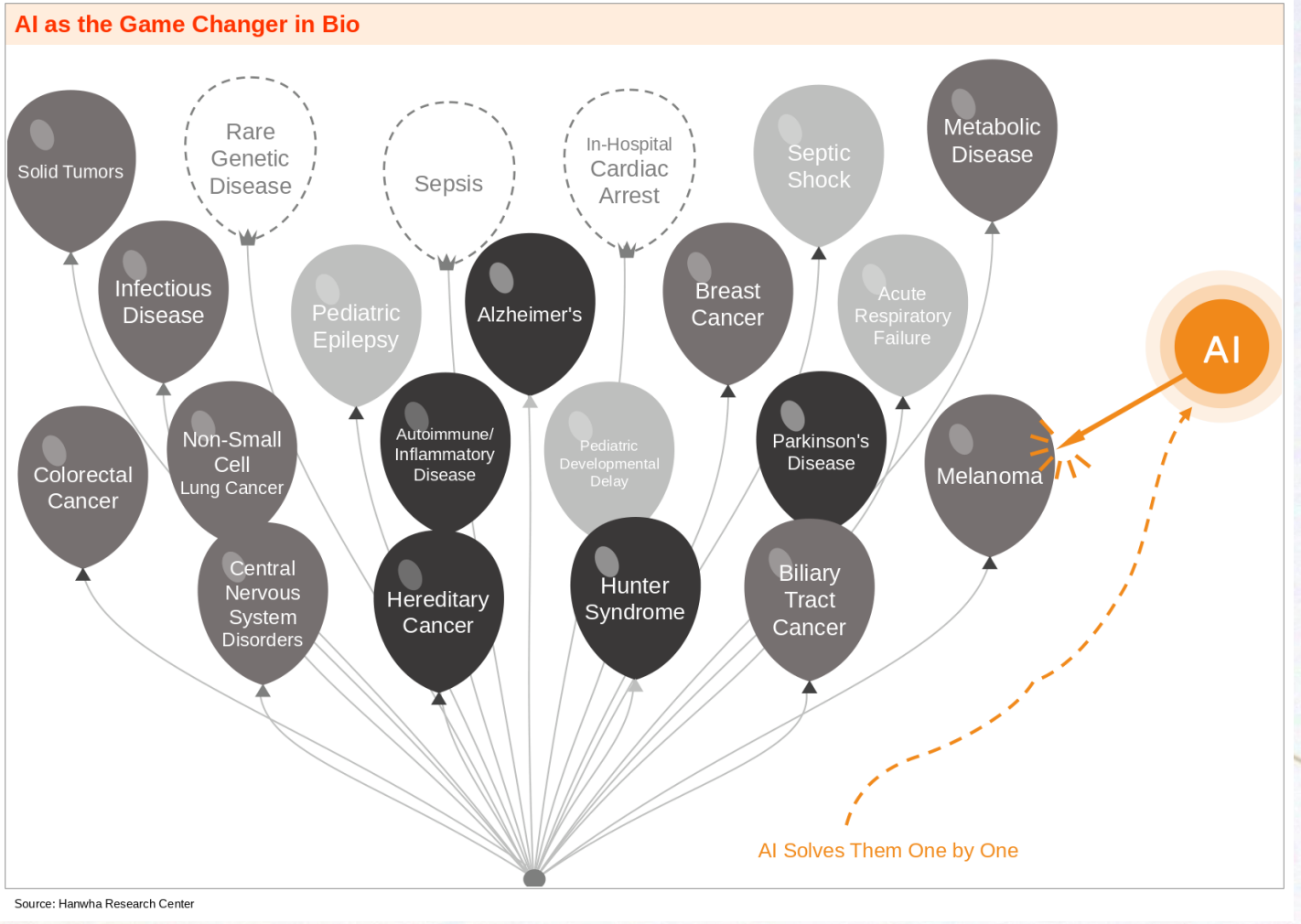
Source: WISEfn, Hanwha Research Center

Credit Balance Trend: Pharma/Bio and Secondary Batteries



Source: WISEfn, Hanwha Research Center

AI, the Game Changer for Bio



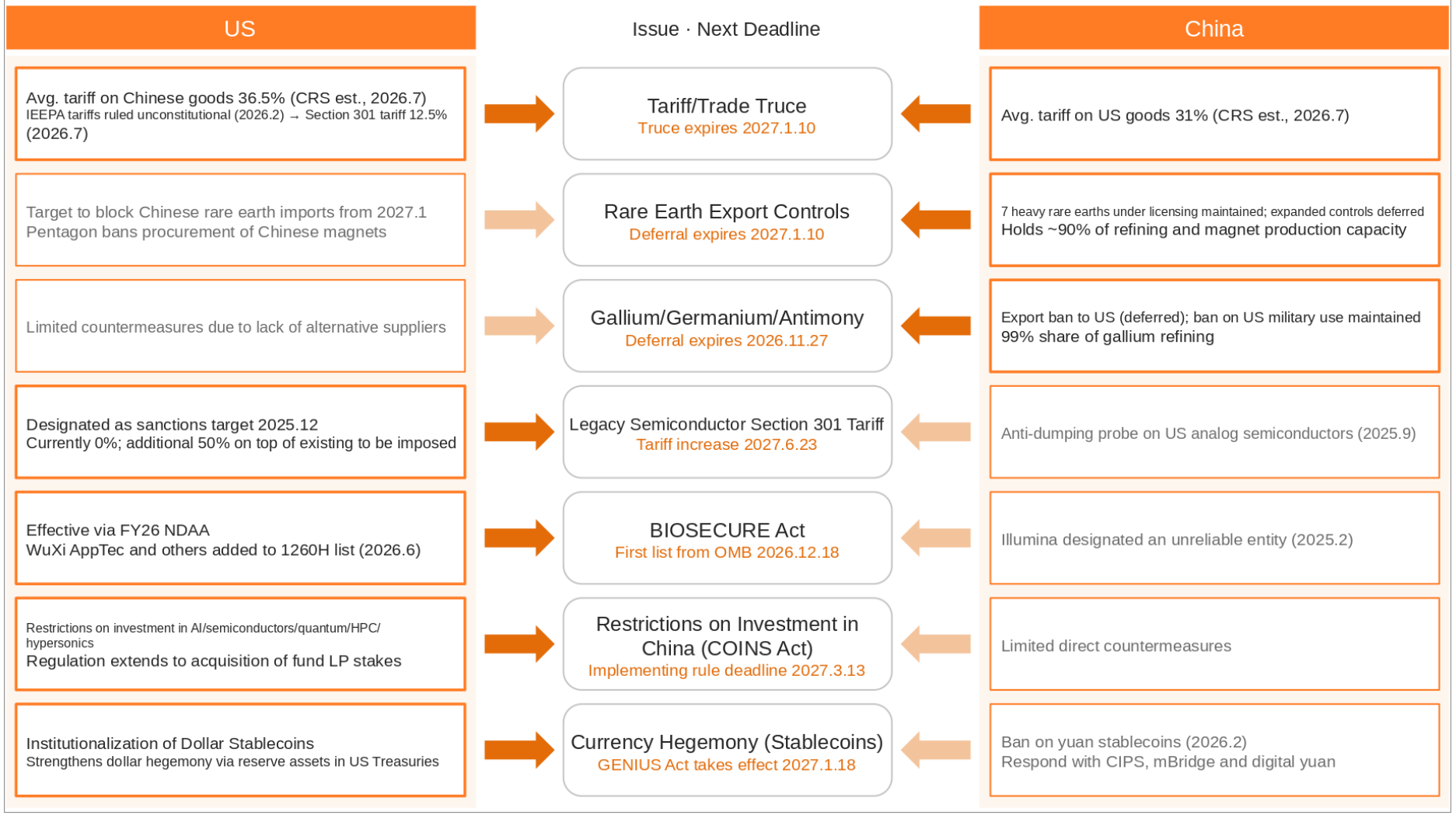
AI, the New Engine for Bio

[Table 13] AI, the New Engine for Bio (Solvability Shown as Filled Dots out of 3)

No.	Challenge	Description	AI Approach and Solution	Related Diseases	Solvability
1	Early warning of clinical deterioration (EWS)	Acute deterioration such as cardiac arrest or sepsis shows signals in vital signs several hours in advance, but it is impossible to continuously monitor every patient in a ward. Sepsis in particular has symptoms similar to other conditions, so it is easy to miss, and it worsens rapidly within hours	Vital signs and test values already accumulating in electronic medical records can be used without additional equipment. Dozens of patients are tracked continuously 24 hours a day using hundreds of time-series features. VUNO predicts the risk of cardiac arrest within 24 hours using only four items: blood pressure, pulse, respiration and body temperature. Prenosis scores sepsis risk with 22 indicators based on its own biobank of more than 100,000 cases; the first sepsis AI to receive FDA approval	Sepsis; septic shock; in-hospital cardiac arrest; acute respiratory failure	●●●
2	Interpreting the pathogenicity of genetic variants	Rare disease patients continue the "diagnostic odyssey", going from hospital to hospital without finding the cause. It takes 5-8 years on average to reach an accurate diagnosis. A single genome has about 4 million variants but no cause is indicated, so most cases end as "cause unknown"	Prioritizes variants by pathogenicity score, shortening diagnoses that took years to weeks. Based on AlphaFold, it predicts the change in protein structure when a single amino acid changes, so variants without records can also be judged. Google DeepMind developed AlphaGenome, which interprets the non-coding regions (98% of the genome), following AlphaMissense (classified 89% of 71 million missense variants)	Rare genetic diseases; genetic diseases of unknown cause; inborn errors of metabolism; pediatric epilepsy	●●●
3	Predicting protein-ligand binding affinity	The problem of calculating in advance, before synthesis, how strongly a candidate substance binds to a disease protein. Until now, molecules were made or calculated by physical simulation, and at 6-24 hours per molecule this cannot be used for thousands of candidates	Boltz developed Boltz-2, a binding affinity prediction model, achieving FEP-level accuracy more than 1,000 times faster. It also developed BoltzMol-1, a small-molecule hit discovery model applying this, which enables hit discovery at the early stage of drug development from a small number of compounds in less time and at lower cost. Pfizer, Takeda and others have introduced Boltz models across their research organizations	Small-molecule drugs overall (cancer, metabolic disease, autoimmune and inflammatory disease, central nervous system disorders)	●●○
4	Reading tumor immune phenotypes	Immuno-oncology drugs work only in "hot tumors", where T cells have entered the tumor. Most patients who are prescribed them do not benefit. Existing tests only look at "how many" and not "where", so the location of immune cells cannot be distinguished	Reading the location of immune cells from tissue images is the image analysis area where AI performs best → reading is standardized and faster. Lunit developed Lunit SCOPE IO, an immune phenotype classification model based on pathology slides. Roche acquired the digital pathology company PathAI and, combining it with its own platform, aims to strengthen biomarker discovery, drug development and precision diagnostics capabilities	Non-small cell lung cancer; melanoma; breast cancer; colorectal cancer	●●○
5	de novo antibody design	Until now, antibodies have been discovered rather than designed, requiring animal immune responses or searches of libraries of billions. The region where an antibody binds the target (CDR loop) has a flexible shape that is hard to predict, and therapeutic antibodies in particular require more precise design	Moving away from the existing trial-and-error approach, molecules can be designed to desired characteristics. A virtuous cycle in which experimental data improves the model and the improved model picks better candidates. Chai Discovery developed an antibody AI that predicts and reprograms molecular interactions. Platform technology license agreements with global pharmaceutical companies such as Eli Lilly, Pfizer and Novartis	Solid tumors; autoimmune disease; infectious disease	●●○
6	Designing blood-brain barrier (BBB) penetration	In the brain, the blood-brain barrier (BBB) blocks even drugs, so only about 0.1% of an administered antibody dose reaches the brain, and increasing the dose increases side effects. The solution is a drug-carrier "shuttle" molecule. However, binding affinity, toxicity and organ selectivity must be matched at the same time, and test-tube results are not reproduced in living bodies. Currently most rely on a single pathway (TfR)	A multivariable optimization problem with conflicting conditions → AI searches for the optimal combination. Manifold Bio developed an AI design platform that measures millions of candidates in living organisms. By repeating design → in vivo measurement → retraining, new brain entry pathways other than TfR can be explored. Joint development agreement with Roche for a next-generation BBB shuttle (aims for clinical entry with molecules verified to reach the brain)	Hunter syndrome; Alzheimer's disease; Parkinson's disease	●○○

Source: Hanwha Research Center

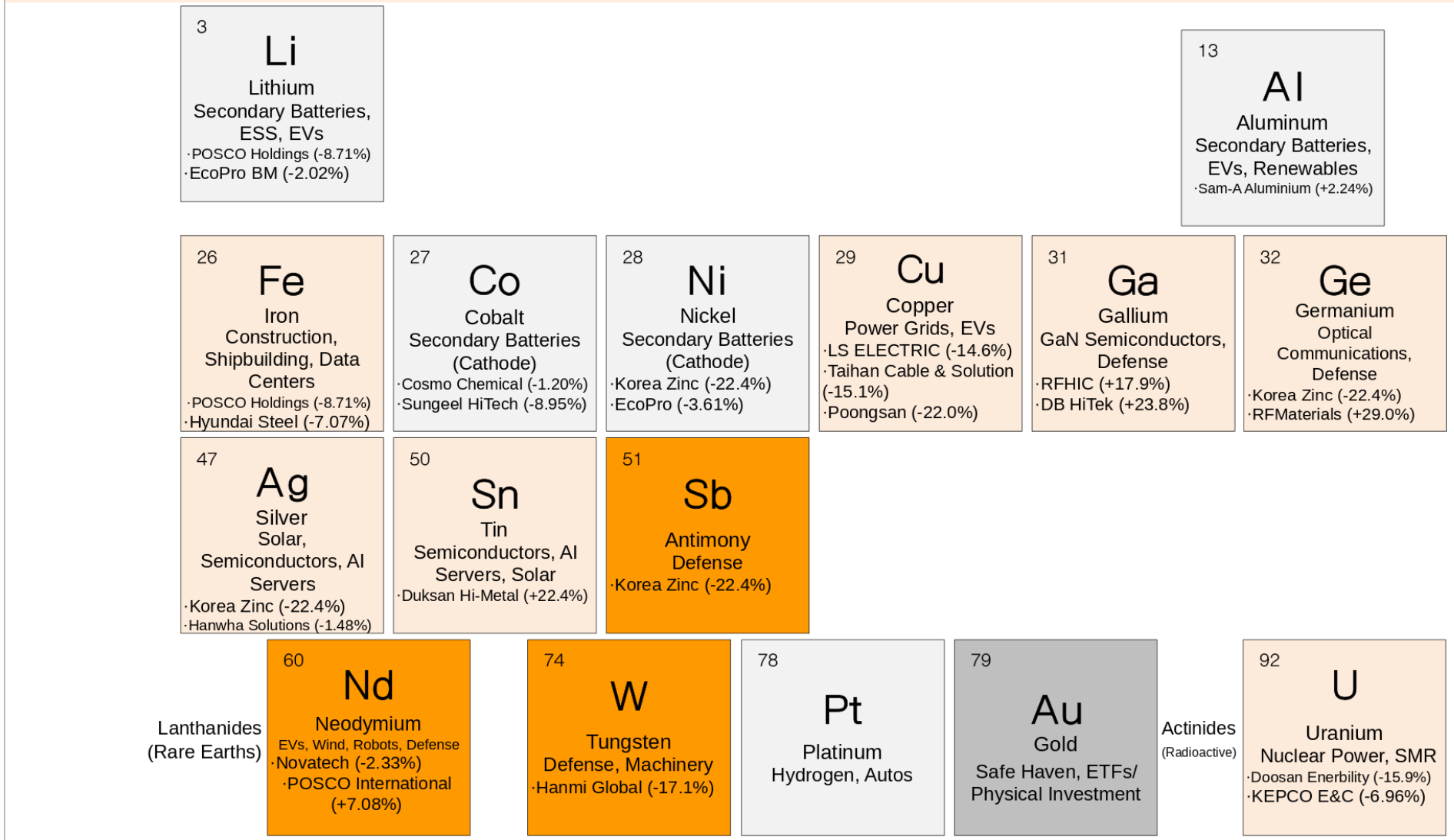
US-China Conflict: Offense and Defense by Issue



Note: Arrows indicate the direction of preemptive actions (dark color); light color is the counterpart's response

Source: Congressional Research Service, U.S. Trade Representative, U.S. Department of the Treasury, U.S. Department of Commerce, Ministry of Commerce, Press Reports, Hanwha Research Center

Key Raw Materials



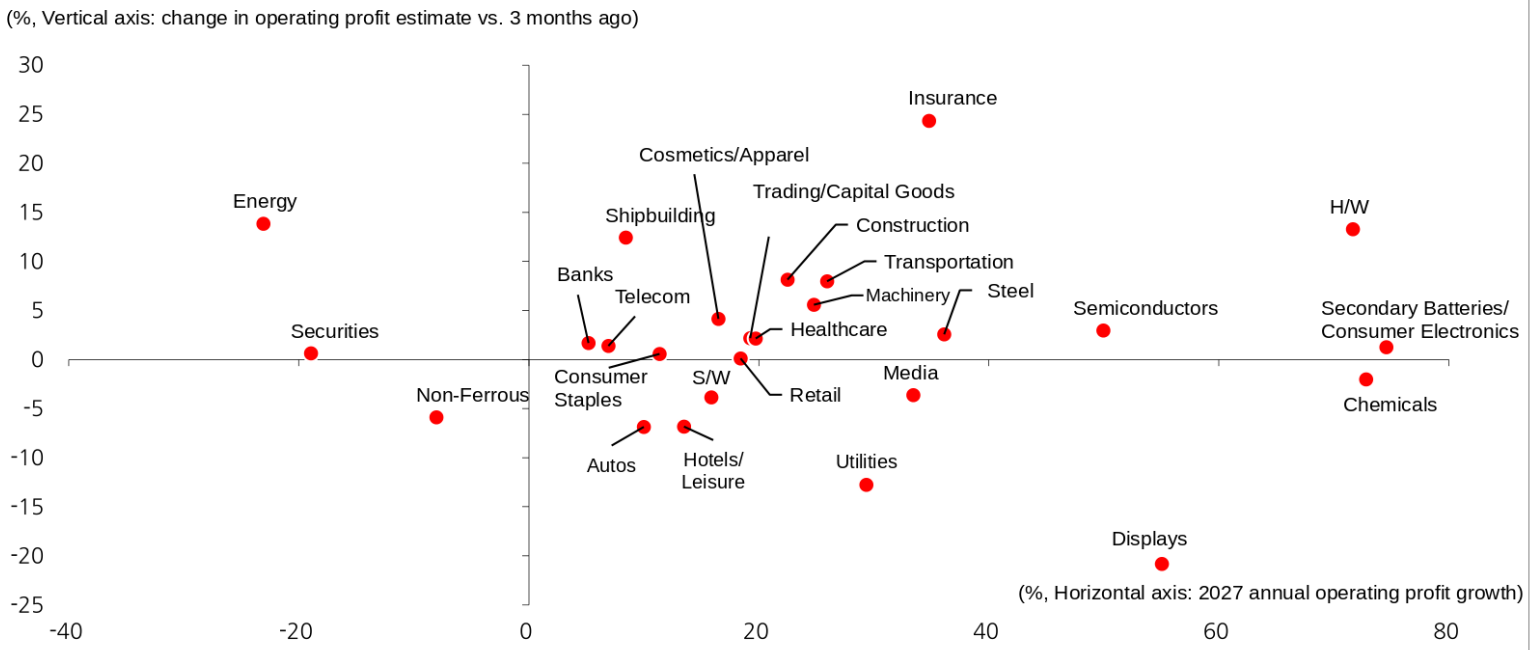
Note: Returns in parentheses are from the 8/31 close to the 10/14 close; shape colors group raw materials whose main industries are similar

Source: WISEfn, Hanwha Research Center

Sectors with a Brighter 2027 Outlook

- Sectors to the right of the origin (0) are those whose 2027 operating profit grows over 2026
- Sectors above the origin (0) are those whose 2027 operating profit estimates have been revised up over the last three months
  - Insurance, H/W, Shipbuilding, Construction and Transportation revised up

Sector 2027 Operating Profit Growth and 3M Change in 2027 Operating Profit Estimate (Net Income for Financials)



Source: WISEfn, Hanwha Research Center

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Hanwha Investment & Securities Research Center

Analyst Directory

Investment Strategy Team				
Hyun-kook Ahn	Team Head	Quant	3772-7646	hg.ahn@hanwha.com
Hye-yoon Lim	Research Fellow	Economy	3772-7728	hylim@hanwha.com
Sung-soo Kim	Research Fellow	Fixed Income	3772-7616	sungsoo.kim@hanwha.com
Jang-kwon Ha	Senior Researcher	Equity Strategy	3772-7717	research.ha@hanwha.com
Gyu-ho Choi	Senior Researcher	Economy	3772-7720	choi.gh@hanwha.com
Si-hwa Han	Senior Researcher	Credit	3772-7737	shhan1229@hanwha.com
Min-song Kim	Researcher	Investment Strategy RA	3772-7385	minsong05@hanwha.com
So-yeon Lim	Researcher	Investment Strategy RA	3772-7812	soyeon.lim@hanwha.com

Global Research Team				
Sang-hi Han, CFA	Team Head	Overseas Equities	3772-7695	sanghi.han@hanwha.com
Yu-jin Park	Senior Researcher	China Equities	3772-8217	yujin.park@hanwha.com
Seung-yun Shin	Researcher	US Equities	3772-7428	seungyun.shin@hanwha.com
Je-in Park	Researcher	Global Markets	3772-7380	jane8314@hanwha.com
Hye-rin Sihm	Researcher	US Equities RA	3772-7374	hyerin.sihm@hanwha.com
Jeong-hyeon Park	Researcher	China Equities RA	3772-7417	jeong.hyeon.park@hanwha.com

Acting Head of Research Center

Sang-hi Han, Overseas Equities

02.3772.7695 · sanghi.han@hanwha.com

Corporate Analysis Team				
So-hye Kim	Team Head	Internet/Games/Media	3772-7404	sohye.kim@hanwha.com
Sung-rae Kim	Principal Research Fellow	Autos/Parts	3772-7751	sr.kim@hanwha.com
Yu-rim Song	Principal Research Fellow	Construction/Utilities	3772-7152	yurim.song@hanwha.com
Se-yeon Park	Principal Research Fellow	ESG	3772-7406	shannon@hanwha.com
Do-ha Kim	Principal Research Fellow	Banks/Insurance	3772-7479	doha.kim@hanwha.com
Yu-jung Han	Research Fellow	Food & Beverage/Cosmetics	3772-7693	yujung.han@hanwha.com
Jin-hyeob Lee	Research Fellow	Retail/Apparel/Holding Cos.	3772-7638	jinhyeob.lee@hanwha.com
Su-jin Eom	Research Fellow	ESG	3772-7407	sujineom@hanwha.com
Su-young Park	Senior Researcher	Entertainment/Leisure/Transportation	3772-7634	suyoung.park.0202@hanwha.com
Jun-young Park	Senior Researcher	Semiconductors	3772-7481	jyp94@hanwha.com
Ji-woo Kwon	Senior Researcher	Steel/Non-Ferrous Metals	3772-7689	jiwoo.kwon@hanwha.com
Un-saem Chae	Senior Researcher	Defense/Machinery	3772-8419	unsam1@hanwha.com
Ye-in Kim	Researcher	Power Equipment/Wind/ESG	3772-8420	yein.kim@hanwha.com
Na-woo Kim	Researcher	Internet/Games/Media/Semiconductors RA	3772-7710	now.kim@hanwha.com

Digital Asset Research Team				
Yun-young Choi	Team Head	Digital Assets	3772-7402	yy.choy@hanwha.com
Jun-young Park	Researcher	Digital Assets RA	3772-8206	junyeong.park@hanwha.com

Hanwha Investment & Securities
Global Markets Sales (Korea Inbound)

Jung Hoon LIM
Sales & Coverage

jung.lim@hanwha.com

Office +82-2 37727467
Mobile +82-10 47106466

Chan Young SEO
Sales & Coverage

cyseo@hanwha.com

Office +82-2 37727279
Mobile +82-10 46172627

Team Contact (24/7)
Global Markets Sales Team

hhglobal@bloomberg.net

Office +82-2 37727400

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